Asbestos Surveys

At

Santa Fe Service Unit
Santa Fe, New Mexico

For

Albuquerque Area Indian Health Service
Division of Health Facilities
5300 Homestead Road, NE
Albuquerque, New Mexico 87110

By

U.S. Public Health Service
Federal Occupational Health
2201 Sixth Avenue, M/S Rx-21
Seattle, Washington 98121-2500

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Asbestos Building Inspections
Albuquerque Area Indian Health Service
Santa Fe Service Unit
Santa Fe, New Mexico

1.0 INTRODUCTION

As requested by the Albuquerque Area Indian Health Service, bulk sampling of suspect asbestos-containing building materials was conducted on eight (8) facility buildings at the Santa Fe Service Unit in Santa Fe, New Mexico. Sampling occurred on December 10 to December 21, 2006. Bulk samples of suspect asbestos-containing materials (ACMs) were collected throughout the interior and exterior of each building. Non-destructive sampling practices were utilized during the inspection.

Federal Occupational Health (FOH) representatives Mr. Donovan Denny, Mr. Landon Johnson, Ms. Stephanie Jury, Mr. James Marshall and Mrs. Suzette Numkena, performed the asbestos inspections. Each inspector holds current Asbestos Hazard Emergency Response Act (AHERA) accreditation as Asbestos Building Inspectors. Certifications for each inspector can be found in Appendix 4.

Sampling of suspect asbestos-containing materials was conducted using procedures similar to the Environmental Protection Agency (EPA) AHERA regulation, 40 CFR 763. Samples were analyzed by Fiberquant Analytical Services in Phoenix, Arizona. Fiberquant is fully accredited by the EPA-required National Voluntary Laboratory Accreditation Program (NVLAP) for analysis of asbestos bulk samples using polarized light microscopy, and is also an American Industrial Hygiene Association (AIHA) accredited laboratory.

2.0 DESCRIPTION OF BUILDINGS INSPECTED

The Santa Fe Service unit consists of eight (8) facility buildings. This includes a main hospital, dental clinic, maintenance building, and other medical and administrative support buildings. The majority of the buildings are modular style buildings. The main hospital (Building 2000) is a large structure built slab on grade. Building 333 is also a former clinic building, which is non-modular. All buildings are located on the same medical complex grounds. Refer to the bulk sample logs, Appendix 2, for a list of individual building materials found at each building. Building floor plans for each type of structure are located in Appendix 1.

Examples of suspect building materials identified and sampled throughout the buildings included floor tiles and associated mastics, sheet vinyl flooring and associated mastics, various carpet mastics, concrete, ceramic tile grout, gypsum drywall materials, wall textures, vinyl wall paper, ceramic wall tile, CMU, CMU mortar, stucco, plaster materials, cement asbestos panels, various ceiling tiles, various mastics, sink undercoating, various insulations, gaskets, fireproofing, thermal pipe insulation, covebase mastic, various caulks, foam, window putty, duct tape, and roof materials.

Suspect roof materials were not sampled due to winter weather conditions for the following buildings 2003, 334, and 336.

3.0 SURVEY PROCEDURES

Each room of each building was assigned a unique functional space number. Each functional space was visually inspected for suspect asbestos-containing materials. Materials were mapped and quantified on maps provided by the IHS or hand drawn maps.

Each suspect ACM was designated as a distinct homogeneous area, which is a single material, uniform in texture and appearance, installed at one time, and unlikely to consist of more than one type or formulation of material. The inspector touched each suspect material in order to determine friability.

A sufficient number of samples were collected of each material to satisfy the Occupational Safety and Health Administration (OSHA) and the National Emission Standard for Hazardous Air Pollutants (NESHAP) regulations for the determination of asbestos content. Bulk sample logs are located in Appendix 1.

Each sample was assigned a unique sample identification number and assessed for damage and friability classifications. The samples, with Chain of Custody documentation were delivered to Fiberquant Analytical Services in Phoenix, Arizona. Fiberquant is fully accredited by the EPA-required National Voluntary Laboratory Accreditation Program (NVLAP) for analysis of asbestos bulk samples using polarized light microscopy, and is also an American Industrial Hygiene Association (AIHA) accredited laboratory. Laboratory analysis reports are located in Appendix 3.

4.0 FINDINGS AND COST ESTIMATES

Table 1 summarizes each building inspected and provides hyperlinks (electronic copy) to Bulk Sample Results & Programming Cost Estimate Tables and Analytical Laboratory Results. The information is also located in the appendices.

Table 1: Santa Fe Service Unit Buildings Inspected for Asbestos Containing Materials and Associated Hyperlinks *(electronic version only)*

Location	Building Number		Unit Type *(Appendix 1)*	Asbestos Present	Bulk Sample Results and Programming Cost Estimate *(Appendix 2)*	Analytical Laboratory Results *(Appendix 3)*
Santa Fe	333		Service	**Yes**	✓	☐
Santa Fe	334		Office	**Yes**	✓	☐
Santa Fe	335		Service	**Yes**	✓	☐
Santa Fe	336		Service	**Yes**	✓	☐
Santa Fe	2000		Hospital	**Yes**	✓	☐
Santa Fe	2001		Service	**Yes**	✓	☐
Santa Fe	2002		Storage	**Yes**	✓	☐
Santa Fe	2003		Clinic	No	✓	☐

The Bulk Sample Results Table (Appendix 2) provides detailed information on materials that tested positive or negative for asbestos content. A separate Bulk Sample Results Table is provided for each building inspected.

Programming Cost Estimates (Appendix 2) were developed for positive asbestos-containing materials found in each Building. The Programming Cost Estimates were derived by use of the Means Cost Guide and by consultation with experienced local contractors. The pricing reflects current industry rates. Future fluctuations in industry pricing, both natural and seasonal, may affect this estimate. In addition, Service Units outside of metropolitan areas may incur additional costs due to travel requirements on the part of abatement contractors. For purposes of this estimate, an 18% mark-up for logistical costs is included.

This estimate is for abatement contractor costs only. Additional costs, which are not included in this estimate, are those of an environmental consultant and replacement of building materials that have been removed. The total estimated cost for the removal of asbestos-containing materials from the Santa Fe Service Unit is **$269,125.85.**

5.0 ACM DESCRIPTIONS

Descriptions of materials found to contain asbestos are as follows:

Floor tile is classified as a non-friable NESHAP Category I material. This material is currently in good condition and intact floor tile poses little to no health threat. Removal or disturbance to this material should be performed by appropriately trained and certified personnel. Removed flooring material should be disposed at a landfill that accepts non-friable asbestos waste. Landfills should be checked for acceptance of this material prior to disposal.

Floor tile mastic is classified as a non-friable NESHAP Category I material. This material is currently in good condition. Floor tile mastic that is beneath floor tile, i.e., inaccessible, poses little to no health threat. Removal or disturbance to this material should be performed by appropriately trained and certified personnel. Removed flooring material should be disposed at a landfill that accepts non-friable asbestos waste. Landfills should be checked for acceptance of this material prior to disposal.

Sheet vinyl flooring and associated materials are classified as friable NESHAP Regulated Asbestos-Containing Material (RACM). The material is currently in good condition and intact sheet vinyl flooring poses little to no health threat. Removal or disturbance to this material should be performed by appropriately trained and certified personnel. Removed sheet vinyl flooring and associated materials should be disposed at a landfill that is approved by the Environmental Protection Agency (EPA) for acceptance of friable asbestos waste. Landfills should be checked for acceptance of this material prior to disposal.

Pipe elbow/fitting insulation is classified as friable NESHAP Regulated Asbestos-Containing Material (RACM). Intact pipe insulation poses little to no health threat. Pipe insulation inside wall cavities also poses little to no health threat. Removal or disturbance to this material is considered an OSHA Class I action, and should be performed by appropriately trained and certified personnel. Removed insulation should be disposed at a landfill that is approved by the Environmental Protection Agency (EPA) for acceptance of friable asbestos waste. Landfills should be checked for acceptance of this material prior to disposal.

Exterior building mastic is classified as non-friable NESHAP Category II material. The material is undamaged; therefore, it poses little to no health threat. Removal of mastic and/or affected materials or other disturbance to this material, should be performed by appropriately trained and certified personnel. Removed materials should be disposed at a landfill that accepts non-friable asbestos waste. Landfills should be checked for acceptance of this material prior to disposal.

Duct seam tape is classified as a friable NESHAP Regulated Asbestos-Containing Material (RACM). The intact duct seam poses little to no health threat. Removal or disturbance to this material should be performed by appropriately trained and certified personnel. Removed duct seam tape should be disposed at a landfill that is approved by the Environmental Protection Agency (EPA) for acceptance of friable asbestos waste. Landfills should be checked for acceptance of this material prior to disposal.

Door and window frame caulk is classified as non-friable NESHAP Category II material. The material is currently in good condition and intact door frame caulk poses little to no health threat. Removal or disturbance to this material should be performed by appropriately trained and certified personnel. Removed caulk material should be disposed at a landfill that accepts non-friable asbestos waste. Landfills should be checked for acceptance of this material prior to disposal.

Sink undercoating is classified as non-friable NESHAP Category II material. The material is currently in good condition and undamaged sink undercoating poses little to no health threat. Removal of this material typically involves the physical removal of the entire sink (whole) and should be performed by appropriately trained and certified personnel. Removed materials should be disposed at a landfill that

accepts non-friable asbestos waste. Landfills should be checked for acceptance of this material prior to disposal.

Window putty is classified as non-friable NESHAP Category II material. The material is currently in good condition and intact window putty poses little to no health threat. Removal of this material should be performed by appropriately trained and certified personnel using intact removal methods. Removed materials should be disposed at a landfill that accepts non-friable asbestos waste. Landfills should be checked for acceptance of this material prior to disposal.

Covebase mastic is classified as non-friable NESHAP Category II material. The material is undamaged and inaccessible; therefore, it poses little to no health threat. Removal of covebase mastic and/or affected materials (e.g. covebase with mastic adhered), or other disturbance to this material, should be performed by appropriately trained and certified personnel. Removed materials should be disposed at a landfill that accepts non-friable asbestos waste. Landfills should be checked for acceptance of this material prior to disposal.

Roofing material is classified as non-friable NESHAP Category I material. Removal or disturbance to this material should be performed by appropriately trained and certified personnel. Removed roofing material should be disposed at a landfill that accepts non-friable asbestos waste. Landfills should be checked for acceptance of this material prior to disposal.

Roof penetration mastic is classified as non-friable NESHAP Category I material. Removal or disturbance to this material should be performed by appropriately trained and certified personnel. Removed penetration mastic should be disposed at a landfill that accepts non-friable asbestos waste. Landfills should be checked for acceptance of this material prior to disposal.

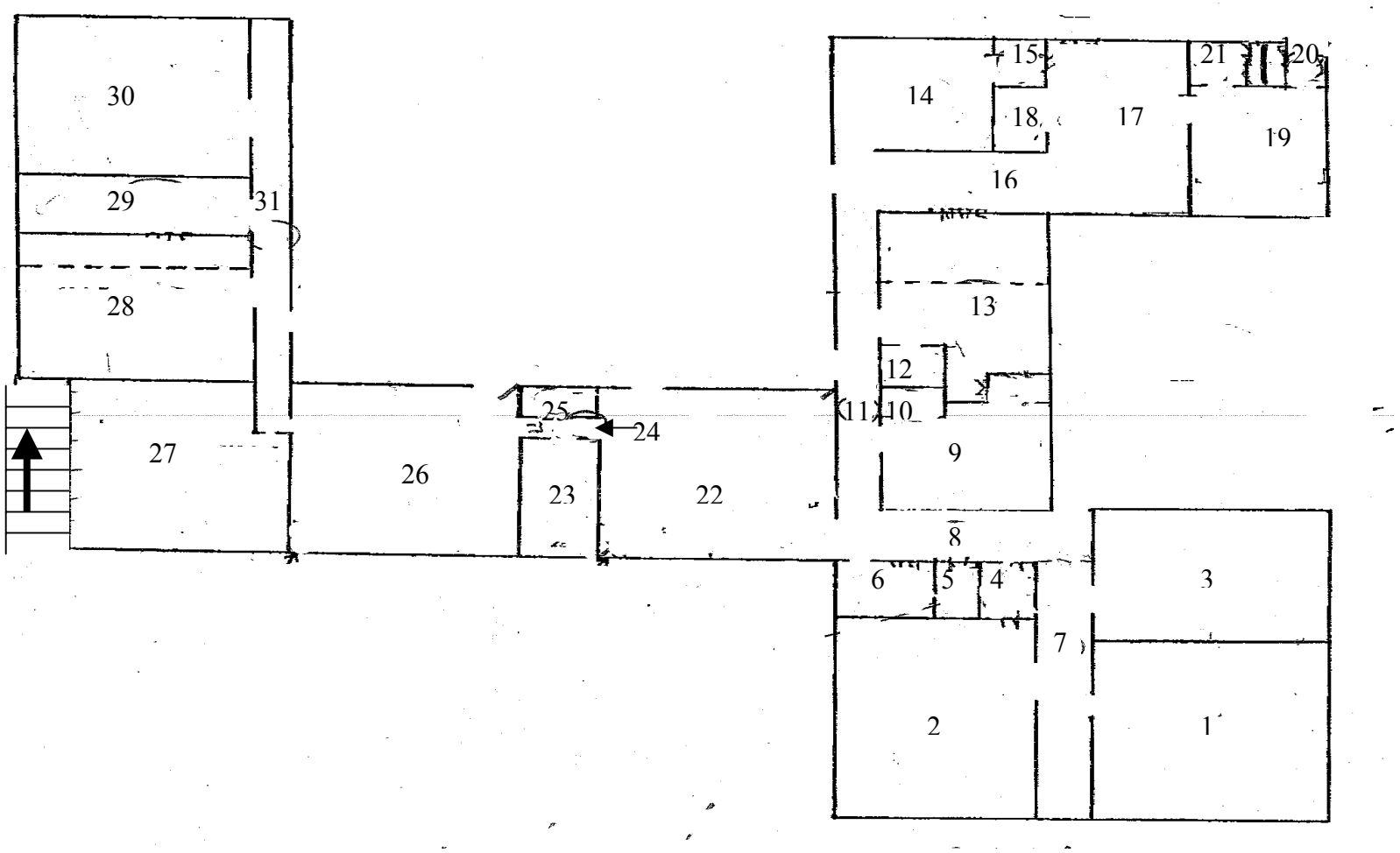
6.0 PROJECT LIMITATIONS

This Project was performed using, as a minimum, practices consistent with standards acceptable within the industry at this time, and a level of diligence typically exercised by industrial hygiene and environmental consultants performing similar services.

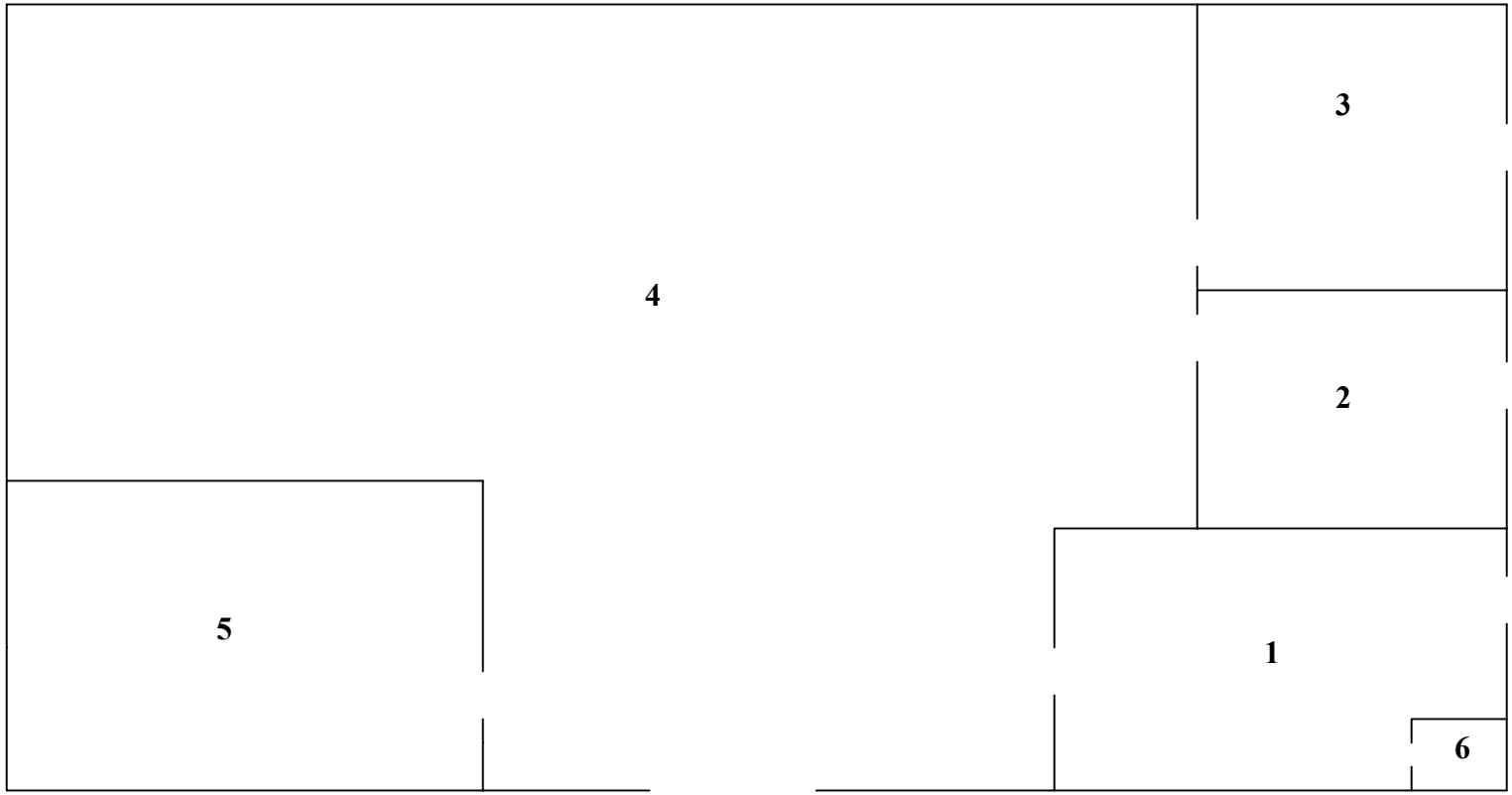
The procedures used in this investigation attempt to establish a balance between the competing goals of limiting investigative and reporting costs and time, and reducing the uncertainty about unknown conditions. Therefore, because the findings of this report were derived from the scope, costs, time, and other limitations, the conclusions should not be construed as a guarantee that all environmental or occupational hazards have been identified and fully evaluated. Where sample collection and testing have been performed, FOH's professional opinions are based in part on the interpretation of data from discrete sampling locations that may not represent conditions at non-sampled locations. FOH assumes no responsibility for omissions or errors resulting from inaccurate information, or data, provided by sources outside of FOH or from omissions or errors in public records.

Furthermore, it is emphasized that the final decision on how much risk to accept always remains with the client since FOH is not in a position to fully understand all of the client's needs. Clients with a greater aversion to risk may want to take additional actions while others, with less aversion to risk, may want to take no further action.

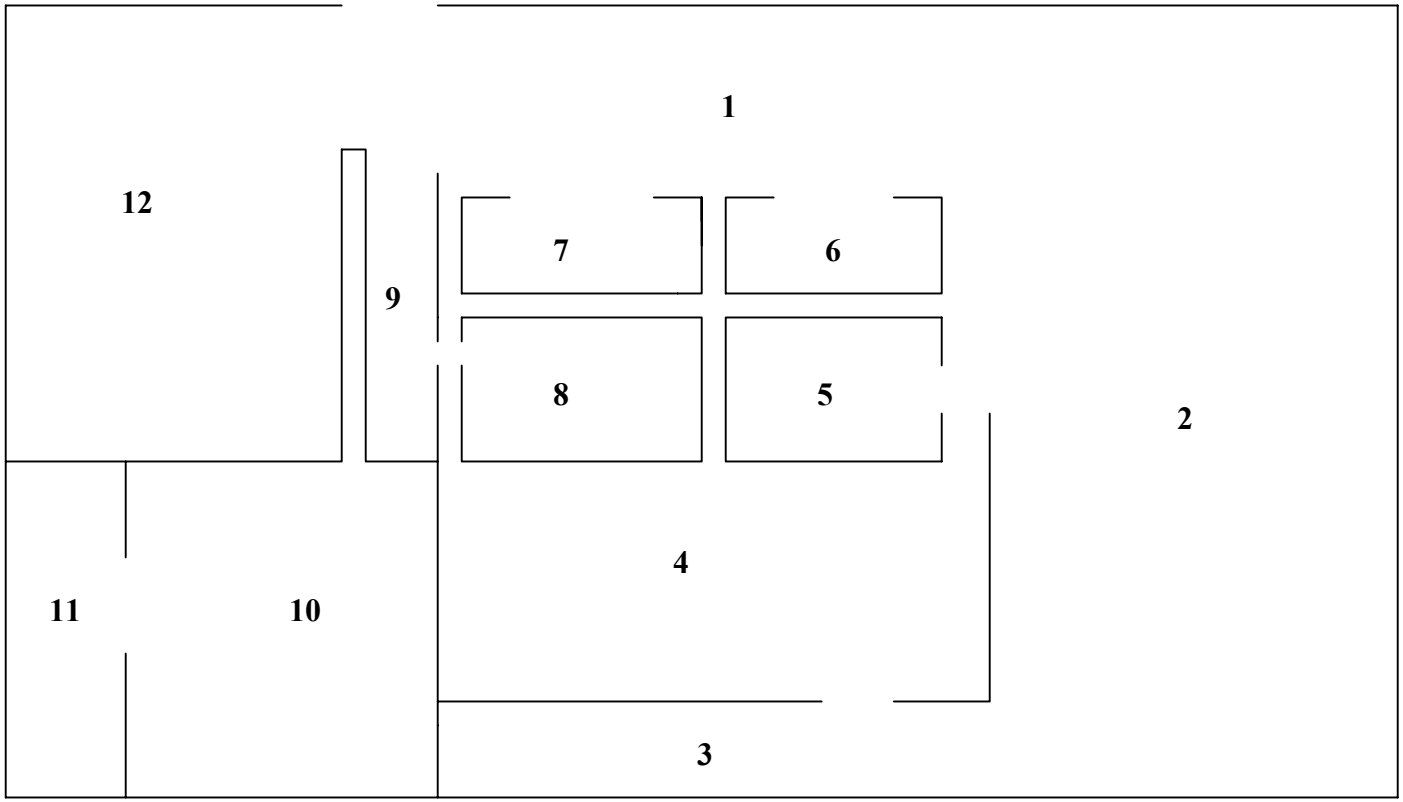
APPENDIX 1
Functional Space Maps
Bulk Sample Logs



U.S. Public Health Service Federal Occupational Health 2201 Sixth Avenue, M/S Rx21 Seattle, Washington 98121-2500	Project: ALBUQUERQUE AREA INDIAN HEALTH SERVICE SANTA FE SERVICE UNIT Santa Fe, New Mexico	**Functional Space Map** Building 333

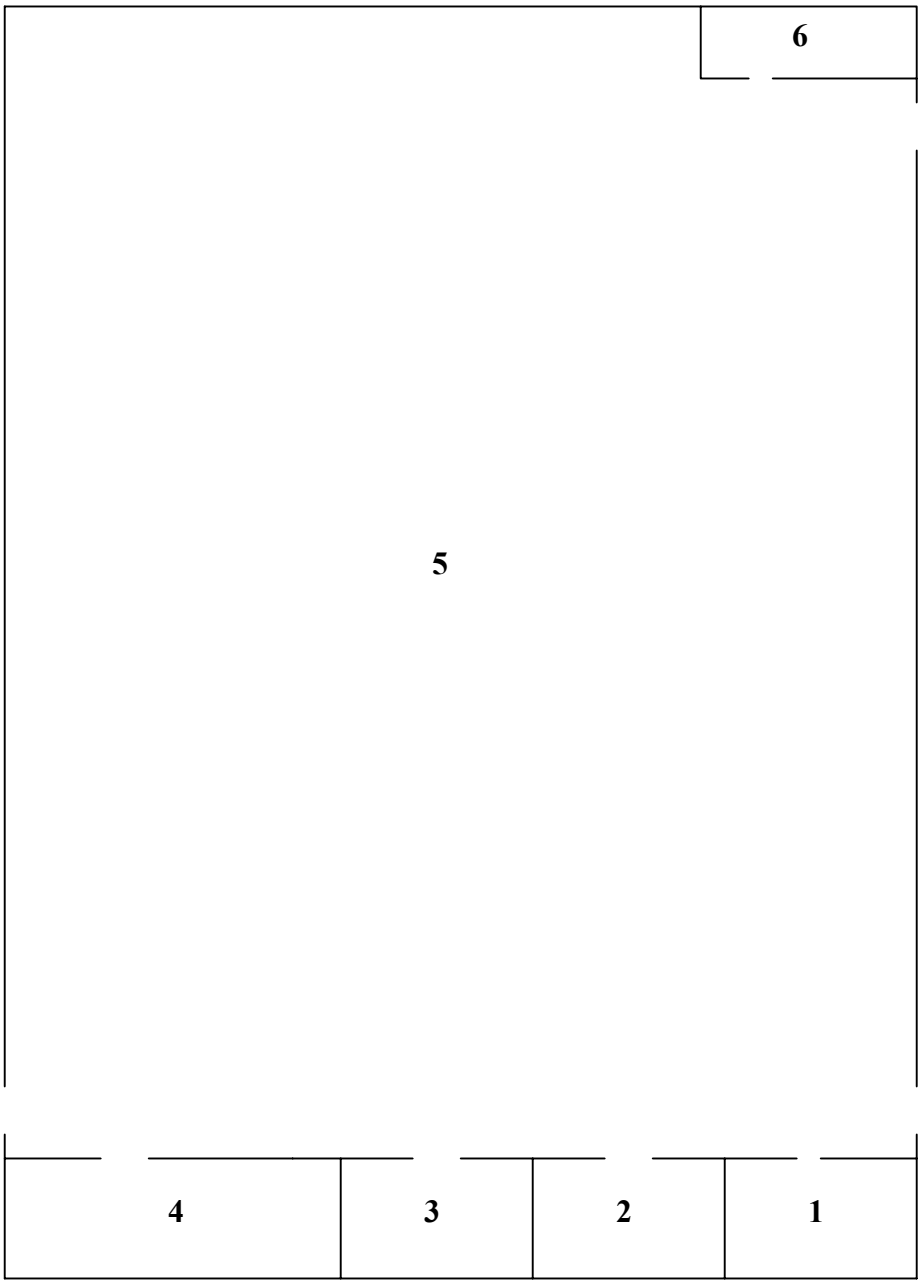


U.S. Public Health Service Federal Occupational Health 2201 Sixth Avenue, M/S Rx21 Seattle, Washington 98121-2500	Project: ALBUQUERQUE AREA INDIAN HEALTH SERVICE SANTA FE SERVICE UNIT Santa Fe, New Mexico	**Functional Space Map** Building 334



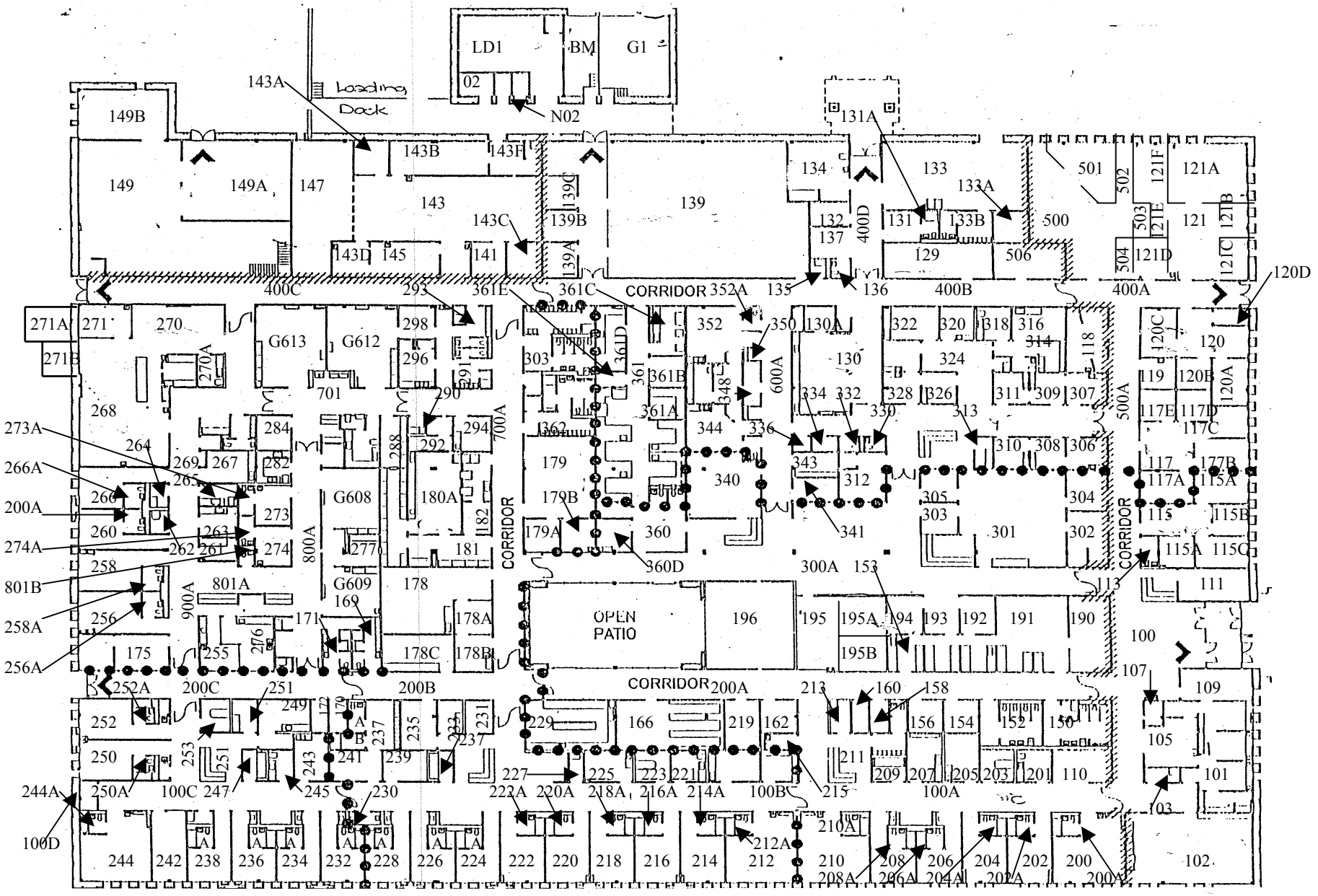
NORTH

U.S. Public Health Service Federal Occupational Health 2201 Sixth Avenue, M/S Rx21 Seattle, Washington 98121-2500	Project: ALBUQUERQUE AREA INDIAN HEALTH SERVICE SANTA FE SERVICE UNIT Santa Fe, New Mexico	**Functional Space Map** Building 335

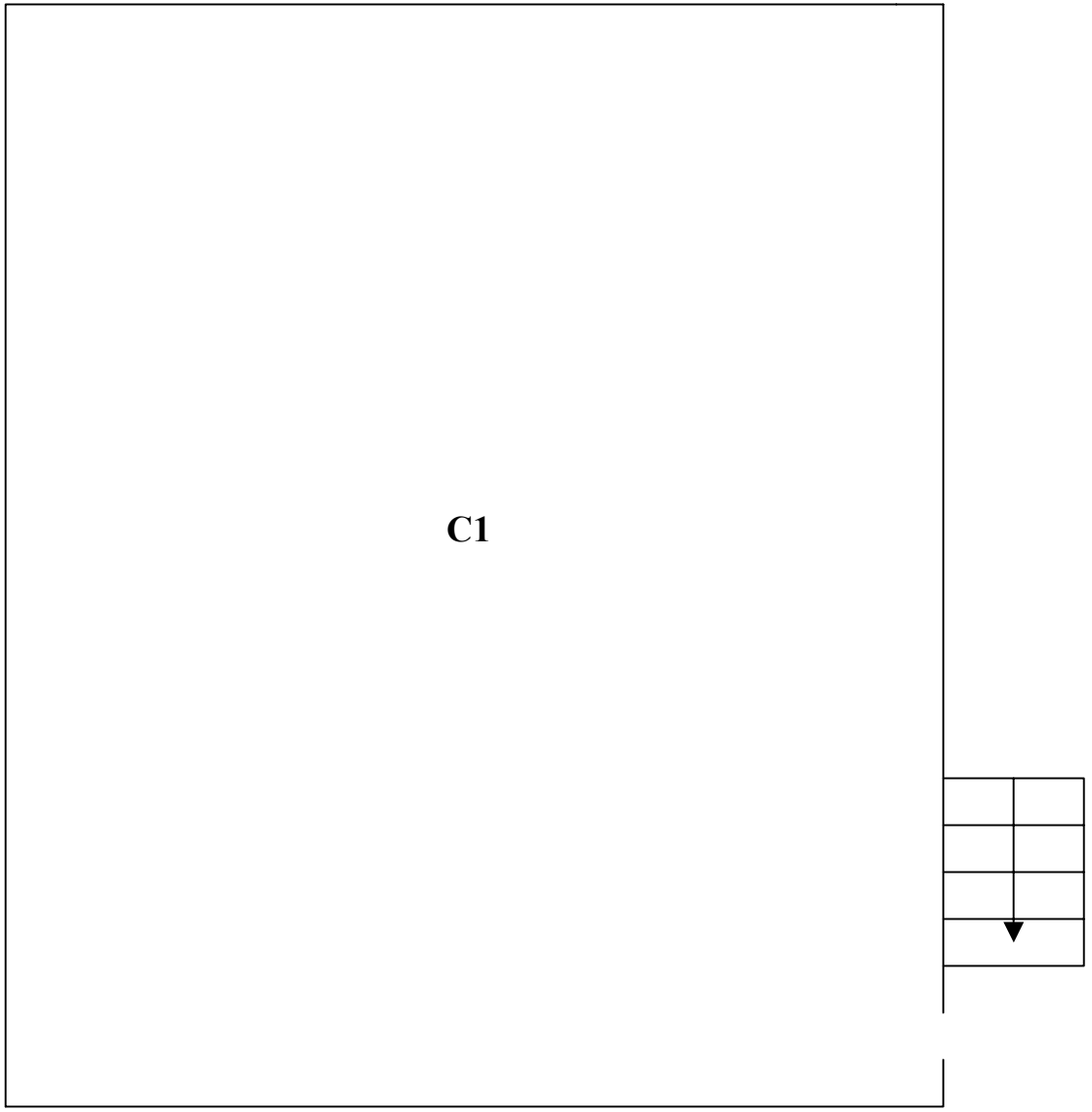


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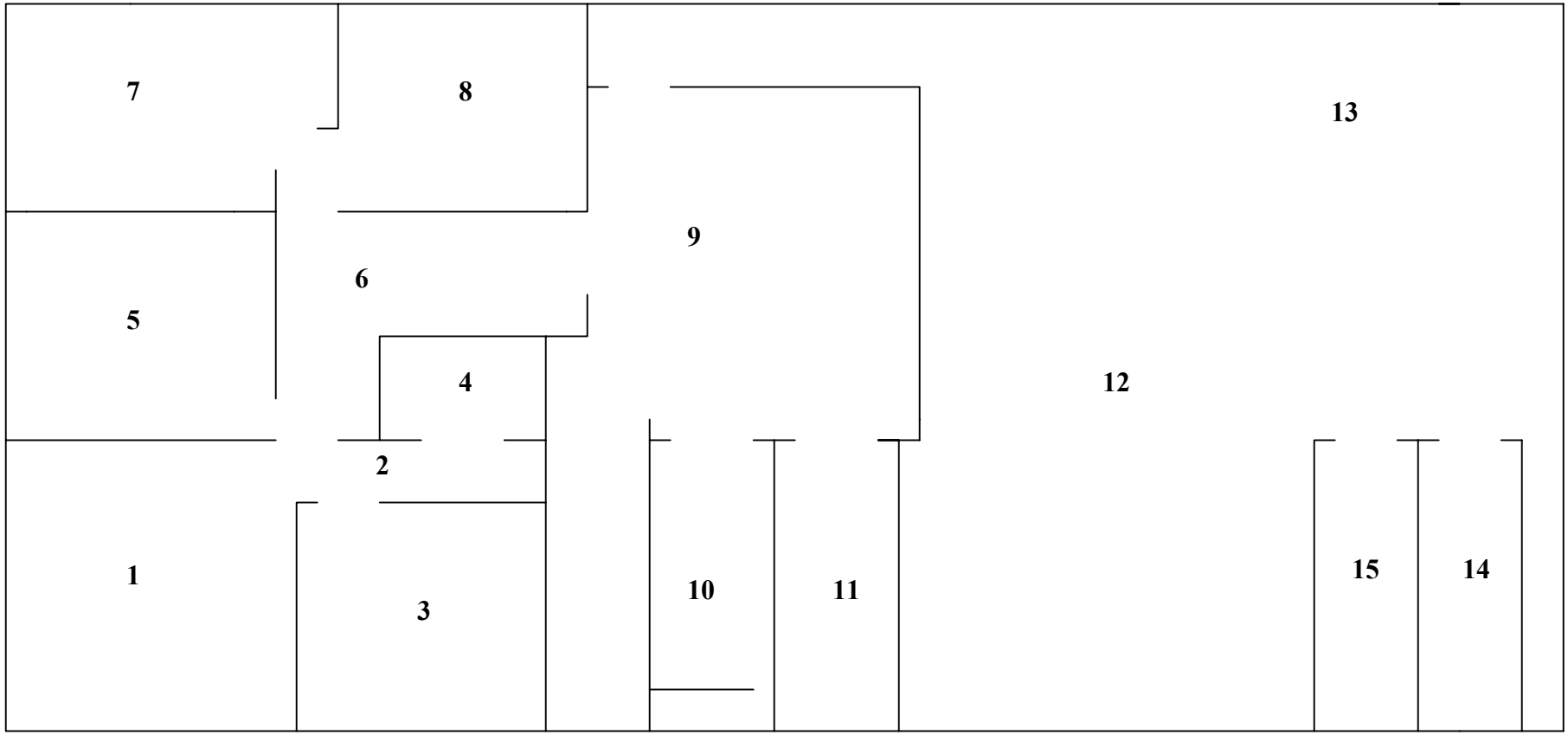
U.S. Public Health Service Federal Occupational Health 2201 Sixth Avenue, M/S Rx21 Seattle, Washington 98121-2500	Project: ALBUQUERQUE AREA INDIAN HEALTH SERVICE SANTA FE SERVICE UNIT Santa Fe, New Mexico	**Functional Space Map** Building 336



U.S. Public Health Service Federal Occupational Health 2201 Sixth Avenue, M/S Rx21 Seattle, Washington 98121-2500	Project: ALBUQUERQUE AREA INDIAN HEALTH SERVICE SANTA FE SERVICE UNIT Santa Fe, New Mexico	**Functional Space Map** Santa Fe Hospital Building 2000

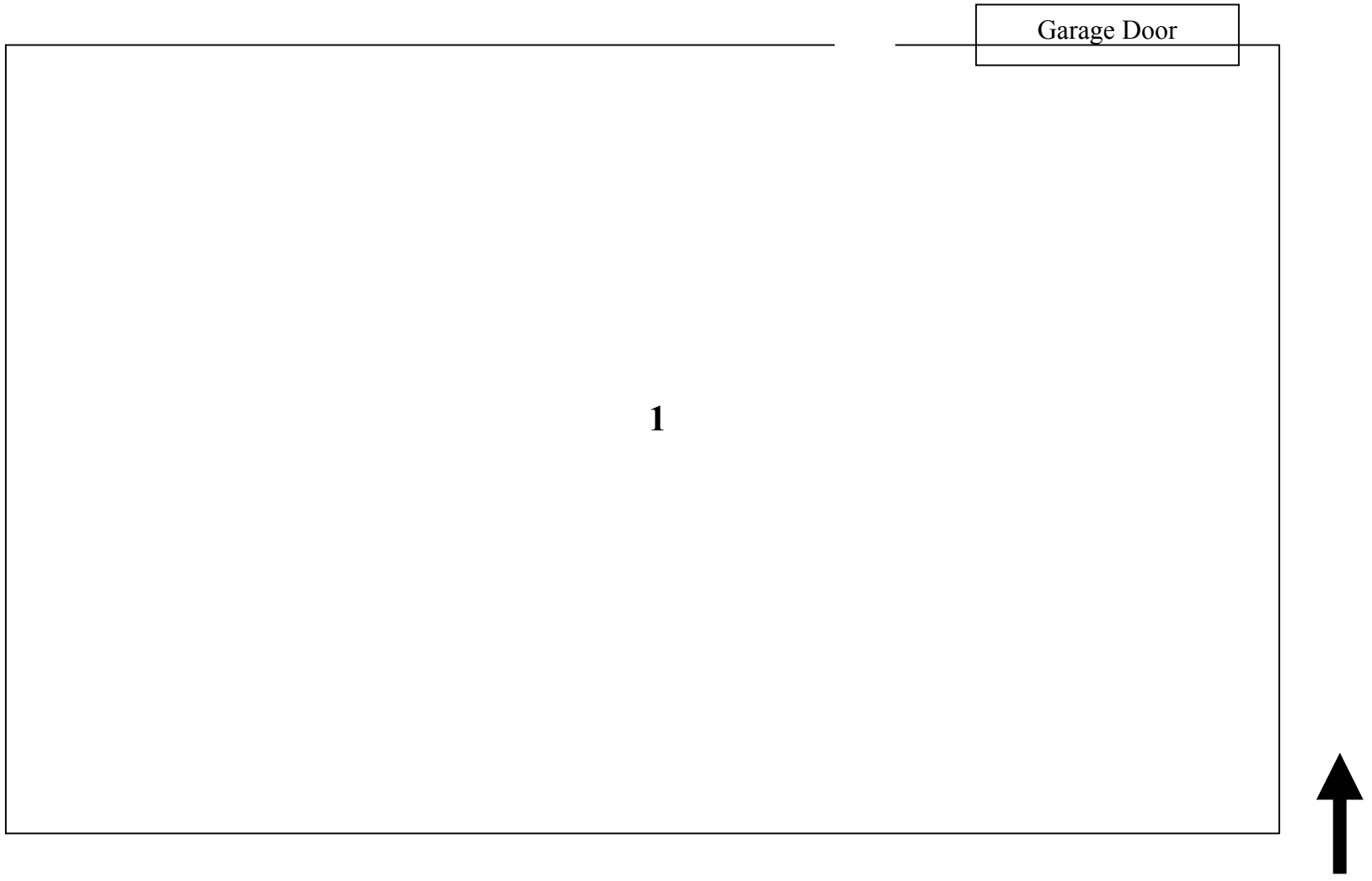


U.S. Public Health Service Federal Occupational Health 2201 Sixth Avenue, M/S Rx21 Seattle, Washington 98121-2500	Project: SANTA FE SERVICE UNIT Santa Fe, New Mexico	**Functional Space Map** Santa Fe Hospital Building 2000 Cooling Tower

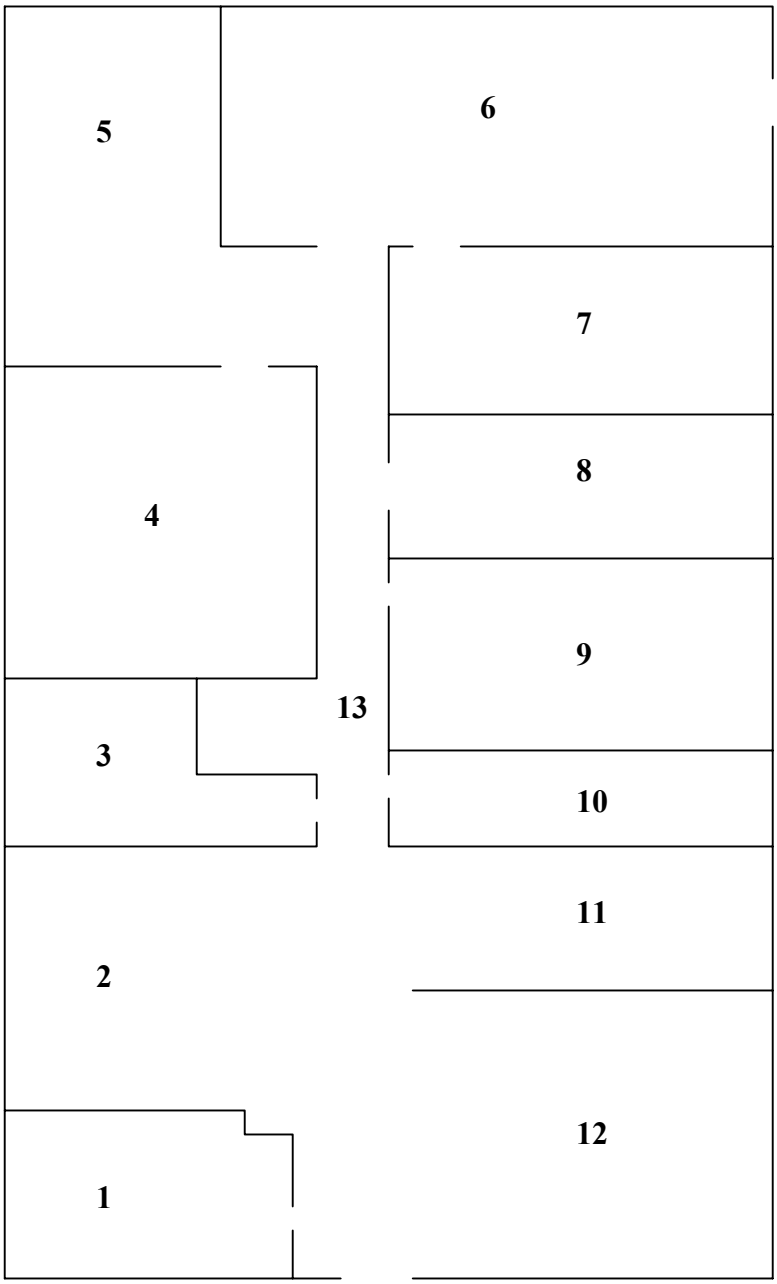


NORTH

U.S. Public Health Service Federal Occupational Health 2201 Sixth Avenue, M/S Rx21 Seattle, Washington 98121-2500	Project: ALBUQUERQUE AREA INDIAN HEALTH SERVICE SANTA FE SERVICE UNIT Santa Fe, New Mexico	**Functional Space Map** Building 2001



U.S. Public Health Service Federal Occupational Health 2201 Sixth Avenue, M/S Rx21 Seattle, Washington 98121-2500	Project: ALBUQUERQUE AREA INDIAN HEALTH SERVICE SANTA FE SERVICE UNIT Santa Fe New Mexico	**Functional Space Map** Building 2002



NORTH ➡

U.S. Public Health Service Federal Occupational Health 2201 Sixth Avenue, M/S Rx21 Seattle, Washington 98121-2500	Project: ALBUQUERQUE AREA INDIAN HEALTH SERVICE SANTA FE SERVICE UNIT Santa Fe, New Mexico	**Functional Space Map** Building 2003

APPENDIX 2
Bulk Sample Results and
Programming Cost Estimates

Building 333
Bulk Sample Results
Santa Fe Service Unit
December 19, 2006

HA #	Material	Functional Space	ACM	Condition	Quantity
1A	**9”x9” White Floor Tile & Mastic**	**21**	**Yes: 5-10% Chrysotile Floor Tile**	**Good**	**40 sf**
1B	Concrete	4, 6, 29	No	Good	240 sf
2A	Plaster Skim Coat	Throughout	No	Good	16,000 sf
2B	Plaster Brown Coat	Throughout	No	Good	16,000 sf
2C	Stucco	Exterior	No	Good	6,540 sf
5A	**TSI Insulation**	**Boiler Room**	**Yes: 10-20% Chrysotile; 5-10% Amosite**	**Good**	**300 sf**
5B	**TSI Elbow**	**Boiler Room**	**Yes: 10-20% Chrysotile**	**Good**	**20 sf**
7A	Caulking	Exterior	No	Good	2 sf
7B	Window Putty	Exterior	No	Damaged	80 sf
7C	**Window Sill Duct Tape**	**Exterior**	**Yes: 5-10% Chrysotile**	**Slightly Damaged**	**20 sf**
8A	**Roof Core**	**Exterior**	**Yes: 30-40% Chrysotile Roof Ply/Bitumen Bottom Layer**	**Good**	**4, 015 sf**
8B	**Black Penetration Mastic**	**Exterior**	**Yes: 5-10% Chrysotile Black Caulk**	**Slightly Damaged**	**8 sf**
8C	Light Gray Penetration Mastic	Exterior	No	Good	10 sf

Programming Cost Estimate
Building 333
Santa Fe Service Unit

Asbestos-Containing Material	Quantity of Material (sf)	Unit Cost (per sf)	Logistical Markup	Price
Floor Tile & Mastic	40 sf	$4.00	18%	$ 188.80
TSI Insulation	300 sf	$25.00	18%	$8,850.00
TSI Elbow	20 sf	$25.00	18%	$ 590.00
Window Sill Duct Tape	20 sf	$40.00	18%	$ 944.00
Roof Core	4,015 sf	$2.25	18%	$10,659.83
Roof Penetration Mastic	8 sf	$20.00	18%	$188.80
Total				**$21,421.43**

Building 334
Bulk Sample Results
Santa Fe Service Unit
December 20, 2006

HA #	Material	Functional Space	ACM	Condition	Quantity
1B	**12"x12" Beige Floor Tile & Mastic**	**4 & 5**	**Yes: 2-5% Chrysotile Floor Tile Only**	**Good**	**50 sf**
1C	12'x12' Brown Floor Tile & Mastic	Throughout	No	Good	800 sf
1D	Foam & Mastic	5	No	Good	10 sf
2A	Gypsum Drywall	6	No	Good	104 sf
3A	2x4 White Ceiling Tile	Throughout Except: 6	No	Good	900 sf
7A	**Brown Covebase Mastic**	**Throughout Except: 6**	**Yes: 2-5% Chrysotile**	**Good**	**60 sf**
7B	**Yellow Covebase Mastic**	**1-4**	**Yes: 2-5% Chrysotile**	**Good**	**45 sf**
7C	Window Caulk	1-4	No	Good	4 sf
7D	Blown-In Insulation	Throughout	No	Good	896 sf

Programming Cost Estimate
Building 334
Santa Fe Service Unit

Asbestos-Containing Material	Quantity of Material (sf)	Unit Cost (per sf)	Logistical Markup	Price
Floor Tile & Mastic	50 sf	$4.00	18%	$ 236.00
Brown Covebase Mastic	60 sf	$4.00	18%	$ 283.20
Yellow Covebase Mastic	45 sf	$4.00	18%	$ 212.40
Total				**$ 731.60**

Building 335
Bulk Sample Results
Santa Fe Service Unit
December 21, 2006

HA #	Material	Functional Space	ACM	Condition	Quantity
1A	**12”x12” Beige Floor Tile & Mastic**	**Throughout Except: 2, 3**	**Yes: 2-5% Chrysotile Floor Tile Only**	**Good**	**877 sf**
2A	Gypsum Drywall	10, 11	No	Good	492 sf
2B	Joint Compound	10, 11	No	Good	492 sf
3A	2’x4’ Ceiling Tile	Throughout	No	Good	1,680 sf
7A	**Brown Covebase Mastic**	**Throughout Except: 10, 11**	**Yes: >1-2% Chrysotile Mastic Only**	**Good**	**83 sf**
7B	Yellow Covebase Mastic	2, 3, 10, 11	No	Good	55 sf
7C	**Sink Undercoating**	**12**	**Yes: >1-2% Chrysotile**	**Good**	**4 sinks**
7D	Gray Window Caulk	Exterior: 2, 10, 12	No	Good	4 sf
7E	Blown-In Insulation	Plenum	No	Good	1,680 sf

Building 335
Santa Fe Service Unit
Programming Cost Estimate

Asbestos-Containing Material	Quantity of Material (sf)	Unit Cost (per sf)	Logistical Markup	Price
Floor Tile & Mastic	877 sf	$4.00	18%	$4,139.44
Brown Covebase Mastic	83 sf	$4.00	18%	$ 391.76
Sink Undercoating	4 sinks	$200.00	18%	$ 944.00
Total				**$5,475.20**

Building 336
Bulk Sample Results
Santa Fe Service Unit
December 20, 2006

HA #	Material	Functional Space	ACM	Condition	Quantity
1A	12”x12” Beige (W/ Brown Lines) Floor Tile & Mastic	Throughout Except: 6	No	Good	1,620 sf
1B	**12”x12” Beige (W/ Brown Specks) Floor Tile & Mastic**	**6**	**Yes: 2-5% Chrysotile Floor Tile Only**	**Good**	**24 sf**
1C	**Yellow Sheet Vinyl Flooring**	**Throughout Except: 6**	**Yes: 20-30% Chrysotile**	**Good**	**1,620 sf**
2A	Gypsum Drywall	4 & 6	No	Good	140 sf
2B	Joint Compound	4 & 6	No	Good	140 sf
3A	2’x4’ Ceiling Tile (Holes)	Throughout Except: 4, 6	No	Good	1,524 sf
3B	2’x4’ Ceiling Tile (Texture)	5	No	Good	56 sf
7A	Covebase Mastic	Throughout Except: 6	No	Good	73 sf
7B	Interior Door Caulk	Throughout	No	Good	5 sf
7C	Black Mastic (On Ducts)	Plenum	No	Good	20 sf
7D	Blow-In Insulation	Plenum	No	Good	1,644 sf

Programming Cost Estimate
Building 336
Santa Fe Service Unit

Asbestos-Containing Material	Quantity of Material (sf)	Unit Cost (per sf)	Logistical Markup	Price
Floor Tile & Mastic	24 sf	$4.00	18%	$ 236.00
Sheet Vinyl	1,620 sf	$4.50	18%	$8,602.20
Total				**$8,838.20**

Building 2000-Santa Fe Hospital
Bulk Sample Results
Santa Fe Service Unit
December 13, 2006

HA #	Material	Functional Space	ACM	Condition	Quantity
1A	**12”x12” White w/ Gray Specks Floor Tile & Mastic**	**109, 291, 293, BM**	**Yes: 2-5% Chrysotile Black Mastic Only**	**Good**	**118 sf**
1B	18”x18” Carpet (Clear Mastic)	102, 120A, 120C, 120D, 121, 121A-F, 141, 143C, 500-504, 506	No	Good	3,501 sf
1C	1’x1’ Light Ceramic Tile Grout	103, 141, 143, 143B-D, 143F, 145, 179B, 200A, 202A, 204A, 206A, 208A, 210A, 212A, 214A, 216A, 218A, 220A, 222A, 223, 224A, 226A, 227, 228A, 232A, 234A, 236A, 238A, 244A, 250A, 252A, 253, 256A, 258A, 260A, 262, 264, 266A, 273, 273A, 274, 274A, 314A, 360B, 801B	No	Good	3,330 sf

HA #	Material	Functional Space	ACM	Condition	Quantity
1D	**12”x12” Beige Floor Tile & Mastic**	**Throughout Except: 102, 103, 109, 113, 115, 115A-D, 117, 117A-E, 118, 130, 131, 131A, 134, 139A-C, 141, 143, 143B, 143D, 143F, 145, 149, 149A, 149B, 150, 152, 158, 160, 180A, 181, 182, 196, 200A, 202A, 203, 204A, 206A, 208A, 208A, 210A, 212A, 214A, 215, 216A, 218A, 220A, 222A, 223, 224A, 226A, 227, 230, 232A, 234A, 236A, 237, 244A, 250A, 252A, 253, 256A, 258A, 260A, 262, 264, 266A, 274A, 277, 288A, 330, 332, 334, 336, 340, 341, 352, 701, G608, G609, G612, G613, 801B**	**Yes: 5-10% Chrysotile Floor Tile Yes: 10-20% Chrysotile Black Mastic**	**Good**	**40,000 sf**
1E	12”x12” Blue & White Floor Tile & Mastic	203, 300A, 340, 341, 361	No	Good	800 sf
1F	Brown Ceramic Tile Grout	113, 131A, 150, 152, 211, 215, 230, 303, 362	No	Good	1,597 sf
1G	12”x12” White Floor Tile & Mastic	223	No	Good	575 sf
1H	12”x12” Orange & Brown Floor Tile & Mastic	196	No	Good	650 sf
1I	Yellow Carpet Mastic	115, 115A-D, 117, 117A-E, 118, 119, 120, 120B, 129, 133, 133A, 133B, 137, 139A-C, 301-303, 305, 341	No	Good	5,440 sf
1J	**Black Mastic**	**139A-C**	**Yes: 5-10% Chrysotile**	**Good**	**270 sf**
1K	12”x12” Light Blue Floor Tile & Mastic	143F	No	Good	25 sf

HA #	Material	Functional Space	ACM	Condition	Quantity
1L	**Brown Floor Tile & Mastic**	**352**	**Yes: 2-5% Chrysotile Black Mastic**	**Good**	**240 sf**
1M	Green & Black Sheet Vinyl Flooring & Mastic	117C-E, 277, 298, 701, G608, G609, G612, G613	No	Good	2,073 sf
2A	Gypsum Drywall	Throughout	No	Good	100,000 sf
2B	Joint Compound	Throughout	No	Good	100,000 sf
2C	Texture 1	Throughout Except: 100A-C, 110, 121A, 121D, 121E, 121F, 149, 149A, 149B, 162, 166, 170, 172, 180A, 181, 182, 200A-C, 201, 203, 219, 250, 251, 252A, 253, 277, 291, 293, 300A, 400A-C, 500, 500A, 501-504, 701, 800A, 801A, 900A, G608, G609, G612, G613	No	Good	40,000 sf
2D	Vinyl Wall Paper	100, 100A-C, 110 171, 196, 200A-C, 211, 300A, 400A-D, 500A, 600A, 700A, 800A, 801A, 900A	No	Good	40,540 sf
2E	Texture 2	100D, 103, 111, 121A, 175, 201, 203, 249, 251, 255, 300A, 500-504, BM, G1, N02, O2	No	Good	4,500 sf
2F	Texture 3	195, 300A, 312	No	Good	1,000 sf
2G	Texture 4	195A	No	Good	320 sf
2H	Texture 5	137	No	Good	60 sf

HA #	Material	Functional Space	ACM	Condition	Quantity
2I	4”x4” Beige Ceramic Wall Tile Grout	103, 131A, 143, 169, 171, 179B, 200A, 202A, 204A, 206A, 208A, 210A, 212A, 214A, 216A, 218A, 220A, 222A, 223, 224A, 226A, 227, 228A, 232A, 234A, 236A, 238A, 244A, 250A, 252A, 253, 256A, 258A, 260A, 262, 264, 266A, 269, 273A, 274A, 277, 291, 293, 314, 330, 332, 352A, 360A, 360B, G608, G609, G612, G613	No	God	2,456 sf
2J	4”x4” White Ceramic Wall Tile Grout	150, 152, 211, 215, 230, 231A, 231B, 303, 362	No	Good	3,000 sf
2K	12”x12” Wall Tile	100	No	Good	100 sf
2L	CMU	C-1	No	Good	300 sf
2M	CMU Mortar	C-1	No	Good	300 sf
2N	**Cement Asbestos Panels**	**Exterior**	**Yes: 10-20% Chrysotile Gray Cem/Asb Board**	**Good**	**4,800 sf**
2O	Exterior Stucco	Exterior	No	Good	62,000 sf

HA #	Material	Functional Space	ACM	Condition	Quantity
3A	Pin Hole 2'x4" Ceiling Tile	100, 100A-C, 101, 102, 105, 109, 110, 115, 115A-D, 117, 117A-E, 118, 119, 120, 120A-D, 121, 121A-D, 129, 132, 133A, 133B, 134, 137, 139A-C, 141, 143A, 143A, 147, 171, 175, 178, 178A-C, 179, 179A, 179B, 190-195, 200A-C, 205, 211, 225, 229, 231, 238, 239, 241-243, 251, 252A, 268, 270, 284, 292, 294, 300A, 301-311, 320, 324, 326, 328, 340, 341, 343, 360, 360D, 361, 361A-D, 400A-C, 500, 500A, 506, 700A, 701, 800A, 801A, 900A	No	Good	40,000 sf
3B	2'x4" Ceiling Tile	121E, 133, 195, 196, 500A, 701	No	Good	1,600 sf
3C	Smooth 2'x4" Ceiling Tile	195A, 200, 202, 204, 206, 208, 210, 212, 214, 216, 218, 220, 224, 226, 228, 232, 234, 236, 244, 250A, 256, 258, 260	No	Good	600 sf
3D	Rough Pinhole 2'x4' Ceiling Tile	501-504	No	Good	2,000 sf
3E	Exterior Texture	Exterior Overhang	No	Good	2,200 sf
5A	Hanger Block Insulation	Penthouse	No	Good	120 Hangers
5B	Green Gaskets	Penthouse	No	Good	2 Gaskets
5C	Hard Pack Pipe Elbow Joints	149, 182	No	Good	6 Elbow
5D	Boiler Gaskets	149A	No	Good	6 Gaskets
5E	Pipe Insulation Casts	C-1	No	Good	300 sf
6A	Fireproofing	Plenum, 172	No	Good	15,000 sf
6B	**Fire Doors**	**Throughout**	**Assume**	**Good**	**60 Doors**
6C	**Fire Door Frame**	**Throughout**	**Assume**	**Good**	**60 Door Frames**
7A	Black Sink Undercoating	101, 102, 316, 318, 320, 322, 326, 328	No	Good	19 sinks

HA #	Material	Functional Space	ACM	Condition	Quantity
7B	**Purple Sink Undercoating**	**205, 294, 361, 361A, 361C, 361D**	**Yes: >1-2% Chrysotile**	**Good**	**8 sinks**
7C	**Black Tar Sink Undercoating**	**100A, 180A, 182, 201, 206A, 207, 219, 221, 226A, 233, 237, 249, 251, 255, 276, 282, 321, 344, 352**	**Yes: 10-20% Chrysotile**	**Good**	**11 sinks**
7D	Yellow & Brown Mix Covebase Mastic	Throughout Except: 130, 130A, 131, 131A, 134-136, 143, 143B, 143D, 143F, 145, 149A, 150, 152, 154, 156, 162, 166, 169, 171, 179B, 180A, 181, 182, 200A, 201, 202A, 204A, 206A, 208A, 210A, 212A, 214, 215, 216A, 218A, 219, 220A, 222A, 223, 224A, 226A, 227, 253, 271, 271A, 271C, 277, 288, 291, 293, 296, 298, 303, 312, 330, 332, 348, 350, 316B, 362, 701, 801A, 900A, G608, G609, G612, G613	No	Good	17,500 sf
7E	Black Window Caulk	Throughout	No	Good	500 sf
7F	**White HVAC Caulk**	**Penthouse**	**Yes: 5-10% Chrysotile White Caulk**	**Good**	**70 sf**
7G	**Silver HVAC Caulk**	**Penthouse**	**Yes: 5-10% Chrysotile Grey Caulk**	**Good**	**70 sf**
7H	White Floor Penetration Caulk	Penthouse	No	Good	400 sf
7I	Fiberglass Duct Insulation	Throughout	No	Good	40,000 sf
7J	Black Tar Duct Mastic	Throughout	No	Good	40,000 sf

HA #	Material	Functional Space	ACM	Condition	Quantity
7K	Tank Insulation (Foam)	C-1	No	Good	200 sf
7L	Exterior Expansion Caulk	Exterior	No	Good	200 sf
7M	Interior White Door Caulk	C-1	No	Good	10 sf
7N	**(Gray) Exterior Vent Caulk**	**Penthouse Exterior**	**Yes: 5-10% Chrysotile**	**Good**	**15 sf**
7O	Vent Mastic	Penthouse Exterior	No	Good	2 sf
8A	**Rolled Roof**	**Cooling Tower Roof**	**Yes: 30-40% Chrysotile Black Roof Ply/Bitumen**	**Good**	**900 sf**
8B	**Cooler Roof Penetration Mastic**	**Cooling Tower Roof**	**Yes: 5-10% Chrysotile Black Caulk**	**Good**	**50 sf**

Programming Cost Estimate
Building 2000-Santa Fe Hospital
Santa Fe Service Unit

Asbestos-Containing Material	Quantity of Material (sf)	Unit Cost (per sf)	Logistical Markup	Price
Floor Tile & Mastic	40,628 sf	$4.00	18%	$191,764.16
Cement Asbestos Panels	4,800 sf	$4.00	18%	$22,656.00
Fire Door & Frames	60 sf	$60.00	18%	$4,248.00
Sink Undercoating	19 sinks	$200.00	18%	$4,484.00
HVAC Caulk	140 sf	$30.00	18%	$4,956.00
Vent Caulk	15 sf	$8.00	18%	$ 141.60
Rolled Roof	900 sf	$2.25	18%	$2389.50
Roof Penetration Mastic	50 sf	$20.00	18%	$1180.00
Total				$231,819.26

Building 2001
Bulk Sample Results
Santa Fe Service Unit
December 15, 2006

HA #	Material	Functional Space	ACM	Condition	Quantity
1A	12”x12” White Floor Tile & Mastic	1, 2, 4, 6, 9	No	Good	529 sf
1B	Green Carpet Tile Mastic	1, 10, 11	No	Good	288 sf
1C	Clear Carpet Tile Mastic	3, 5, 7, 8	No	Good	464 sf
1D	**12”x12” Beige Floor Tile & Mastic**	**14**	**Yes: 2-5% Chrysotile Floor Tile Only**	**Good**	**84 sf**
1E	Blue Sheet Vinyl	15	No	Good	60 sf
2A	Gypsum Drywall	3, 4, 7, 9, 10, 11, 12, 13, 14, 15	No	Good	2,369 sf
2B	Joint Compound	3, 4, 7, 9, 10, 11, 12, 13, 14, 15	No	Good	2,369 sf
2C	Drywall Texture 1	4, 9, 10, 11, 12	No	Good	1,768 sf
2D	Drywall Texture 2	3	No	Good	320 sf
3A	1’x1’ Ceiling Tile w/ Holes	1, 2, 3, 5, 6, 7, 8	No	Good	975 sf
3B	1’x1’ Ceiling Tile Mastic	1, 2, 3, 5, 6, 7, 8	No	Good	975 sf
3C	2’x4’ Ceiling Tile w/ Pinholes	14, 15	No	Good	144 sf
7A	Covebase Mastic	1, 2, 5, 6, 7, 8	No	Good	107 sf
7B	Interior Window Caulk	14	No	Good	2 sf
7C	Gray Penetration Mastic	Exterior	No	Good	10 sf
8A	**Roof Penetration Mastic**	**Roof**	**Yes: 5-10% Chrysotile Black Mastic**	**Good**	**8 sf**

Santa Fe Service Unit
Building 2001
Programming Cost Estimate

Asbestos-Containing Material	Quantity of Material (sf)	Unit Cost (per sf)	Logistical Markup	Price
Floor Tile	84 sf	$4.00	18%	$ 396.48
Roof Penetration Mastic	8 sf	$20.00	18%	$ 188.80
Total				**$ 585.28**

Building 2002
Bulk Sample Results
Santa Fe Service Unit
December 14, 2006

HA #	Material	Functional Space	ACM	Condition	Quantity
7A	Fiberglass Insulation	1	No	Good	1,800 sf
7B	**Interior Window Caulk**	**1**	**Yes: 2-5% Chrysotile White Caulk**	**Good**	**5 sf**
8A	**Gray/Black Roof Penetration Mastic**	**Roof**	**Yes: 10-20% Chrysotile Black Caulk**	**Good**	**12 sf**
8B	**Gray Flue Mastic**	**Roof**	**Yes: >1-2% Chrysotile Gray Caulk**	**Good**	**4 sf**

Santa Fe Service Unit
Building 2002
Programming Cost Estimate

Asbestos-Containing Material	Quantity of Material (sf)	Unit Cost (per sf)	Logistical Markup	Price
Interior Window Caulk	5 sf	$8.00	18%	$ 47.20
Roof Penetration Mastic	12 sf	$20.00	18%	$ 283.20
Flue Mastic	4 sf	$10.00	18%	$ 47.20
Total				$ 377.60

Building 2003
Bulk Sample Results
Santa Fe Service Unit
December 20, 2006

HA #	Material	Functional Space	ACM	Condition	Quantity
1A	Pink Sheet Vinyl Flooring	Throughout Except: 4, 5, 6	No	Good	1,323 sf
1B	Yellow Carpet Mastic	4, 5, 6	No	Good	521 sf
1C	Yellow Sheet Vinyl Flooring	8, 9	No	Good	83 sf
1D	Blue Sheet Vinyl Flooring	8	No	Good	240 sf
1E	Green Carpet Mastic	5	No	Good	10 sf
2A	Gypsum Drywall	Throughout	No	Good	4,072 sf
3A	2'x4' Ceiling Tile	Throughout	No	Good	1,960 sf
7A	Caulking	Throughout	No	Good	10 sf
7B	Covebase Mastic	Throughout	No	Good	14 sf
7C	White Sink Undercoating	3, 11	No	Good	3 sinks
7D	Gray Sink Undercoating	11	No	Good	1 Sink

APPENDIX 3
Laboratory Results

FIBERQUANT ANALYTICAL SERVICES

Polarized Light Microscope (PLM) Analysis for Asbestos

JobNumber:	**200700081**

Client: **IHI ENVIRONMENTAL**

(PHOENIX OFFICE)
4527 N 16th ST STE 105
PHOENIX, AZ 85016-0000
Office Phone: (602) 776-0300
FAX: (602) 776-0301

Samples: 51 PLM **Rec:** 1/4/2007 **Method:** Interim (EPA/600/M4-82-020) PLM analysis for asbestos in bulk smp
Client Job: 07P-3031A / IHS-333 **PO Number:** 07P-3031A
Report Date: 1/23/2007 **Date Analyzed:** 1/23/2007 **Routing Number:** -

Method and Analysis Information: **Fiberquant Internal SOP:** PLMi

Each bulk sample is first dissected under a 7-30x magnification stereo-microscope. This examination is used to determine the general type of sample, how many and what type of layers it has, and initial estimates of fiber types and quantities. Second, liquid media mounts are made of each layer - such mounts may be of selected fibers (used solely for identification purposes) or may be representative of the layer as a whole (used for quantitation purposes). The mounts may be made in a synthetic Canadian balsam, one of several solvents, or in refractive index oils (media of known refractive index). Generally, a variety of different mounts are made: some optimized for fiber visibility, some optimized for fiber identification, and some optimized for fiber quantitation. The mounted slides are then examined at 50-400x magnification on a Nikon Labphot-pol microscope. Optical characteristics are used to identify each observed fiber type; the optical data are contained for each sample on its detail analysis sheet, attached.

Current EPA, NESHAP and OSHA regulations designate a result of <=1 % asbestos as "negative" and >1 % asbestos as "positive". Samples containing layers that have been determined to be "positive" may have to be handled differently during a renovation or demolition than samples whose layers have been determined to be "negative."

The method of fiber analysis and identification is the EPA Interim Method (EPA Method 600/M4-82-020). The method of fiber quantitation is an estimation technique in which the analysts quantitation is routinely calibrated by reference quantitation standards, and which has been shown to be equivalent in precision and accuracy to point counting. Friability is estimated for the purposes of deciding when to point count. Friabilities determined in the field take precedence over those determined in the laboratory. Those sample layers which are friable and estimated by the analyst to contain <= 1% asbestos are point counted using 400 points, and the results listed on the detail analysis sheet. Such point counting is required by NESHAP (National Emission Standards for Hazardous Air Polutants, Nov. 1990) in order to rely on analytical results that are <= 1%. The coefficient of variation for the estimation quantitation technique is 100% in the range 0-5%. This means that PLM analysis is not capable of conclusively determining whether a layer containing close to 1% asbestos is actually "positive" or "negative". For this reason, Fiberquant refers to results where asbestos was detected but <= 1% as "borderline negative", and results where asbestos was >1 % but <= 2% as "borderline positive" to indicate the uncertainty in assigning a "positive" or "negative" label. In the sample summary, a "ND" means that no asbestos was detected during the analysis. A "Tr" or "Trace" of asbestos is defined for our purposes as the detection of several asbestos asbestos fibers during the analysis; an amount essentially at the limit of detection for the method. The limit of detection (the smallest % of asbestos that can be detected) varies greatly depending on the matrix in which the asbestos is found. As little as 0.001% asbestos can be detected in favorable samples, while detection in unfavorable samples may approach the detection limit of 1% stated in the method. During the analysis, the analyst, for Fiberquant identification purposes only, determines the "apparent sample type" and "apparent layer types." It must be emphasized that these types are only what is apparent. Often, different materials appear similar or identical after sampling, so the analyst may assign a type other than what was sampled.

Floor tiles present a special problem for PLM asbestos analysis. Floor tile can contain chrysotile fibers so thin that they cannot be resolved by optical methods. In such a case, we may observe a percentage of asbestos which is lower than the actual percentage, or not observe asbestos at all when some is present. For this reason, floor tiles reported as negative should be confirmed to be negative using transmission electron microscope (TEM) analysis. Likewise, vermiculite insulation materials containing traces of asbestiform asbestos present a problem for routine PLM analysis - the amphiboles are sometimes present in trace amounts inhomogeneously distributed. We recommend a hydro-separation technique for such samples.

Vermiculite-containing samples may contain trace amounts of asbestiform amphibole that may or may not be detected during routine PLM analysis. For this reason, loose vermiculite samples reported as negative should be confirmed to contain no amphibole using hydroseparation techniques.

The samples were analyzed under the following ongoing quality assurance program: Blank samples are routinely analyzed to maintain contamination-free materials. Each analyst is a degreed geologist or mineralogist, and has also completed extensive training specific to asbestos analysis for 1-3 months before being allowed to analyze client samples. Qualitative reference samples are routinely analyzed to assure that analysts can identify asbestos and asbestos-look-alike fibers. Quantitative reference samples are routinely analyzed to calibrate and characterize the estimation procedure. Microscope alignment is checked each day. Refractive index oils are calibrated at least monthly. At least 10% of client samples are re-analyzed from scratch by a different analyst than the original, and any discrepancies are resolved for the sample and similar sample types before the results are reported. All quality checks performed for these samples were in control except as detailed in the "Analytical Notes" below. All analysts participate in interlab round robins and proficiency testing to assure competence. Fiberquant is accredited by NVLAP (#101031) for the analysis of bulk samples for asbestos using PLM. Accreditation does not imply endorsement by the EPA, any other United States governmental agency or any private agency or association. Each lab analysis refers only to the sample tested, and may not, due to the sampling process, be

representative of the material sampled. This report may not be reproduced except in full, without the approval of Fiberquant Analytical Services.

The "Analysis Results" below are the averages for all layers of each sample listed. The "+ Layer" column indicates whether any of the layers in the sample were >1% asbestos. Please refer to each detailed analysis sheet to determine which layer(s) of a sample was(were) positive.

Some results may have been calculated using client supplied data, such as volume or area sampled, for which Fiberquant assumes no liability for accuracy.

Job Analysis Notes:

Single layer sample analysis as per client request. Any material or layer other than that indicated on the chain of custody was not analyzed, even if a suspect material.

PLM Analysis Summary: **Job Number: 200700081** 07P-3031A / IHS-333

Sample Number / Layer	Color	Lab Number / Apparent Layer Type *	Apparent Sample Type * / *Asbestos Results*	Positive Layer Yes or No
Sample IHS-333-1A1-1		2007-00081- 1	Flooring	Positive Layer? Yes
Layer # 1	off-white	floor tile	*5-10% chrysotile asbestos*	
Sample IHS-333-1A2-2		2007-00081- 2	Flooring	Positive Layer? Yes
Layer # 1	off-white	floor tile	*5-10% chrysotile asbestos*	
Layer # 2	Yellow	mastic	*no asbestos detected*	
Sample IHS-333-1A3-3		2007-00081- 3	Flooring	Positive Layer? Yes
Layer # 1	off-white	floor tile	*5-10% chrysotile asbestos*	
Layer # 2	Yellow	mastic	*no asbestos detected*	
Sample IHS-333-1B1-4		2007-00081- 4	Cementitious	Positive Layer? No
Layer # 1	Gray	concrete	*no asbestos detected*	
Sample IHS-333-1B2-5		2007-00081- 5	Cementitious	Positive Layer? No
Layer # 1	Gray	concrete	*no asbestos detected*	
Sample IHS-333-1B3-6		2007-00081- 6	Cementitious	Positive Layer? No
Layer # 1	Gray	concrete	*no asbestos detected*	
Sample IHS-333-2A1-7		2007-00081- 7	Wall System	Positive Layer? No
Layer # 1	White	plaster (top coat)	*no asbestos detected*	
Sample IHS-333-2A2-8		2007-00081- 8	Wall System	Positive Layer? No
Layer # 1	White	plaster (top coat)	*no asbestos detected*	
Sample IHS-333-2A3-9		2007-00081- 9	Wall System	Positive Layer? No
Layer # 1	White	plaster (top coat)	*no asbestos detected*	
Sample IHS-333-2A4-10		2007-00081- 10	Wall System	Positive Layer? No
Layer # 1	White	plaster (top coat)	*no asbestos detected*	
Sample IHS-333-2A5-11		2007-00081- 11	Wall System	Positive Layer? No
Layer # 1	White	plaster (top coat)	*no asbestos detected*	
Sample IHS-333-2A6-12		2007-00081- 12	Wall System	Positive Layer? No
Layer # 1	White	plaster (top coat)	*no asbestos detected*	
Sample IHS-333-2A7-13		2007-00081- 13	Wall System	Positive Layer? No
Layer # 1	White	plaster (top coat)	*no asbestos detected*	
Sample IHS-333-2B1-14		2007-00081- 14	Wall System	Positive Layer? No
Layer # 1	Tan	plaster (scratch coat)	*no asbestos detected*	
Sample IHS-333-2B2-15		2007-00081- 15	Wall System	Positive Layer? No
Layer # 1	Tan	plaster (scratch coat)	*no asbestos detected*	
Sample IHS-333-2B3-16		2007-00081- 16	Wall System	Positive Layer? No
Layer # 1	Tan	plaster (scratch coat)	*no asbestos detected*	
Sample IHS-333-2B4-17		2007-00081- 17	Wall System	Positive Layer? No
Layer # 1	Tan	plaster (scratch coat)	*no asbestos detected*	
Sample IHS-333-2B5-18		2007-00081- 18	Wall System	Positive Layer? No
Layer # 1	Tan	plaster (scratch coat)	*no asbestos detected*	
Sample IHS-333-2B6-19		2007-00081- 19	Wall System	Positive Layer? No
Layer # 1	Tan	plaster (scratch coat)	*no asbestos detected*	
Sample IHS-333-2B7-20		2007-00081- 20	Wall System	Positive Layer? No
Layer # 1	Tan	plaster (scratch coat)	*no asbestos detected*	
Sample IHS-333-2C1-21		2007-00081- 21	Wall System	Positive Layer? No
Layer # 1	Tan	stucco	*no asbestos detected*	
Sample IHS-333-2C2-22		2007-00081- 22	Wall System	Positive Layer? No
Layer # 1	Tan	stucco	*no asbestos detected*	
Sample IHS-333-2C3-23		2007-00081- 23	Wall System	Positive Layer? No
Layer # 1	Tan	stucco	*no asbestos detected*	
Sample IHS-333-2C4-24		2007-00081- 24	Wall System	Positive Layer? No
Layer # 1	Tan	stucco	*no asbestos detected*	
Sample IHS-333-2C5-25		2007-00081- 25	Wall System	Positive Layer? No
Layer # 1	Tan	stucco	*no asbestos detected*	
Sample IHS-333-2C6-26		2007-00081- 26	Wall System	Positive Layer? No
Layer # 1	Tan	stucco	*no asbestos detected*	
Sample IHS-333-2C7-27		2007-00081- 27	Wall System	Positive Layer? No
Layer # 1	Tan	stucco	*no asbestos detected*	
Sample IHS-333-5A1-28		2007-00081- 28	TSI	Positive Layer? Yes
Layer # 1	White	insulation	*10-20% chrysotile asbestos*	

Sample	**IHS-333-5A2-29**		2007-00081- 29	TSI		Positive Layer? Yes
	Layer # 1	White	insulation	*10-20% chrysotile asbestos*	*2-5% amosite (grunerite) asbestos*	
Sample	**IHS-333-5A3-30**		2007-00081- 30	TSI		Positive Layer? Yes
	Layer # 1	White	insulation	*5-10% chrysotile asbestos*	*5-10% amosite (grunerite) asbestos*	
Sample	**IHS-333-5B1-31**		2007-00081- 31	TSI		Positive Layer? Yes
	Layer # 1	White	insulation	*10-20% chrysotile asbestos*		
Sample	**IHS-333-5B2-32**		2007-00081- 32	TSI		Positive Layer? Yes
	Layer # 1	White	insulation	*10-20% chrysotile asbestos*		
Sample	**IHS-333-5B3-33**		2007-00081- 33	TSI		Positive Layer? Yes
	Layer # 1	White	insulation	*10-20% chrysotile asbestos*		
Sample	**IHS-333-7A1-34**		2007-00081- 34	Adhesive/caulk		Positive Layer? No
	Layer # 1	white	caulk	*no asbestos detected*		
Sample	**IHS-333-7A2-35**		2007-00081- 35	Adhesive/caulk		Positive Layer? No
	Layer # 1	white	caulk	*no asbestos detected*		
Sample	**IHS-333-7A3-36**		2007-00081- 36	Adhesive/caulk		Positive Layer? No
	Layer # 1	white	caulk	*no asbestos detected*		
Sample	**IHS-333-7B1-37**		2007-00081- 37	Adhesive/caulk		Positive Layer? No
	Layer # 1	Off-white	caulk	*no asbestos detected*		
Sample	**IHS-333-7B2-38**		2007-00081- 38	Adhesive/caulk		Positive Layer? No
	Layer # 1	Off-white	caulk	*no asbestos detected*		
Sample	**IHS-333-7B3-39**		2007-00081- 39	Adhesive/caulk		Positive Layer? No
	Layer # 1	Off-white	caulk	*<=1% chrysotile asbestos*		
Sample	**IHS-333-7C1-40**		2007-00081- 40	Miscellaneous		Positive Layer? No
	Layer # 1	White	duct tape	*no asbestos detected*		
Sample	**IHS-333-7C2-41**		2007-00081- 41	Miscellaneous		Positive Layer? Yes
	Layer # 1	Off-white	duct tape	*5-10% chrysotile asbestos*		
Sample	**IHS-333-7C3-42**		2007-00081- 42	Miscellaneous		Positive Layer? Yes
	Layer # 1	Off-white	duct tape	*5-10% chrysotile asbestos*		
Sample	**IHS-333-8A1-43**		2007-00081- 43	Roofing		Positive Layer? Yes
	Layer # 1	Off-white	paint	*no asbestos detected*		
	Layer # 2	White	paint	*no asbestos detected*		
	Layer # 3	Yellow	foam	*no asbestos detected*		
	Layer # 4	Tan	paper/cardboard	*no asbestos detected*		
	Layer # 5	Black	roof ply/bitumen	*no asbestos detected*		
	Layer # 6	Black	roof ply/bitumen	*no asbestos detected*		
	Layer # 7	Black	roof ply/bitumen	*no asbestos detected*		
	Layer # 8	Black	roof ply/bitumen	*30-40% chrysotile asbestos*		
	Layer # 9	Black	roof ply/bitumen	*30-40% chrysotile asbestos*		
	Layer # 10	Black	roof ply/bitumen	*30-40% chrysotile asbestos*		
	Layer # 11	Black	roof ply/bitumen	*30-40% chrysotile asbestos*		
	Layer # 12	Black	roof ply/bitumen	*no asbestos detected*		
	Layer # 13	Black	roof ply/bitumen	*no asbestos detected*		
	Layer # 14	Black	roof ply/bitumen	*no asbestos detected*		
Sample	**IHS-333-8A2-44**		2007-00081- 44	Roofing		Positive Layer? Yes
	Layer # 1	Off-white	paint	*no asbestos detected*		
	Layer # 2	White	paint	*no asbestos detected*		
	Layer # 3	Yellow	foam	*no asbestos detected*		
	Layer # 4	Black	bitumen	*10-20% chrysotile asbestos*		
	Layer # 5	Black	roof ply/bitumen	*no asbestos detected*		
	Layer # 6	Black	roof ply/bitumen	*no asbestos detected*		
	Layer # 7	Black	roof ply/bitumen	*no asbestos detected*		
	Layer # 8	Black	roof ply/bitumen	*30-40% chrysotile asbestos*		
	Layer # 9	Black	roof ply/bitumen	*30-40% chrysotile asbestos*		
	Layer # 10	Black	roof ply/bitumen	*30-40% chrysotile asbestos*		
Sample	**IHS-333-8A3-45**		2007-00081- 45	Roofing		Positive Layer? Yes
	Layer # 1	Off-white	paint	*no asbestos detected*		
	Layer # 2	White	paint	*no asbestos detected*		
	Layer # 3	Yellow	foam	*no asbestos detected*		
	Layer # 4	Black	bitumen	*10-20% chrysotile asbestos*		
	Layer # 5	Black	roof ply/bitumen	*30-40% chrysotile asbestos*		
	Layer # 6	Black	roof ply/bitumen	*30-40% chrysotile asbestos*		
	Layer # 7	Black	roof ply/bitumen	*30-40% chrysotile asbestos*		
Sample	**IHS-333-8B1-46**		2007-00081- 46	Adhesive/caulk		Positive Layer? Yes
	Layer # 1	Black	caulk	*5-10% chrysotile asbestos*		
Sample	**IHS-333-8B2-47**		2007-00081- 47	Adhesive/caulk		Positive Layer? Yes
	Layer # 1	Black	caulk	*5-10% chrysotile asbestos*		
Sample	**IHS-333-8B3-48**		2007-00081- 48	Adhesive/caulk		Positive Layer? Yes
	Layer # 1	Black	caulk	*5-10% chrysotile asbestos*		
Sample	**IHS-333-8C1-49**		2007-00081- 49	Adhesive/caulk		Positive Layer? No
	Layer # 1	Gray	caulk	*no asbestos detected*		
	Layer # 2	White	caulk	*no asbestos detected*		
Sample	**IHS-333-8C2-50**		2007-00081- 50	Adhesive/caulk		Positive Layer? No
	Layer # 1	Gray	caulk	*no asbestos detected*		
	Layer # 2	White	caulk	*no asbestos detected*		

Sample	**IHS-333-8C3-51**		2007-00081- 51	Adhesive/caulk		Positive Layer? No
	Layer # 1	Gray	caulk	*no asbestos detected*		
	Layer # 2	White	caulk	*no asbestos detected*		

* Apparent Sample Types and Apparent Layer Types are as they appeared to the analyst. Since many types of materials appear similar after sampling damage, the apparent type of material may not be the actual type of material.

Sample IHS-333-1A1-1 **Lab Number** 2007-00081- 1 **Sampled:** 12/19/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	floor tile	100	off-white	1	5-10%	-	-	-	-	-
	Total %	100		**Average %**	5-10%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent. No mastic was observed.

Sample IHS-333-1A2-2 **Lab Number** 2007-00081- 2 **Sampled:** 12/19/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** Yes **# Sub-Samples** 5
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	floor tile	97	off-white	1	5-10%	-	-	-	-	-
2	mastic	3	Yellow	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	5-10%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-333-1A3-3 **Lab Number** 2007-00081- 3 **Sampled:** 12/19/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** Yes **# Sub-Samples** 5
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	floor tile	97	off-white	1	5-10%	n.d.	-	-	-	-
2	mastic	3	Yellow	1	n.d.	<=1%	-	-	-	-
	Total %	100		**Average %**	5-10%	<=1%	-	-	-	-
				Fiber Identification:	chrysotile asbestos	cellulose fiber				

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2	cellulose fiber	W	F	N	N	H	+	U					
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-333-1B1-4 **Lab Number** 2007-00081- 4 **Sampled:** 12/19/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Cementitious Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, rock,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	concrete	100	Gray	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: dissolution of matrix using dilute HCl acid. Procedure: tweased apart using forceps.

Sample IHS-333-1B2-5 **Lab Number** 2007-00081- 5 **Sampled:** 12/19/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Cementitious Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, rock,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	concrete	100	Gray	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: dissolution of matrix using dilute HCl acid. Procedure: tweased apart using forceps.

Sample IHS-333-1B3-6 **Lab Number** 2007-00081- 6 **Sampled:** 12/19/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Cementitious Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, rock,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	concrete	100	Gray	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: dissolution of matrix using dilute HCl acid. Procedure: tweased apart using forceps.

Sample IHS-333-2A1-7 **Lab Number** 2007-00081- 7 **Sampled:** 12/19/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, rock,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	plaster (top coat)	100	White	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: dissolution of matrix using dilute HCl acid. Procedure: tweased apart using forceps.

Sample IHS-333-2A2-8 **Lab Number** 2007-00081- 8 **Sampled:** 12/19/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, rock,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	plaster (top coat)	100	White	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: dissolution of matrix using dilute HCl acid. Procedure: tweased apart using forceps.

Sample IHS-333-2A3-9 **Lab Number** 2007-00081- 9 **Sampled:** 12/19/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, rock,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	plaster (top coat)	100	White	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: dissolution of matrix using dilute HCl acid. Procedure: tweased apart using forceps.

Sample IHS-333-2A4-10 **Lab Number** 2007-00081- 10 **Sampled:** 12/19/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, rock,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	plaster (top coat)	100	White	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: dissolution of matrix using dilute HCl acid. Procedure: tweased apart using forceps.

Sample IHS-333-2A5-11 **Lab Number** 2007-00081- 11 **Sampled:** 12/19/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, rock,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	plaster (top coat)	100	White	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: dissolution of matrix using dilute HCl acid. Procedure: tweased apart using forceps.

Sample IHS-333-2A6-12 **Lab Number** 2007-00081- 12 **Sampled:** 12/19/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, rock,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	plaster (top coat)	100	White	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: dissolution of matrix using dilute HCl acid. Procedure: tweased apart using forceps.

Sample IHS-333-2A7-13 **Lab Number** 2007-00081- 13 **Sampled:** 12/19/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, rock,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	plaster (top coat)	100	White	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: dissolution of matrix using dilute HCl acid. Procedure: tweased apart using forceps.

Sample IHS-333-2B1-14 **Lab Number** 2007-00081- 14 **Sampled:** 12/19/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, rock,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	plaster (scratch coat)	100	Tan	2	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: dissolution of matrix using dilute HCl acid. Procedure: tweased apart using forceps.

Sample IHS-333-2B2-15 **Lab Number** 2007-00081- 15 **Sampled:** 12/19/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, rock,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	plaster (scratch coat)	100	Tan	2	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: dissolution of matrix using dilute HCl acid. Procedure: tweased apart using forceps.

Sample IHS-333-2B3-16 **Lab Number** 2007-00081- 16 **Sampled:** 12/19/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, rock,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	plaster (scratch coat)	100	Tan	2	<=1%	-	-	-	-	-
	Total %	100		**Average %**	<=1%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: dissolution of matrix using dilute HCl acid. Procedure: tweased apart using forceps.

Sample IHS-333-2B4-17 **Lab Number** 2007-00081- 17 **Sampled:** 12/19/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, rock,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	plaster (scratch coat)	100	Tan	2	<=1%	-	-	-	-	-
	Total %	100		**Average %**	<=1%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: dissolution of matrix using dilute HCl acid. Procedure: tweased apart using forceps.

Sample IHS-333-2B5-18 **Lab Number** 2007-00081- 18 **Sampled:** 12/19/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, rock,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	plaster (scratch coat)	100	Tan	2	<=1%	-	-	-	-	-
	Total %	100		**Average %**	<=1%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: dissolution of matrix using dilute HCl acid. Procedure: tweased apart using forceps.

Sample IHS-333-2B6-19 **Lab Number** 2007-00081- 19 **Sampled:** 12/19/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, rock,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	plaster (scratch coat)	100	Tan	2	<=1%	-	-	-	-	-
	Total %	100		**Average %**	<=1%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: dissolution of matrix using dilute HCl acid. Procedure: tweased apart using forceps.

Sample IHS-333-2B7-20 **Lab Number** 2007-00081- 20 **Sampled:** 12/19/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, rock,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	plaster (scratch coat)	100	Tan	2	<=1%	-	-	-	-	-
	Total %	100		**Average %**	<=1%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: dissolution of matrix using dilute HCl acid. Procedure: tweased apart using forceps.

Sample IHS-333-2C1-21 **Lab Number** 2007-00081- 21 **Sampled:** 12/19/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, rock,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	stucco	100	Tan	2	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: dissolution of matrix using dilute HCl acid. Procedure: tweased apart using forceps.

Sample IHS-333-2C2-22 **Lab Number** 2007-00081- 22 **Sampled:** 12/19/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, rock,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	stucco	100	Tan	2	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: dissolution of matrix using dilute HCl acid. Procedure: tweased apart using forceps.

Sample IHS-333-2C3-23 **Lab Number** 2007-00081- 23 **Sampled:** 12/19/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, rock,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	stucco	100	Tan	2	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: dissolution of matrix using dilute HCl acid. Procedure: tweased apart using forceps.

Sample IHS-333-2C4-24 **Lab Number** 2007-00081- 24 **Sampled:** 12/19/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, rock,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	stucco	100	Tan	2	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: dissolution of matrix using dilute HCl acid. Procedure: tweased apart using forceps.

Sample IHS-333-2C5-25 **Lab Number** 2007-00081- 25 **Sampled:** 12/19/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, rock,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	stucco	100	Tan	2	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: dissolution of matrix using dilute HCl acid. Procedure: tweased apart using forceps.

Sample IHS-333-2C6-26 **Lab Number** 2007-00081- 26 **Sampled:** 12/19/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, rock,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	stucco	100	Tan	2	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: dissolution of matrix using dilute HCl acid. Procedure: tweased apart using forceps.

Sample IHS-333-2C7-27 **Lab Number** 2007-00081- 27 **Sampled:** 12/19/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, rock,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	stucco	100	Tan	2	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: dissolution of matrix using dilute HCl acid. Procedure: tweased apart using forceps.

Sample IHS-333-5A1-28 **Lab Number** 2007-00081- 28 **Sampled:** 12/19/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** TSI Powder
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	insulation	100	White	4	10-20%	<=1%	-	-	-	-
	Total %	100		**Average %**	10-20%	<=1%	-	-	-	-
				Fiber Identification:	chrysotile asbestos	cellulose fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2	cellulose fiber	W	F	N	N	H	+	U					
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-333-5A2-29 **Lab Number** 2007-00081- 29 **Sampled:** 12/19/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** TSI Powder
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	insulation	100	White	4	10-20%	2-5%	-	-	-	-
	Total %	100		**Average %**	10-20%	2-5%	-	-	-	-
				Fiber Identification:	chrysotile asbestos	amosite (grunerite)				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2	amosite (grunerite) asbestos	W	B	N	N	M	+	P	1.680	db/ly	sb/o	1.697	1.685
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-333-5A3-30 **Lab Number** 2007-00081- 30 **Sampled:** 12/19/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** TSI Powder
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	insulation	100	White	4	5-10%	5-10%	-	-	-	-
	Total %	100		**Average %**	5-10%	5-10%	-	-	-	-
				Fiber Identification:	chrysotile asbestos	amosite (grunerite)				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2	amosite (grunerite) asbestos	W	B	N	N	M	+	P	1.680	db/ly	sb/o	1.697	1.685
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-333-5B1-31 **Lab Number** 2007-00081- 31 **Sampled:** 12/19/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** TSI Powder
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	insulation	100	White	4	10-20%	<=1%	-	-	-	-
	Total %	100		**Average %**	10-20%	<=1%	-	-	-	-
				Fiber Identification:	chrysotile asbestos	cellulose fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2	cellulose fiber	W	F	N	N	H	+	U					
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-333-5B2-32 **Lab Number** 2007-00081- 32 **Sampled:** 12/19/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** TSI Powder
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	insulation	100	White	4	10-20%	<=1%	-	-	-	-
	Total %	100		**Average %**	10-20%	<=1%	-	-	-	-
				Fiber Identification:	chrysotile asbestos	cellulose fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2	cellulose fiber	W	F	N	N	H	+	U					
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-333-5B3-33 **Lab Number** 2007-00081- 33 **Sampled:** 12/19/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** TSI Powder
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	insulation	100	White	4	10-20%	<=1%	-	-	-	-
	Total %	100		**Average %**	10-20%	<=1%	-	-	-	-
				Fiber Identification:	chrysotile asbestos	cellulose fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2	cellulose fiber	W	F	N	N	H	+	U					
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-333-7A1-34 **Lab Number** 2007-00081- 34 **Sampled:** 12/19/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	white	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-333-7A2-35 **Lab Number** 2007-00081- 35 **Sampled:** 12/19/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	white	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-333-7A3-36 **Lab Number** 2007-00081- 36 **Sampled:** 12/19/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	white	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-333-7B1-37 **Lab Number** 2007-00081- 37 **Sampled:** 12/19/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Off-white	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-333-7B2-38 **Lab Number** 2007-00081- 38 **Sampled:** 12/19/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Off-white	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-333-7B3-39 **Lab Number** 2007-00081- 39 **Sampled:** 12/19/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Off-white	1	<=1%	-	-	-	-	-
	Total %	100		**Average %**	<=1%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-333-7C1-40 **Lab Number** 2007-00081- 40 **Sampled:** 12/19/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Miscellaneous Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): binder, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	duct tape	100	White	2	10-20%	-	-	-	-	-
	Total %	100		**Average %**	10-20%	-	-	-	-	-
				Fiber Identification:	glass fiber					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	glass fiber	CL	D	Y									
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-333-7C2-41 **Lab Number** 2007-00081- 41 **Sampled:** 12/19/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Miscellaneous Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 4
Non-Fibrous Components (in approx. decreasing order): binder, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	duct tape	100	Off-white	2	5-10%	5-10%	-	-	-	-
	Total %	100		**Average %**	5-10%	5-10%	-	-	-	-
				Fiber Identification:	glass fiber	chrysotile asbestos				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	glass fiber	CL	D	Y									
2	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent. The asbestos is in the caulk portion of the duct tape.

Sample IHS-333-7C3-42 **Lab Number** 2007-00081- 42 **Sampled:** 12/19/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Miscellaneous Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 4
Non-Fibrous Components (in approx. decreasing order): binder, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	duct tape	100	Off-white	2	5-10%	5-10%	-	-	-	-
	Total %	100		**Average %**	5-10%	5-10%	-	-	-	-
				Fiber Identification:	glass fiber	chrysotile asbestos				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	glass fiber	CL	D	Y									
2	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent. The asbestos is in the caulk portion of the duct tape.

Sample IHS-333-8A1-43 **Lab Number** 2007-00081- 43 **Sampled:** 12/19/2006 **Condition:** acceptable

Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Roofing Fibrous Solid

Homogeneous No **# Layers** 14 **Pos Layer?** Yes **# Sub-Samples** 28

Non-Fibrous Components (in approx. decreasing order): bitumen, polymer foam, binder

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	paint	3	Off-white	1	n.d.	n.d.	n.d.	-	-	-
2	paint	5	White	1	10-20%	n.d.	n.d.	-	-	-
3	foam	10	Yellow	3	n.d.	n.d.	n.d.	-	-	-
4	paper/cardboard	2	Tan	2	n.d.	n.d.	n.d.	-	-	-
5	roof ply/bitumen	9	Black	1	>1-2%	50-60%	n.d.	-	-	-
6	roof ply/bitumen	9	Black	1	>1-2%	50-60%	n.d.	-	-	-
7	roof ply/bitumen	9	Black	1	>1-2%	50-60%	n.d.	-	-	-
8	roof ply/bitumen	8	Black	1	n.d.	10-20%	30-40%	-	-	-
9	roof ply/bitumen	8	Black	1	n.d.	10-20%	30-40%	-	-	-
10	roof ply/bitumen	8	Black	1	n.d.	10-20%	30-40%	-	-	-
11	roof ply/bitumen	8	Black	1	n.d.	10-20%	30-40%	-	-	-
12	roof ply/bitumen	7	Black	1	>1-2%	50-60%	n.d.	-	-	-
13	roof ply/bitumen	7	Black	1	>1-2%	50-60%	n.d.	-	-	-
14	roof ply/bitumen	7	Black	1	>1-2%	50-60%	n.d.	-	-	-
	Total %	100		**Average %**	>1-2%	30-40%	10-20%	-	-	-
				Fiber Identification:	synthetic fiber (extr	cellulose fiber	chrysotile asbestos			

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	synthetic fiber (extruded)	W	E	N	N	H	+	P					
2	cellulose fiber	W	F	N	N	H	+	U					
3	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
4													
5													
6													

Sample Analytical Note

Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent. Layer order is probably incorrect.

Sample IHS-333-8A2-44 **Lab Number** 2007-00081- 44 **Sampled:** 12/19/2006 **Condition:** acceptable

Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Roofing Fibrous Solid

Homogeneous No **# Layers** 10 **Pos Layer?** Yes **# Sub-Samples** 22

Non-Fibrous Components (in approx. decreasing order): bitumen, polymer foam, binder

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	paint	5	Off-white	1	n.d.	n.d.	n.d.	-	-	-
2	paint	10	White	1	10-20%	n.d.	n.d.	-	-	-
3	foam	16	Yellow	3	n.d.	n.d.	n.d.	-	-	-
4	bitumen	15	Black	1	n.d.	20-30%	10-20%	-	-	-
5	roof ply/bitumen	9	Black	1	>1-2%	50-60%	n.d.	-	-	-
6	roof ply/bitumen	9	Black	1	>1-2%	50-60%	n.d.	-	-	-
7	roof ply/bitumen	9	Black	1	>1-2%	50-60%	n.d.	-	-	-
8	roof ply/bitumen	9	Black	1	n.d.	10-20%	30-40%	-	-	-
9	roof ply/bitumen	9	Black	1	n.d.	10-20%	30-40%	-	-	-
10	roof ply/bitumen	9	Black	1	n.d.	10-20%	30-40%	-	-	-
	Total %	100		**Average %**	2-5%	20-30%	10-20%	-	-	-
				Fiber Identification:	synthetic fiber (extr	cellulose fiber	chrysotile asbestos			

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	synthetic fiber (extruded)	W	E	N	N	H	+	P					
2	cellulose fiber	W	F	N	N	H	+	U					
3	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
4													
5													
6													

Sample Analytical Note

Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent. Layer order is probably incorrect.

Sample IHS-333-8A3-45 **Lab Number** 2007-00081- 45 **Sampled:** 12/19/2006 **Condition:** acceptable

Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Roofing Fibrous Solid

Homogeneous No **# Layers** 7 **Pos Layer?** Yes **# Sub-Samples** 15

Non-Fibrous Components (in approx. decreasing order): bitumen, polymer foam, binder

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	paint	5	Off-white	1	n.d.	n.d.	n.d.	-	-	-
2	paint	10	White	1	10-20%	n.d.	n.d.	-	-	-
3	foam	16	Yellow	3	n.d.	n.d.	n.d.	-	-	-
4	bitumen	15	Black	1	n.d.	n.d.	10-20%	-	-	-
5	roof ply/bitumen	18	Black	1	n.d.	10-20%	30-40%	-	-	-
6	roof ply/bitumen	18	Black	1	n.d.	10-20%	30-40%	-	-	-
7	roof ply/bitumen	18	Black	1	n.d.	10-20%	30-40%	-	-	-
	Total %	100		**Average %**	>1-2%	5-10%	20-30%	-	-	-
				Fiber Identification:	synthetic fiber (extr	cellulose fiber	chrysotile asbestos			

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	synthetic fiber (extruded)	W	E	N	N	H	+	P					
2	cellulose fiber	W	F	N	N	H	+	U					
3	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
4													
5													
6													

Sample Analytical Note

Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent. Layer order is probably incorrect. Note: Sample was in very poor condition. Exact number of layers could not be determined.

Sample IHS-333-8B1-46 **Lab Number** 2007-00081- 46 **Sampled:** 12/19/2006 **Condition:** acceptable

Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid

Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3

Non-Fibrous Components (in approx. decreasing order): bitumen, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Black	1	5-10%	-	-	-	-	-
	Total %	100		**Average %**	5-10%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2													
3													
4													
5													
6													

Sample Analytical Note

Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-333-8B2-47 **Lab Number** 2007-00081- 47 **Sampled:** 12/19/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): bitumen, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Black	1	5-10%	-	-	-	-	-
	Total %	100		**Average %**	5-10%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers	Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1 chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2												
3												
4												
5												
6												

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-333-8B3-48 **Lab Number** 2007-00081- 48 **Sampled:** 12/19/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): bitumen, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Black	1	5-10%	-	-	-	-	-
	Total %	100		**Average %**	5-10%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers	Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1 chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2												
3												
4												
5												
6												

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-333-8C1-49 **Lab Number** 2007-00081- 49 **Sampled:** 12/19/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Fibrous Mat
Homogeneous No **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 6
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	60	Gray	1	n.d.	-	-	-	-	-
2	caulk	40	White	1	20-30%	-	-	-	-	-
	Total %	100		**Average %**	5-10%	-	-	-	-	-
				Fiber Identification:	synthetic fiber (extr					

Fibers	Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1 synthetic fiber (extruded)	W	E	N	N	H	+	P					
2												
3												
4												
5												
6												

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-333-8C2-50 **Lab Number** 2007-00081- 50 **Sampled:** 12/19/2006 **Condition:** acceptable

Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Fibrous Mat

Homogeneous No **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 6

Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	70	Gray	1	n.d.	-	-	-	-	-
2	caulk	30	White	1	20-30%	-	-	-	-	-
	Total %	100		**Average %**	5-10%	-	-	-	-	-
				Fiber Identification:	synthetic fiber (extr					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	synthetic fiber (extruded)	W	E	N	N	H	+	P					
2													
3													
4													
5													
6													

Sample Analytical Note

Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-333-8C3-51 **Lab Number** 2007-00081- 51 **Sampled:** 12/19/2006 **Condition:** acceptable

Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Fibrous Mat

Homogeneous No **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 6

Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	60	Gray	1	n.d.	-	-	-	-	-
2	caulk	40	White	1	20-30%	-	-	-	-	-
	Total %	100		**Average %**	5-10%	-	-	-	-	-
				Fiber Identification:	synthetic fiber (extr					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	synthetic fiber (extruded)	W	E	N	N	H	+	P					
2													
3													
4													
5													
6													

Sample Analytical Note

Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Fr=Friability: 1=very non-friable; 2= non-friable; 3=friable; 4=highly friable
Colors: B=black;BL=blue;BR=brown;CL=clear;G=Green;GY=gray;OR=orange;OW=off-white;PN=pink;PU=purple;R=red;TN=tan;W=white;Y=yellow;V=various
Fiber Morphology: A=fine fibers/bundles, white, sinewy, flexible; B=fine fibers/bundles, w-br, straight, broomed ends; C=fine fibers/bundles, blue, straight, broomed ends; D=fine to coarse fibers, CL-B, brittle; E=coarse fibers,CL or dyed, striated; F=coarse fibers or splinters, W-BR, ribbon-like; G=lath-like or shards, low aspect ratio, may taper
Iso=isotropism - may be yes or no; Pleo=pleochroism - may be yes or no; Bi=birefringence - may be None, Low, Medium or High
Elg=sign of elongation - may be + or -; Ext=extinction - may be Parallel, Oblique, None or Undulating; Oil=medium used to for dispersion staining
Col Par=dispersion staining colors parallel to the fiber (fiber/halo): b/w=black/white; dg/py=dark gray/pale yellow; vg/y=violet gray/yellow; db/ly=dark blue/lemon yellow; vb/g= vivid blue/gold; sb/o=sky blue/orange; pb/r=pale blue/red; gb/dr=gray blue/dark red; w/b=white/black. Col Perp=same only perpendicular to fiber.
RI Par=refractive index parallel to fiber; RI Perp=refractive index perpendicular to fiber

Analyst: ROBERT A. McCORMICK

Printed: 23-Jan-07

Original Print Date: 23-Jan-07

Larry S. Pierce, Approved Accreditation Signatory

FIBERQUANT ANALYTICAL SERVICES

Polarized Light Microscope (PLM) Analysis for Asbestos

JobNumber:	**200700085**

Client: **IHI ENVIRONMENTAL**

(PHOENIX OFFICE)
4527 N 16th ST STE 105
PHOENIX, AZ 85016-0000
Office Phone: (602) 776-0300
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Samples: 27 PLM **Rec:** 1/4/2007 **Method:** Interim (EPA/600/M4-82-020) PLM analysis for asbestos in bulk smp
Client Job: 07P-3031A / IHS-SF334 **PO Number:** 07P-3031A
Report Date: 1/24/2007 **Date Analyzed:** 1/24/2007 **Routing Number:** -

Method and Analysis Information: **Fiberquant Internal SOP:** PLMi

Each bulk sample is first dissected under a 7-30x magnification stereo-microscope. This examination is used to determine the general type of sample, how many and what type of layers it has, and initial estimates of fiber types and quantities. Second, liquid media mounts are made of each layer - such mounts may be of selected fibers (used solely for identification purposes) or may be representative of the layer as a whole (used for quantitation purposes). The mounts may be made in a synthetic Canadian balsam, one of several solvents, or in refractive index oils (media of known refractive index). Generally, a variety of different mounts are made: some optimized for fiber visibility, some optimized for fiber identification, and some optimized for fiber quantitation. The mounted slides are then examined at 50-400x magnification on a Nikon Labphot-pol microscope. Optical characteristics are used to identify each observed fiber type; the optical data are contained for each sample on its detail analysis sheet, attached.

Current EPA, NESHAP and OSHA regulations designate a result of <=1 % asbestos as "negative" and >1 % asbestos as "positive". Samples containing layers that have been determined to be "positive" may have to be handled differently during a renovation or demolition than samples whose layers have been determined to be "negative."

The method of fiber analysis and identification is the EPA Interim Method (EPA Method 600/M4-82-020). The method of fiber quantitation is an estimation technique in which the analysts quantitation is routinely calibrated by reference quantitation standards, and which has been shown to be equivalent in precision and accuracy to point counting. Friability is estimated for the purposes of deciding when to point count. Friabilities determined in the field take precedence over those determined in the laboratory. Those sample layers which are friable and estimated by the analyst to contain <= 1% asbestos are point counted using 400 points, and the results listed on the detail analysis sheet. Such point counting is required by NESHAP (National Emission Standards for Hazardous Air Polutants, Nov. 1990) in order to rely on analytical results that are <= 1%. The coefficient of variation for the estimation quantitation technique is 100% in the range 0-5%. This means that PLM analysis is not capable of conclusively determining whether a layer containing close to 1% asbestos is actually "positive" or "negative". For this reason, Fiberquant refers to results where asbestos was detected but <= 1% as "borderline negative", and results where asbestos was >1 % but <= 2% as "borderline positive" to indicate the uncertainty in assigning a "positive" or "negative" label. In the sample summary, a "ND" means that no asbestos was detected during the analysis. A "Tr" or "Trace" of asbestos is defined for our purposes as the detection of several asbestos asbestos fibers during the analysis; an amount essentially at the limit of detection for the method. The limit of detection (the smallest % of asbestos that can be detected) varies greatly depending on the matrix in which the asbestos is found. As little as 0.001% asbestos can be detected in favorable samples, while detection in unfavorable samples may approach the detection limit of 1% stated in the method. During the analysis, the analyst, for Fiberquant identification purposes only, determines the "apparent sample type" and "apparent layer types." It must be emphasized that these types are only what is apparent. Often, different materials appear similar or identical after sampling, so the analyst may assign a type other than what was sampled.

Floor tiles present a special problem for PLM asbestos analysis. Floor tile can contain chrysotile fibers so thin that they cannot be resolved by optical methods. In such a case, we may observe a percentage of asbestos which is lower than the actual percentage, or not observe asbestos at all when some is present. For this reason, floor tiles reported as negative should be confirmed to be negative using transmission electron microscope (TEM) analysis. Likewise, vermiculite insulation materials containing traces of asbestiform asbestos present a problem for routine PLM analysis - the amphiboles are sometimes present in trace amounts inhomogeneously distributed. We recommend a hydro-separation technique for such samples.

Vermiculite-containing samples may contain trace amounts of asbestiform amphibole that may or may not be detected during routine PLM analysis. For this reason, loose vermiculite samples reported as negative should be confirmed to contain no amphibole using hydroseparation techniques.

The samples were analyzed under the following ongoing quality assurance program: Blank samples are routinely analyzed to maintain contamination-free materials. Each analyst is a degreed geologist or mineralogist, and has also completed extensive training specific to asbestos analysis for 1-3 months before being allowed to analyze client samples. Qualitative reference samples are routinely analyzed to assure that analysts can identify asbestos and asbestos-look-alike fibers. Quantitative reference samples are routinely analyzed to calibrate and characterize the estimation procedure. Microscope alignment is checked each day. Refractive index oils are calibrated at least monthly. At least 10% of client samples are re-analyzed from scratch by a different analyst than the original, and any discrepancies are resolved for the sample and similar sample types before the results are reported. All quality checks performed for these samples were in control except as detailed in the "Analytical Notes" below. All analysts participate in interlab round robins and proficiency testing to assure competence. Fiberquant is accredited by NVLAP (#101031) for the analysis of bulk samples for asbestos using PLM. Accreditation does not imply endorsement by the EPA, any other United States governmental agency or any private agency or association. Each lab analysis refers only to the sample tested, and may not, due to the sampling process, be

representative of the material sampled. This report may not be reproduced except in full, without the approval of Fiberquant Analytical Services.

The "Analysis Results" below are the averages for all layers of each sample listed. The "+ Layer" column indicates whether any of the layers in the sample were >1% asbestos. Please refer to each detailed analysis sheet to determine which layer(s) of a sample was(were) positive.

Some results may have been calculated using client supplied data, such as volume or area sampled, for which Fiberquant assumes no liability for accuracy.

Job Analysis Notes:

Single layer sample analysis as per client request. Any material or layer other than that indicated on the chain of custody was not analyzed, even if a suspect material.

PLM Analysis Summary: **Job Number: 200700085** 07P-3031A / IHS-SF334

Sample Number / Layer	Color	Lab Number / Apparent Layer Type *	Apparent Sample Type * / *Asbestos Results*	Positive Layer Yes or No
Sample **IHS-SF334-1B1-1**		2007-00085- 1	Flooring	Positive Layer? Yes
Layer # 1	off-white floor tile		*2-5% chrysotile asbestos*	
Sample **IHS-SF334-1B2-2**		2007-00085- 2	Flooring	Positive Layer? Yes
Layer # 1	off-white floor tile		*2-5% chrysotile asbestos*	
Layer # 2	Yellow mastic		*no asbestos detected*	
Sample **IHS-SF334-1B3-3**		2007-00085- 3	Flooring	Positive Layer? Yes
Layer # 1	off-white floor tile		*2-5% chrysotile asbestos*	
Layer # 2	Yellow mastic		*no asbestos detected*	
Sample **IHS-SF334-1C1-4**		2007-00085- 4	Flooring	Positive Layer? No
Layer # 1	off-white floor tile		*no asbestos detected*	
Layer # 2	Yellow mastic		*no asbestos detected*	
Sample **IHS-SF334-1C2-5**		2007-00085- 5	Flooring	Positive Layer? No
Layer # 1	off-white floor tile		*no asbestos detected*	
Layer # 2	Black mastic		*no asbestos detected*	
Sample **IHS-SF334-1C3-6**		2007-00085- 6	Flooring	Positive Layer? No
Layer # 1	off-white floor tile		*no asbestos detected*	
Layer # 2	Various mastic		*no asbestos detected*	
Sample **IHS-SF334-1D1-7**		2007-00085- 7	Adhesive/caulk	Positive Layer? No
Layer # 1	Black foam		*no asbestos detected*	
Layer # 2	Yellow mastic		*no asbestos detected*	
Sample **IHS-SF334-1D2-8**		2007-00085- 8	Adhesive/caulk	Positive Layer? No
Layer # 1	Black foam		*no asbestos detected*	
Layer # 2	Yellow mastic		*no asbestos detected*	
Sample **IHS-SF334-1D3-9**		2007-00085- 9	Adhesive/caulk	Positive Layer? No
Layer # 1	Black foam		*no asbestos detected*	
Layer # 2	Yellow mastic		*no asbestos detected*	
Sample **IHS-SF334-2A1-10**		2007-00085- 10	Wall System	Positive Layer? No
Layer # 1	white drywall core		*no asbestos detected*	
Sample **IHS-SF334-2A2-11**		2007-00085- 11	Wall System	Positive Layer? No
Layer # 1	white drywall core		*no asbestos detected*	
Sample **IHS-SF334-2A3-12**		2007-00085- 12	Wall System	Positive Layer? No
Layer # 1	white drywall core		*no asbestos detected*	
Sample **IHS-SF334-3A1-13**		2007-00085- 13	Acoustic Tile	Positive Layer? No
Layer # 1	Off-white acoustical tile		*no asbestos detected*	
Sample **IHS-SF334-3A2-14**		2007-00085- 14	Acoustic Tile	Positive Layer? No
Layer # 1	Off-white acoustical tile		*no asbestos detected*	
Sample **IHS-SF334-3A3-15**		2007-00085- 15	Acoustic Tile	Positive Layer? No
Layer # 1	Off-white acoustical tile		*no asbestos detected*	
Sample **IHS-SF334-7A1-16**		2007-00085- 16	Adhesive/caulk	Positive Layer? Yes
Layer # 1	Yellow mastic		*no asbestos detected*	
Layer # 2	Brown mastic		*2-5% chrysotile asbestos*	
Sample **IHS-SF334-7A2-17**		2007-00085- 17	Adhesive/caulk	Positive Layer? No
Layer # 1	Yellow mastic		*no asbestos detected*	
Sample **IHS-SF334-7A3-18**		2007-00085- 18	Adhesive/caulk	Positive Layer? Yes
Layer # 1	Yellow mastic		*no asbestos detected*	
Layer # 2	Brown mastic		*2-5% chrysotile asbestos*	
Sample **IHS-SF334-7B1-19**		2007-00085- 19	Adhesive/caulk	Positive Layer? No
Layer # 1	Yellow mastic		*no asbestos detected*	
Sample **IHS-SF334-7B2-20**		2007-00085- 20	Adhesive/caulk	Positive Layer? Yes
Layer # 1	Yellow mastic		*no asbestos detected*	
Layer # 2	Brown mastic		*2-5% chrysotile asbestos*	
Sample **IHS-SF334-7B3-21**		2007-00085- 21	Adhesive/caulk	Positive Layer? No
Layer # 1	Yellow caulk		*no asbestos detected*	
Sample **IHS-SF334-7C1-22**		2007-00085- 22	Adhesive/caulk	Positive Layer? No
Layer # 1	Gray caulk		*no asbestos detected*	
Sample **IHS-SF334-7C2-23**		2007-00085- 23	Adhesive/caulk	Positive Layer? No
Layer # 1	Gray caulk		*no asbestos detected*	
Sample **IHS-SF334-7C3-24**		2007-00085- 24	Adhesive/caulk	Positive Layer? No
Layer # 1	Gray caulk		*no asbestos detected*	

Sample	**IHS-SF334-7D1-25**	2007-00085- 25	Insulation	Positive Layer? No
	Layer # 1 Gray insulation		*no asbestos detected*	
Sample	**IHS-SF334-7D2-26**	2007-00085- 26	Insulation	Positive Layer? No
	Layer # 1 Gray insulation		*no asbestos detected*	
Sample	**IHS-SF334-7D3-27**	2007-00085- 27	Insulation	Positive Layer? No
	Layer # 1 Gray insulation		*no asbestos detected*	

* Apparent Sample Types and Apparent Layer Types are as they appeared to the analyst. Since many types of materials appear similar after sampling damage, the apparent type of material may not be the actual type of material.

Sample IHS-SF334-1B1-1 **Lab Number** 2007-00085- 1 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/24/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	floor tile	100	off-white	1	2-5%	<=1%	-	-	-	-
	Total %	100		**Average %**	2-5%	<=1%	-	-	-	-
				Fiber Identification:	chrysotile asbestos	cellulose fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2	cellulose fiber	W	F	N	N	H	+	U					
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF334-1B2-2 **Lab Number** 2007-00085- 2 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/24/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** Yes **# Sub-Samples** 5
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	floor tile	98	off-white	1	2-5%	<=1%	-	-	-	-
2	mastic	2	Yellow	1	n.d.	>1-2%	-	-	-	-
	Total %	100		**Average %**	2-5%	>1-2%	-	-	-	-
				Fiber Identification:	chrysotile asbestos	cellulose fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2	cellulose fiber	W	F	N	N	H	+	U					
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF334-1B3-3 **Lab Number** 2007-00085- 3 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/24/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** Yes **# Sub-Samples** 5
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	floor tile	98	off-white	1	2-5%	<=1%	-	-	-	-
2	mastic	2	Yellow	1	n.d.	>1-2%	-	-	-	-
	Total %	100		**Average %**	2-5%	>1-2%	-	-	-	-
				Fiber Identification:	chrysotile asbestos	cellulose fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2	cellulose fiber	W	F	N	N	H	+	U					
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF334-1C1-4 **Lab Number** 2007-00085- 4 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/24/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 5
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	floor tile	98	off-white	1	n.d.	-	-	-	-	-
2	mastic	2	Yellow	1	2-5%	-	-	-	-	-
	Total %	100		**Average %**	<=1%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF334-1C2-5 **Lab Number** 2007-00085- 5 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/24/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 5
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	floor tile	98	off-white	1	n.d.	n.d.	-	-	-	-
2	mastic	2	Black	1	2-5%	<=1%	-	-	-	-
	Total %	100		**Average %**	<=1%	<=1%	-	-	-	-
				Fiber Identification:	cellulose fiber	glass fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2	glass fiber	CL	D	Y									
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent. Mastic was a mix of more than one mastic.

Sample IHS-SF334-1C3-6 **Lab Number** 2007-00085- 6 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/24/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 5
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	floor tile	97	off-white	1	n.d.	n.d.	-	-	-	-
2	mastic	3	Various	1	2-5%	<=1%	-	-	-	-
	Total %	100		**Average %**	<=1%	<=1%	-	-	-	-
				Fiber Identification:	cellulose fiber	glass fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2	glass fiber	CL	D	Y									
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent. Mastic was a mix of more than one mastic.

Sample IHS-SF334-1D1-7 **Lab Number** 2007-00085- 7 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/24/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 6
Non-Fibrous Components (in approx. decreasing order): polymer foam, binder, filler

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	foam	60	Black	3	n.d.	-	-	-	-	-
2	mastic	40	Yellow	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF334-1D2-8 **Lab Number** 2007-00085- 8 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/24/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 6
Non-Fibrous Components (in approx. decreasing order): polymer foam, binder, filler

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	foam	80	Black	3	n.d.	-	-	-	-	-
2	mastic	20	Yellow	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF334-1D3-9 **Lab Number** 2007-00085- 9 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/24/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 6
Non-Fibrous Components (in approx. decreasing order): polymer foam, binder, filler

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	foam	80	Black	3	n.d.	-	-	-	-	-
2	mastic	20	Yellow	1	2-5%	-	-	-	-	-
	Total %	100		**Average %**	<=1%	-	-	-	-	-
				Fiber Identification:	synthetic fiber (extr					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	synthetic fiber (extruded)	W	E	N	N	H	+	P					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF334-2A1-10 **Lab Number** 2007-00085- 10 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/24/2007 **An?** OK **Apparent Smp Type** Wall System Fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	drywall core	100	white	3	<=1%	-	-	-	-	-
	Total %	100		**Average %**	<=1%	-	-	-	-	-
				Fiber Identification:	cellulose					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps.

Sample IHS-SF334-2A2-11 **Lab Number** 2007-00085- 11 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/24/2007 **An?** OK **Apparent Smp Type** Wall System Fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	drywall core	100	white	3	<=1%	-	-	-	-	-
	Total %	100		**Average %**	<=1%	-	-	-	-	-
				Fiber Identification:	cellulose					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps.

Sample IHS-SF334-2A3-12 **Lab Number** 2007-00085- 12 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/24/2007 **An?** OK **Apparent Smp Type** Wall System Fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	drywall core	100	white	3	<=1%	-	-	-	-	-
	Total %	100		**Average %**	<=1%	-	-	-	-	-
				Fiber Identification:	cellulose					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps.

Sample IHS-SF334-3A1-13 **Lab Number** 2007-00085- 13 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/24/2007 **An?** OK **Apparent Smp Type** Acoustic Tile Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): perlite, glass,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	acoustical tile	100	Off-white	3	40-50%	20-30%	-	-	-	-
	Total %	100		**Average %**	40-50%	20-30%	-	-	-	-
				Fiber Identification:	cellulose fiber	glass fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2	glass fiber	CL	D	Y									
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF334-3A2-14 **Lab Number** 2007-00085- 14 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/24/2007 **An?** OK **Apparent Smp Type** Acoustic Tile Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): perlite, glass,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	acoustical tile	100	Off-white	3	40-50%	20-30%	-	-	-	-
	Total %	100		**Average %**	40-50%	20-30%	-	-	-	-
				Fiber Identification:	cellulose fiber	glass fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2	glass fiber	CL	D	Y									
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF334-3A3-15 **Lab Number** 2007-00085- 15 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/24/2007 **An?** OK **Apparent Smp Type** Acoustic Tile Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): perlite, glass,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	acoustical tile	100	Off-white	3	40-50%	20-30%	-	-	-	-
	Total %	100		**Average %**	40-50%	20-30%	-	-	-	-
				Fiber Identification:	cellulose fiber	glass fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2	glass fiber	CL	D	Y									
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF334-7A1-16 **Lab Number** 2007-00085- 16 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/24/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** Yes **# Sub-Samples** 5
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	15	Yellow	1	n.d.	-	-	-	-	-
2	mastic	85	Brown	1	2-5%	-	-	-	-	-
	Total %	100		**Average %**	2-5%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF334-7A2-17 **Lab Number** 2007-00085- 17 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/24/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	100	Yellow	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF334-7A3-18 **Lab Number** 2007-00085- 18 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/24/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** Yes **# Sub-Samples** 5
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	50	Yellow	1	n.d.	-	-	-	-	-
2	mastic	50	Brown	1	2-5%	-	-	-	-	-
	Total %	100		**Average %**	>1-2%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF334-7B1-19 **Lab Number** 2007-00085- 19 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/24/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	100	Yellow	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF334-7B2-20 **Lab Number** 2007-00085- 20 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/24/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** Yes **# Sub-Samples** 5
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	80	Yellow	1	n.d.	-	-	-	-	-
2	mastic	20	Brown	1	2-5%	-	-	-	-	-
	Total %	100		**Average %**	<=1%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF334-7B3-21 **Lab Number** 2007-00085- 21 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/24/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Yellow	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF334-7C1-22 **Lab Number** 2007-00085- 22 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/24/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Rubbery
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Gray	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF334-7C2-23 **Lab Number** 2007-00085- 23 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/24/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Rubbery
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Gray	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF334-7C3-24 **Lab Number** 2007-00085- 24 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/24/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Rubbery
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Gray	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF334-7D1-25 **Lab Number** 2007-00085- 25 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/24/2007 **An?** OK **Apparent Smp Type** Insulation Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): binder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	insulation	100	Gray	3	90-100%	-	-	-	-	-
	Total %	100		**Average %**	90-100%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps.

Sample IHS-SF334-7D2-26 **Lab Number** 2007-00085- 26 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/24/2007 **An?** OK **Apparent Smp Type** Insulation Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): binder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	insulation	100	Gray	3	90-100%	-	-	-	-	-
	Total %	100		**Average %**	90-100%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps.

Sample IHS-SF334-7D3-27 **Lab Number** 2007-00085- 27 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/24/2007 **An?** OK **Apparent Smp Type** Insulation Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): binder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	insulation	100	Gray	3	90-100%	-	-	-	-	-
	Total %	100		**Average %**	90-100%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps.

Fr=Friability: 1=very non-friable; 2= non-friable; 3=friable; 4=highly friable
Colors: B=black;BL=blue;BR=brown;CL=clear;G=Green;GY=gray;OR=orange;OW=off-white;PN=pink;PU=purple;R=red;TN=tan;W=white;Y=yellow;V=various
Fiber Morphology: A=fine fibers/bundles, white, sinewy, flexible; B=fine fibers/bundles, w-br, straight, broomed ends; C=fine fibers/bundles, blue, straight, broomed ends; D=fine to coarse fibers, CL-B, brittle; E=coarse fibers,CL or dyed, striated; F=coarse fibers or splinters, W-BR, ribbon-like; G=lath-like or shards, low aspect ratio, may taper
Iso=isotropism - may be yes or no; Pleo=pleochroism - may be yes or no; Bi=birefringence - may be None, Low, Medium or High
Elg=sign of elongation - may be + or -; Ext=extinction - may be Parallel, Oblique, None or Undulating; Oil=medium used to for dispersion staining
Col Par=dispersion staining colors parallel to the fiber (fiber/halo): b/w=black/white; dg/py=dark gray/pale yellow; vg/y=violet gray/yellow; db/ly=dark blue/lemon yellow; vb/g= vivid blue/gold; sb/o=sky blue/orange; pb/r=pale blue/red; gb/dr=gray blue/dark red; w/b=white/black. Col Perp=same only perpendicular to fiber.
RI Par=refractive index parallel to fiber; RI Perp=refractive index perpendicular to fiber

Analyst: ROBERT A. McCORMICK

Printed: 24-Jan-07

Original Print Date: 24-Jan-07

Larry S. Pierce, Approved Accreditation Signatory

FIBERQUANT ANALYTICAL SERVICES

Polarized Light Microscope (PLM) Analysis for Asbestos

JobNumber:	**200700084**

Client: **IHI ENVIRONMENTAL**

(PHOENIX OFFICE)
4527 N 16th ST STE 105
PHOENIX, AZ 85016-0000
Office Phone: (602) 776-0300
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Samples: 27 PLM **Rec:** 1/4/2007 **Method:** Interim (EPA/600/M4-82-020) PLM analysis for asbestos in bulk smp
Client Job: 07P-3031A / IHS-SF335 **PO Number:** 07P-3031A
Report Date: 1/24/2007 **Date Analyzed:** 1/23/2007 **Routing Number:** -

Method and Analysis Information: **Fiberquant Internal SOP:** PLMi

Each bulk sample is first dissected under a 7-30x magnification stereo-microscope. This examination is used to determine the general type of sample, how many and what type of layers it has, and initial estimates of fiber types and quantities. Second, liquid media mounts are made of each layer - such mounts may be of selected fibers (used solely for identification purposes) or may be representative of the layer as a whole (used for quantitation purposes). The mounts may be made in a synthetic Canadian balsam, one of several solvents, or in refractive index oils (media of known refractive index). Generally, a variety of different mounts are made: some optimized for fiber visibility, some optimized for fiber identification, and some optimized for fiber quantitation. The mounted slides are then examined at 50-400x magnification on a Nikon Labphot-pol microscope. Optical characteristics are used to identify each observed fiber type; the optical data are contained for each sample on its detail analysis sheet, attached.

Current EPA, NESHAP and OSHA regulations designate a result of <=1 % asbestos as "negative" and >1 % asbestos as "positive". Samples containing layers that have been determined to be "positive" may have to be handled differently during a renovation or demolition than samples whose layers have been determined to be "negative."

The method of fiber analysis and identification is the EPA Interim Method (EPA Method 600/M4-82-020). The method of fiber quantitation is an estimation technique in which the analysts quantitation is routinely calibrated by reference quantitation standards, and which has been shown to be equivalent in precision and accuracy to point counting. Friability is estimated for the purposes of deciding when to point count. Friabilities determined in the field take precedence over those determined in the laboratory. Those sample layers which are friable and estimated by the analyst to contain <= 1% asbestos are point counted using 400 points, and the results listed on the detail analysis sheet. Such point counting is required by NESHAP (National Emission Standards for Hazardous Air Polutants, Nov. 1990) in order to rely on analytical results that are <= 1%. The coefficient of variation for the estimation quantitation technique is 100% in the range 0-5%. This means that PLM analysis is not capable of conclusively determining whether a layer containing close to 1% asbestos is actually "positive" or "negative". For this reason, Fiberquant refers to results where asbestos was detected but <= 1% as "borderline negative", and results where asbestos was >1 % but <= 2% as "borderline positive" to indicate the uncertainty in assigning a "positive" or "negative" label. In the sample summary, a "ND" means that no asbestos was detected during the analysis. A "Tr" or "Trace" of asbestos is defined for our purposes as the detection of several asbestos asbestos fibers during the analysis; an amount essentially at the limit of detection for the method. The limit of detection (the smallest % of asbestos that can be detected) varies greatly depending on the matrix in which the asbestos is found. As little as 0.001% asbestos can be detected in favorable samples, while detection in unfavorable samples may approach the detection limit of 1% stated in the method. During the analysis, the analyst, for Fiberquant identification purposes only, determines the "apparent sample type" and "apparent layer types." It must be emphasized that these types are only what is apparent. Often, different materials appear similar or identical after sampling, so the analyst may assign a type other than what was sampled.

Floor tiles present a special problem for PLM asbestos analysis. Floor tile can contain chrysotile fibers so thin that they cannot be resolved by optical methods. In such a case, we may observe a percentage of asbestos which is lower than the actual percentage, or not observe asbestos at all when some is present. For this reason, floor tiles reported as negative should be confirmed to be negative using transmission electron microscope (TEM) analysis. Likewise, vermiculite insulation materials containing traces of asbestiform asbestos present a problem for routine PLM analysis - the amphiboles are sometimes present in trace amounts inhomogeneously distributed. We recommend a hydro-separation technique for such samples.

Vermiculite-containing samples may contain trace amounts of asbestiform amphibole that may or may not be detected during routine PLM analysis. For this reason, loose vermiculite samples reported as negative should be confirmed to contain no amphibole using hydroseparation techniques.

The samples were analyzed under the following ongoing quality assurance program: Blank samples are routinely analyzed to maintain contamination-free materials. Each analyst is a degreed geologist or mineralogist, and has also completed extensive training specific to asbestos analysis for 1-3 months before being allowed to analyze client samples. Qualitative reference samples are routinely analyzed to assure that analysts can identify asbestos and asbestos-look-alike fibers. Quantitative reference samples are routinely analyzed to calibrate and characterize the estimation procedure. Microscope alignment is checked each day. Refractive index oils are calibrated at least monthly. At least 10% of client samples are re-analyzed from scratch by a different analyst than the original, and any discrepancies are resolved for the sample and similar sample types before the results are reported. All quality checks performed for these samples were in control except as detailed in the "Analytical Notes" below. All analysts participate in interlab round robins and proficiency testing to assure competence. Fiberquant is accredited by NVLAP (#101031) for the analysis of bulk samples for asbestos using PLM. Accreditation does not imply endorsement by the EPA, any other United States governmental agency or any private agency or association. Each lab analysis refers only to the sample tested, and may not, due to the sampling process, be

representative of the material sampled. This report may not be reproduced except in full, without the approval of Fiberquant Analytical Services.

The "Analysis Results" below are the averages for all layers of each sample listed. The "+ Layer" column indicates whether any of the layers in the sample were >1% asbestos. Please refer to each detailed analysis sheet to determine which layer(s) of a sample was(were) positive.

Some results may have been calculated using client supplied data, such as volume or area sampled, for which Fiberquant assumes no liability for accuracy.

Job Analysis Notes:

Single layer sample analysis as per client request. Any material or layer other than that indicated on the chain of custody was not analyzed, even if a suspect material.

PLM Analysis Summary: **Job Number: 200700084** 07P-3031A / IHS-SF335

Sample Number / Layer	Color	Lab Number / Apparent Layer Type *	Apparent Sample Type * / *Asbestos Results*	Positive Layer Yes or No
Sample **IHS-SF335-1A1-1**		2007-00084- 1	Flooring	Positive Layer? Yes
Layer # 1	off-white	floor tile	*2-5% chrysotile asbestos*	
Layer # 2	Tan	mastic	*no asbestos detected*	
Sample **IHS-SF335-1A2-2**		2007-00084- 2	Flooring	Positive Layer? Yes
Layer # 1	off-white	floor tile	*2-5% chrysotile asbestos*	
Layer # 2	Tan	mastic	*no asbestos detected*	
Sample **IHS-SF335-1A3-3**		2007-00084- 3	Flooring	Positive Layer? Yes
Layer # 1	off-white	floor tile	*2-5% chrysotile asbestos*	
Layer # 2	Tan	mastic	*no asbestos detected*	
Sample **IHS-SF335-2A1-4**		2007-00084- 4	Wall System	Positive Layer? No
Layer # 1	white	texture/joint compound	*no asbestos detected*	
Sample **IHS-SF335-2A2-5**		2007-00084- 5	Wall System	Positive Layer? No
Layer # 1	white	drywall core	*no asbestos detected*	
Sample **IHS-SF335-2A3-6**		2007-00084- 6	Wall System	Positive Layer? No
Layer # 1	white	drywall core	*no asbestos detected*	
Sample **IHS-SF335-2B1-7**		2007-00084- 7	Wall System	Positive Layer? No
Layer # 1	white	texture/joint compound	*no asbestos detected*	
Sample **IHS-SF335-2B2-8**		2007-00084- 8	Wall System	Positive Layer? No
Layer # 1	white	texture/joint compound	*no asbestos detected*	
Sample **IHS-SF335-2B3-9**		2007-00084- 9	Wall System	Positive Layer? No
Layer # 1	white	texture/joint compound	*no asbestos detected*	
Sample **IHS-SF335-3A1-10**		2007-00084- 10	Acoustic Tile	Positive Layer? Yes
Layer # 1	Tan	acoustical tile	*no asbestos detected*	
Sample **IHS-SF335-3A2-11**		2007-00084- 11	Acoustic Tile	Positive Layer? Yes
Layer # 1	Tan	acoustical tile	*no asbestos detected*	
Sample **IHS-SF335-3A3-12**		2007-00084- 12	Acoustic Tile	Positive Layer? Yes
Layer # 1	Tan	acoustical tile	*no asbestos detected*	
Sample **IHS-SF335-7A1-13**		2007-00084- 13	Adhesive/caulk	Positive Layer? Yes
Layer # 1	Brown	mastic	*>1-2% chrysotile asbestos*	
Layer # 2	Tan	mastic	*no asbestos detected*	
Sample **IHS-SF335-7A2-14**		2007-00084- 14	Adhesive/caulk	Positive Layer? Yes
Layer # 1	Brown	mastic	*>1-2% chrysotile asbestos*	
Sample **IHS-SF335-7A3-15**		2007-00084- 15	Adhesive/caulk	Positive Layer? Yes
Layer # 1	Brown	mastic	*>1-2% chrysotile asbestos*	
Sample **IHS-SF335-7B1-16**		2007-00084- 16	Adhesive/caulk	Positive Layer? No
Layer # 1	Tan	mastic	*no asbestos detected*	
Sample **IHS-SF335-7B2-17**		2007-00084- 17	Adhesive/caulk	Positive Layer? No
Layer # 1	Tan	mastic	*no asbestos detected*	
Sample **IHS-SF335-7B3-18**		2007-00084- 18	Adhesive/caulk	Positive Layer? No
Layer # 1	Tan	mastic	*no asbestos detected*	
Sample **IHS-SF335-7C1-19**		2007-00084- 19	Miscellaneous	Positive Layer? Yes
Layer # 1	Black	coating	*>1-2% chrysotile asbestos*	
Sample **IHS-SF335-7C2-20**		2007-00084- 20	Miscellaneous	Positive Layer? Yes
Layer # 1	Black	coating	*>1-2% chrysotile asbestos*	
Sample **IHS-SF335-7C3-21**		2007-00084- 21	Miscellaneous	Positive Layer? Yes
Layer # 1	Black	coating	*>1-2% chrysotile asbestos*	
Sample **IHS-SF335-7D1-22**		2007-00084- 22	Adhesive/caulk	Positive Layer? No
Layer # 1	Gray	caulk	*no asbestos detected*	
Sample **IHS-SF335-7D2-23**		2007-00084- 23	Adhesive/caulk	Positive Layer? No
Layer # 1	Gray	caulk	*no asbestos detected*	
Sample **IHS-SF335-7D3-24**		2007-00084- 24	Adhesive/caulk	Positive Layer? No
Layer # 1	Gray	caulk	*no asbestos detected*	
Sample **IHS-SF335-7E1-25**		2007-00084- 25	Insulation	Positive Layer? No
Layer # 1	Tan	insulation	*no asbestos detected*	
Sample **IHS-SF335-7E2-26**		2007-00084- 26	Insulation	Positive Layer? No
Layer # 1	Tan	insulation	*no asbestos detected*	
Sample **IHS-SF335-7E3-27**		2007-00084- 27	Insulation	Positive Layer? No
Layer # 1	Tan	insulation	*no asbestos detected*	

* Apparent Sample Types and Apparent Layer Types are as they appeared to the analyst. Since many types of materials appear similar after sampling damage, the apparent type of material may not be the actual type of material.

Sample IHS-SF335-1A1-1 **Lab Number** 2007-00084- 1 **Sampled:** 12/21/2006 **Condition:** acceptable
Analyzed By MAC 1/23/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** Yes **# Sub-Samples** 6
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	floor tile	88	off-white	1	2-5%	n.d.	n.d.	-	-	-
2	mastic	12	Tan	1	n.d.	<=1%	<=1%	-	-	-
	Total %	100		**Average %**	2-5%	<=1%	<=1%	-	-	-
				Fiber Identification:	chrysotile asbestos	cellulose fiber	synthetic fiber (extr			

Fibers	Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1 chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2 cellulose fiber	W	F	N	N	H	+	U					
3 synthetic fiber (extruded)	W	E	N	N	H	+	P					
4												
5												
6												

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF335-1A2-2 **Lab Number** 2007-00084- 2 **Sampled:** 12/21/2006 **Condition:** acceptable
Analyzed By MAC 1/23/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** Yes **# Sub-Samples** 6
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	floor tile	88	off-white	1	2-5%	n.d.	n.d.	-	-	-
2	mastic	12	Tan	1	n.d.	<=1%	<=1%	-	-	-
	Total %	100		**Average %**	2-5%	<=1%	<=1%	-	-	-
				Fiber Identification:	chrysotile asbestos	cellulose fiber	synthetic fiber (extr			

Fibers	Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1 chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2 cellulose fiber	W	F	N	N	H	+	U					
3 synthetic fiber (extruded)	W	E	N	N	H	+	P					
4												
5												
6												

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF335-1A3-3 **Lab Number** 2007-00084- 3 **Sampled:** 12/21/2006 **Condition:** acceptable
Analyzed By MAC 1/23/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** Yes **# Sub-Samples** 6
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	floor tile	98	off-white	1	2-5%	n.d.	-	-	-	-
2	mastic	2	Tan	1	n.d.	<=1%	-	-	-	-
	Total %	100		**Average %**	2-5%	<=1%	-	-	-	-
				Fiber Identification:	chrysotile asbestos	cellulose fiber				

Fibers	Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1 chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2 cellulose fiber	W	F	N	N	H	+	U					
3												
4												
5												
6												

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF335-2A1-4 **Lab Number** 2007-00084- 4 **Sampled:** 12/21/2006 **Condition:** acceptable
Analyzed By MAC 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	texture/joint compound	100	white	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
No drywall core present, only texture/joint compound. Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF335-2A2-5 **Lab Number** 2007-00084- 5 **Sampled:** 12/21/2006 **Condition:** acceptable
Analyzed By MAC 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	drywall core	100	white	3	<=1%	<=1%	-	-	-	-
	Total %	100		**Average %**	<=1%	<=1%	-	-	-	-
				Fiber Identification:	cellulose	glass fiber				

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose	W	F	N	N	H	+	U					
2	glass fiber	CL	D	Y									
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps.

Sample IHS-SF335-2A3-6 **Lab Number** 2007-00084- 6 **Sampled:** 12/21/2006 **Condition:** acceptable
Analyzed By MAC 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	drywall core	100	white	3	<=1%	<=1%	-	-	-	-
	Total %	100		**Average %**	<=1%	<=1%	-	-	-	-
				Fiber Identification:	cellulose	glass fiber				

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose	W	F	N	N	H	+	U					
2	glass fiber	CL	D	Y									
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps.

Sample IHS-SF335-2B1-7 **Lab Number** 2007-00084- 7 **Sampled:** 12/21/2006 **Condition:** acceptable
Analyzed By MAC 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	texture/joint compound	100	white	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF335-2B2-8 **Lab Number** 2007-00084- 8 **Sampled:** 12/21/2006 **Condition:** acceptable
Analyzed By MAC 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	texture/joint compound	100	white	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF335-2B3-9 **Lab Number** 2007-00084- 9 **Sampled:** 12/21/2006 **Condition:** acceptable
Analyzed By MAC 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	texture/joint compound	100	white	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF335-3A1-10 **Lab Number** 2007-00084- 10 **Sampled:** 12/21/2006 **Condition:** acceptable
Analyzed By MAC 1/23/2007 **An?** OK **Apparent Smp Type** Acoustic Tile Fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): perlite, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	acoustical tile	100	Tan	3	30-40%	20-30%	-	-	-	-
	Total %	100		**Average %**	30-40%	20-30%	-	-	-	-
				Fiber Identification:	cellulose fiber	glass fiber				

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2	glass fiber	CL	D	Y									
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF335-3A2-11 **Lab Number** 2007-00084- 11 **Sampled:** 12/21/2006 **Condition:** acceptable
Analyzed By MAC 1/24/2007 **An?** OK **Apparent Smp Type** Acoustic Tile Fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): perlite, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	acoustical tile	100	Tan	3	30-40%	20-30%	-	-	-	-
	Total %	100		**Average %**	30-40%	20-30%	-	-	-	-
				Fiber Identification:	cellulose fiber	glass fiber				

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2	glass fiber	CL	D	Y									
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF335-3A3-12 **Lab Number** 2007-00084- 12 **Sampled:** 12/21/2006 **Condition:** acceptable
Analyzed By MAC 1/24/2007 **An?** OK **Apparent Smp Type** Acoustic Tile Fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): perlite, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	acoustical tile	100	Tan	3	30-40%	20-30%	-	-	-	-
	Total %	100		**Average %**	30-40%	20-30%	-	-	-	-
				Fiber Identification:	cellulose fiber	glass fiber				

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2	glass fiber	CL	D	Y									
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF335-7A1-13 **Lab Number** 2007-00084- 13 **Sampled:** 12/21/2006 **Condition:** acceptable
Analyzed By MAC 1/24/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** Yes **# Sub-Samples** 6
Non-Fibrous Components (in approx. decreasing order): binder, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	80	Brown	1	>1-2%	-	-	-	-	-
2	mastic	20	Tan	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	>1-2%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF335-7A2-14 **Lab Number** 2007-00084- 14 **Sampled:** 12/21/2006 **Condition:** acceptable
Analyzed By MAC 1/24/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): binder, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	100	Brown	1	>1-2%	-	-	-	-	-
	Total %	100		**Average %**	>1-2%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF335-7A3-15 **Lab Number** 2007-00084- 15 **Sampled:** 12/21/2006 **Condition:** acceptable
Analyzed By MAC 1/24/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): binder, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	100	Brown	1	>1-2%	-	-	-	-	-
	Total %	100		**Average %**	>1-2%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF335-7B1-16 **Lab Number** 2007-00084- 16 **Sampled:** 12/21/2006 **Condition:** acceptable
Analyzed By MAC 1/24/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	100	Tan	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF335-7B2-17 **Lab Number** 2007-00084- 17 **Sampled:** 12/21/2006 **Condition:** acceptable
Analyzed By MAC 1/24/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	100	Tan	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF335-7B3-18 **Lab Number** 2007-00084- 18 **Sampled:** 12/21/2006 **Condition:** acceptable
Analyzed By MAC 1/24/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	100	Tan	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF335-7C1-19 **Lab Number** 2007-00084- 19 **Sampled:** 12/21/2006 **Condition:** acceptable
Analyzed By MAC 1/24/2007 **An?** OK **Apparent Smp Type** Miscellaneous Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	coating	100	Black	1	>1-2%	-	-	-	-	-
	Total %	100		**Average %**	>1-2%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.556
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF335-7C2-20 **Lab Number** 2007-00084- 20 **Sampled:** 12/21/2006 **Condition:** acceptable
Analyzed By MAC 1/24/2007 **An?** OK **Apparent Smp Type** Miscellaneous Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	coating	100	Black	1	>1-2%	-	-	-	-	-
	Total %	100		**Average %**	>1-2%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.556
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF335-7C3-21 **Lab Number** 2007-00084- 21 **Sampled:** 12/21/2006 **Condition:** acceptable
Analyzed By MAC 1/24/2007 **An?** OK **Apparent Smp Type** Miscellaneous Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	coating	100	Black	1	>1-2%	-	-	-	-	-
	Total %	100		**Average %**	>1-2%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.556
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF335-7D1-22 **Lab Number** 2007-00084- 22 **Sampled:** 12/21/2006 **Condition:** acceptable
Analyzed By MAC 1/24/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): binder, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Gray	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF335-7D2-23 **Lab Number** 2007-00084- 23 **Sampled:** 12/21/2006 **Condition:** acceptable
Analyzed By MAC 1/24/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): binder, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Gray	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF335-7D3-24 **Lab Number** 2007-00084- 24 **Sampled:** 12/21/2006 **Condition:** acceptable
Analyzed By MAC 1/24/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): binder, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Gray	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF335-7E1-25 **Lab Number** 2007-00084- 25 **Sampled:** 12/21/2006 **Condition:** acceptable
Analyzed By MAC 1/24/2007 **An?** OK **Apparent Smp Type** Insulation Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): binder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	insulation	100	Tan	4	90-100%	-	-	-	-	-
	Total %	100		**Average %**	90-100%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps.

Sample IHS-SF335-7E2-26 **Lab Number** 2007-00084- 26 **Sampled:** 12/21/2006 **Condition:** acceptable
Analyzed By MAC 1/24/2007 **An?** OK **Apparent Smp Type** Insulation Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): binder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	insulation	100	Tan	4	90-100%	-	-	-	-	-
	Total %	100		**Average %**	90-100%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps.

Sample IHS-SF335-7E3-27 **Lab Number** 2007-00084- 27 **Sampled:** 12/21/2006 **Condition:** acceptable
Analyzed By MAC 1/24/2007 **An?** OK **Apparent Smp Type** Insulation Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): binder, ,

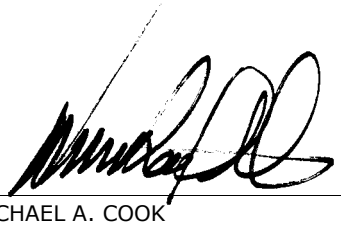
Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	insulation	100	Tan	4	90-100%	-	-	-	-	-
	Total %	100		**Average %**	90-100%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps.

Fr=Friability: 1=very non-friable; 2= non-friable; 3=friable; 4=highly friable
Colors: B=black;BL=blue;BR=brown;CL=clear;G=Green;GY=gray;OR=orange;OW=off-white;PN=pink;PU=purple;R=red;TN=tan;W=white;Y=yellow;V=various
Fiber Morphology: A=fine fibers/bundles, white, sinewy, flexible; B=fine fibers/bundles, w-br, straight, broomed ends; C=fine fibers/bundles, blue, straight, broomed ends; D=fine to coarse fibers, CL-B, brittle; E=coarse fibers,CL or dyed, striated; F=coarse fibers or splinters, W-BR, ribbon-like; G=lath-like or shards, low aspect ratio, may taper
Iso=isotropism - may be yes or no; Pleo=pleochroism - may be yes or no; Bi=birefringence - may be None, Low, Medium or High
Elg=sign of elongation - may be + or -; Ext=extinction - may be Parallel, Oblique, None or Undulating; Oil=medium used to for dispersion staining
Col Par=dispersion staining colors parallel to the fiber (fiber/halo): b/w=black/white; dg/py=dark gray/pale yellow; vg/y=violet gray/yellow; db/ly=dark blue/lemon yellow; vb/g= vivid blue/gold; sb/o=sky blue/orange; pb/r=pale blue/red; gb/dr=gray blue/dark red; w/b=white/black. Col Perp=same only perpendicular to fiber.
RI Par=refractive index parallel to fiber; RI Perp=refractive index perpendicular to fiber



Analyst: MICHAEL A. COOK

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Larry S. Pierce, Approved Accreditation Signatory

FIBERQUANT ANALYTICAL SERVICES

Polarized Light Microscope (PLM) Analysis for Asbestos

JobNumber:	**200700082**

Client: **IHI ENVIRONMENTAL**

(PHOENIX OFFICE)
4527 N 16th ST STE 105
PHOENIX, AZ 85016-0000
Office Phone: (602) 776-0300
FAX: (602) 776-0301

Samples: 33 PLM **Rec:** 1/4/2007 **Method:** Interim (EPA/600/M4-82-020) PLM analysis for asbestos in bulk smp
Client Job: 07P-3031A / IHS-SF336 **PO Number:** 07P-3031A
Report Date: 1/23/2007 **Date Analyzed:** 1/23/2007 **Routing Number:** -

Method and Analysis Information: **Fiberquant Internal SOP:** PLMi

Each bulk sample is first dissected under a 7-30x magnification stereo-microscope. This examination is used to determine the general type of sample, how many and what type of layers it has, and initial estimates of fiber types and quantities. Second, liquid media mounts are made of each layer - such mounts may be of selected fibers (used solely for identification purposes) or may be representative of the layer as a whole (used for quantitation purposes). The mounts may be made in a synthetic Canadian balsam, one of several solvents, or in refractive index oils (media of known refractive index). Generally, a variety of different mounts are made: some optimized for fiber visibility, some optimized for fiber identification, and some optimized for fiber quantitation. The mounted slides are then examined at 50-400x magnification on a Nikon Labphot-pol microscope. Optical characteristics are used to identify each observed fiber type; the optical data are contained for each sample on its detail analysis sheet, attached.

Current EPA, NESHAP and OSHA regulations designate a result of <=1 % asbestos as "negative" and >1 % asbestos as "positive". Samples containing layers that have been determined to be "positive" may have to be handled differently during a renovation or demolition than samples whose layers have been determined to be "negative."

The method of fiber analysis and identification is the EPA Interim Method (EPA Method 600/M4-82-020). The method of fiber quantitation is an estimation technique in which the analysts quantitation is routinely calibrated by reference quantitation standards, and which has been shown to be equivalent in precision and accuracy to point counting. Friability is estimated for the purposes of deciding when to point count. Friabilities determined in the field take precedence over those determined in the laboratory. Those sample layers which are friable and estimated by the analyst to contain <= 1% asbestos are point counted using 400 points, and the results listed on the detail analysis sheet. Such point counting is required by NESHAP (National Emission Standards for Hazardous Air Polutants, Nov. 1990) in order to rely on analytical results that are <= 1%. The coefficient of variation for the estimation quantitation technique is 100% in the range 0-5%. This means that PLM analysis is not capable of conclusively determining whether a layer containing close to 1% asbestos is actually "positive" or "negative". For this reason, Fiberquant refers to results where asbestos was detected but <= 1% as "borderline negative", and results where asbestos was >1 % but <= 2% as "borderline positive" to indicate the uncertainty in assigning a "positive" or "negative" label. In the sample summary, a "ND" means that no asbestos was detected during the analysis. A "Tr" or "Trace" of asbestos is defined for our purposes as the detection of several asbestos asbestos fibers during the analysis; an amount essentially at the limit of detection for the method. The limit of detection (the smallest % of asbestos that can be detected) varies greatly depending on the matrix in which the asbestos is found. As little as 0.001% asbestos can be detected in favorable samples, while detection in unfavorable samples may approach the detection limit of 1% stated in the method. During the analysis, the analyst, for Fiberquant identification purposes only, determines the "apparent sample type" and "apparent layer types." It must be emphasized that these types are only what is apparent. Often, different materials appear similar or identical after sampling, so the analyst may assign a type other than what was sampled.

Floor tiles present a special problem for PLM asbestos analysis. Floor tile can contain chrysotile fibers so thin that they cannot be resolved by optical methods. In such a case, we may observe a percentage of asbestos which is lower than the actual percentage, or not observe asbestos at all when some is present. For this reason, floor tiles reported as negative should be confirmed to be negative using transmission electron microscope (TEM) analysis. Likewise, vermiculite insulation materials containing traces of asbestiform asbestos present a problem for routine PLM analysis - the amphiboles are sometimes present in trace amounts inhomogeneously distributed. We recommend a hydro-separation technique for such samples.

Vermiculite-containing samples may contain trace amounts of asbestiform amphibole that may or may not be detected during routine PLM analysis. For this reason, loose vermiculite samples reported as negative should be confirmed to contain no amphibole using hydroseparation techniques.

The samples were analyzed under the following ongoing quality assurance program: Blank samples are routinely analyzed to maintain contamination-free materials. Each analyst is a degreed geologist or mineralogist, and has also completed extensive training specific to asbestos analysis for 1-3 months before being allowed to analyze client samples. Qualitative reference samples are routinely analyzed to assure that analysts can identify asbestos and asbestos-look-alike fibers. Quantitative reference samples are routinely analyzed to calibrate and characterize the estimation procedure. Microscope alignment is checked each day. Refractive index oils are calibrated at least monthly. At least 10% of client samples are re-analyzed from scratch by a different analyst than the original, and any discrepancies are resolved for the sample and similar sample types before the results are reported. All quality checks performed for these samples were in control except as detailed in the "Analytical Notes" below. All analysts participate in interlab round robins and proficiency testing to assure competence. Fiberquant is accredited by NVLAP (#101031) for the analysis of bulk samples for asbestos using PLM. Accreditation does not imply endorsement by the EPA, any other United States governmental agency or any private agency or association. Each lab analysis refers only to the sample tested, and may not, due to the sampling process, be

representative of the material sampled. This report may not be reproduced except in full, without the approval of Fiberquant Analytical Services.

The "Analysis Results" below are the averages for all layers of each sample listed. The "+ Layer" column indicates whether any of the layers in the sample were >1% asbestos. Please refer to each detailed analysis sheet to determine which layer(s) of a sample was(were) positive.

Some results may have been calculated using client supplied data, such as volume or area sampled, for which Fiberquant assumes no liability for accuracy.

Job Analysis Notes:

Single layer sample analysis as per client request. Any material or layer other than that indicated on the chain of custody was not analyzed, even if a suspect material.

PLM Analysis Summary: **Job Number: 200700082** 07P-3031A / IHS-SF336

Sample Number / Layer	Color	Lab Number / Apparent Layer Type *	Apparent Sample Type * / *Asbestos Results*	Positive Layer Yes or No
Sample **IHS-SF336-1A1-1**		2007-00082- 1	Flooring	Positive Layer? No
Layer # 1	off-white	floor tile	*no asbestos detected*	
Layer # 2	Yellow	mastic	*no asbestos detected*	
Sample **IHS-SF336-1A2-2**		2007-00082- 2	Flooring	Positive Layer? No
Layer # 1	off-white	floor tile	*no asbestos detected*	
Layer # 2	Yellow	mastic	*no asbestos detected*	
Sample **IHS-SF336-1A3-3**		2007-00082- 3	Flooring	Positive Layer? No
Layer # 1	off-white	floor tile	*no asbestos detected*	
Layer # 2	Yellow	mastic	*no asbestos detected*	
Sample **IHS-SF336-1B1-4**		2007-00082- 4	Flooring	Positive Layer? No
Layer # 1	Gray	floor tile	*2-5% chrysotile asbestos*	
Layer # 2	Clear	mastic	*no asbestos detected*	
Sample **IHS-SF336-1B2-5**		2007-00082- 5	Flooring	Positive Layer? No
Layer # 1	Gray	floor tile	*2-5% chrysotile asbestos*	
Layer # 2	Clear	mastic	*no asbestos detected*	
Sample **IHS-SF336-1B3-6**		2007-00082- 6	Flooring	Positive Layer? No
Layer # 1	Gray	floor tile	*2-5% chrysotile asbestos*	
Layer # 2	Clear	mastic	*no asbestos detected*	
Sample **IHS-SF336-1C1-7**		2007-00082- 7	Flooring	Positive Layer? No
Layer # 1	Off-white	sheet flooring	*20-30% chrysotile asbestos*	
Sample **IHS-SF336-1C2-8**		2007-00082- 8	Flooring	Positive Layer? No
Layer # 1	Off-white	sheet flooring	*20-30% chrysotile asbestos*	
Layer # 2	White	mastic	*no asbestos detected*	
Sample **IHS-SF336-1C3-9**		2007-00082- 9	Flooring	Positive Layer? No
Layer # 1	Off-white	sheet flooring	*20-30% chrysotile asbestos*	
Sample **IHS-SF336-2A1-10**		2007-00082- 10	Wall System	Positive Layer? No
Layer # 1	white	drywall core	*no asbestos detected*	
Sample **IHS-SF336-2A2-11**		2007-00082- 11	Wall System	Positive Layer? No
Layer # 1	white	drywall core	*no asbestos detected*	
Sample **IHS-SF336-2A3-12**		2007-00082- 12	Wall System	Positive Layer? No
Layer # 1	white	drywall core	*no asbestos detected*	
Sample **IHS-SF336-2B1-13**		2007-00082- 13	Wall System	Positive Layer? No
Layer # 1	white	texture/joint compound	*no asbestos detected*	
Sample **IHS-SF336-2B2-14**		2007-00082- 14	Wall System	Positive Layer? No
Layer # 1	white	texture/joint compound	*no asbestos detected*	
Sample **IHS-SF336-2B3-15**		2007-00082- 15	Wall System	Positive Layer? No
Layer # 1	white	texture/joint compound	*no asbestos detected*	
Sample **IHS-SF336-3A1-16**		2007-00082- 16	Acoustic Tile	Positive Layer? No
Layer # 1	Gray	acoustical tile	*no asbestos detected*	
Sample **IHS-SF336-3A2-17**		2007-00082- 17	Acoustic Tile	Positive Layer? No
Layer # 1	Gray	acoustical tile	*no asbestos detected*	
Sample **IHS-SF336-3A3-18**		2007-00082- 18	Acoustic Tile	Positive Layer? No
Layer # 1	Gray	acoustical tile	*no asbestos detected*	
Sample **IHS-SF336-3B1-19**		2007-00082- 19	Acoustic Tile	Positive Layer? No
Layer # 1	Off-white	acoustical tile	*no asbestos detected*	
Sample **IHS-SF336-3B2-20**		2007-00082- 20	Acoustic Tile	Positive Layer? No
Layer # 1	Off-white	acoustical tile	*no asbestos detected*	
Sample **IHS-SF336-3B3-21**		2007-00082- 21	Acoustic Tile	Positive Layer? No
Layer # 1	Off-white	acoustical tile	*no asbestos detected*	
Sample **IHS-SF336-7A1-22**		2007-00082- 22	Adhesive/caulk	Positive Layer? No
Layer # 1	Brown	mastic	*no asbestos detected*	
Sample **IHS-SF336-7A2-23**		2007-00082- 23	Adhesive/caulk	Positive Layer? No
Layer # 1	Brown	mastic	*no asbestos detected*	
Sample **IHS-SF336-7A3-24**		2007-00082- 24	Adhesive/caulk	Positive Layer? No
Layer # 1	Brown	mastic	*no asbestos detected*	
Layer # 2	Yellow	mastic	*no asbestos detected*	
Sample **IHS-SF336-7B1-25**		2007-00082- 25	Adhesive/caulk	Positive Layer? No
Layer # 1	white	caulk	*no asbestos detected*	

Sample	**IHS-SF336-7B2-26**	2007-00082- 26	Adhesive/caulk	Positive Layer? No
	Layer # 1 white caulk		*no asbestos detected*	
Sample	**IHS-SF336-7B3-27**	2007-00082- 27	Adhesive/caulk	Positive Layer? No
	Layer # 1 white caulk		*no asbestos detected*	
Sample	**IHS-SF336-7C1-28**	2007-00082- 28	Adhesive/caulk	Positive Layer? No
	Layer # 1 Black caulk		*no asbestos detected*	
Sample	**IHS-SF336-7C2-29**	2007-00082- 29	Adhesive/caulk	Positive Layer? No
	Layer # 1 Black caulk		*no asbestos detected*	
Sample	**IHS-SF336-7C3-30**	2007-00082- 30	Adhesive/caulk	Positive Layer? No
	Layer # 1 Black caulk		*no asbestos detected*	
Sample	**IHS-SF336-7D1-31**	2007-00082- 31	Insulation	Positive Layer? No
	Layer # 1 Tan insulation		*no asbestos detected*	
Sample	**IHS-SF336-7D2-32**	2007-00082- 32	Insulation	Positive Layer? No
	Layer # 1 Tan insulation		*no asbestos detected*	
Sample	**IHS-SF336-7D3-33**	2007-00082- 33	Insulation	Positive Layer? No
	Layer # 1 Tan insulation		*no asbestos detected*	

* Apparent Sample Types and Apparent Layer Types are as they appeared to the analyst. Since many types of materials appear similar after sampling damage, the apparent type of material may not be the actual type of material.

Sample IHS-SF336-1A1-1 **Lab Number** 2007-00082- 1 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 6
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	floor tile	98	off-white	1	n.d.	-	-	-	-	-
2	mastic	2	Yellow	1	<=1%	-	-	-	-	-
	Total %	100		**Average %**	<=1%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF336-1A2-2 **Lab Number** 2007-00082- 2 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 6
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	floor tile	97	off-white	1	n.d.	-	-	-	-	-
2	mastic	3	Yellow	1	2-5%	-	-	-	-	-
	Total %	100		**Average %**	<=1%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF336-1A3-3 **Lab Number** 2007-00082- 3 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 6
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	floor tile	95	off-white	1	n.d.	-	-	-	-	-
2	mastic	5	Yellow	1	>1-2%	-	-	-	-	-
	Total %	100		**Average %**	<=1%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF336-1B1-4 **Lab Number** 2007-00082- 4 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 5
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	floor tile	98	Gray	1	2-5%	n.d.	-	-	-	-
2	mastic	2	Clear	1	n.d.	5-10%	-	-	-	-
	Total %	100		**Average %**	2-5%	<=1%	-	-	-	-
				Fiber Identification:	chrysotile asbestos	cellulose fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2	cellulose fiber	W	F	N	N	H	+	U					
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF336-1B2-5 **Lab Number** 2007-00082- 5 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 5
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	floor tile	97	Gray	1	2-5%	n.d.	-	-	-	-
2	mastic	3	Clear	1	n.d.	5-10%	-	-	-	-
	Total %	100		**Average %**	2-5%	<=1%	-	-	-	-
				Fiber Identification:	chrysotile asbestos	cellulose fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2	cellulose fiber	W	F	N	N	H	+	U					
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF336-1B3-6 **Lab Number** 2007-00082- 6 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 5
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	floor tile	97	Gray	1	2-5%	n.d.	-	-	-	-
2	mastic	3	Clear	1	n.d.	5-10%	-	-	-	-
	Total %	100		**Average %**	2-5%	<=1%	-	-	-	-
				Fiber Identification:	chrysotile asbestos	cellulose fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2	cellulose fiber	W	F	N	N	H	+	U					
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF336-1C1-7 **Lab Number** 2007-00082- 7 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Flooring Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	sheet flooring	100	Off-white	2	20-30%	2-5%	-	-	-	-
	Total %	100		**Average %**	20-30%	2-5%	-	-	-	-
				Fiber Identification:	chrysotile asbestos	cellulose fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2	cellulose fiber	W	F	N	N	H	+	U					
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF336-1C2-8 **Lab Number** 2007-00082- 8 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Flooring Fibrous Mat
Homogeneous No **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 5
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	sheet flooring	90	Off-white	2	20-30%	2-5%	-	-	-	-
2	mastic	10	White	1	n.d.	n.d.	-	-	-	-
	Total %	100		**Average %**	20-30%	2-5%	-	-	-	-
				Fiber Identification:	chrysotile asbestos	cellulose fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2	cellulose fiber	W	F	N	N	H	+	U					
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF336-1C3-9 **Lab Number** 2007-00082- 9 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Flooring Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	sheet flooring	100	Off-white	2	20-30%	2-5%	-	-	-	-
	Total %	100		**Average %**	20-30%	2-5%	-	-	-	-
				Fiber Identification:	chrysotile asbestos	cellulose fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2	cellulose fiber	W	F	N	N	H	+	U					
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF336-2A1-10 **Lab Number** 2007-00082- 10 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	drywall core	100	white	3	>1-2%	-	-	-	-	-
	Total %	100		**Average %**	>1-2%	-	-	-	-	-
				Fiber Identification:	cellulose					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps.

Sample IHS-SF336-2A2-11 **Lab Number** 2007-00082- 11 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	drywall core	100	white	3	>1-2%	-	-	-	-	-
	Total %	100		**Average %**	>1-2%	-	-	-	-	-
				Fiber Identification:	cellulose					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps.

Sample IHS-SF336-2A3-12 **Lab Number** 2007-00082- 12 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	drywall core	100	white	3	>1-2%	-	-	-	-	-
	Total %	100		**Average %**	>1-2%	-	-	-	-	-
				Fiber Identification:	cellulose					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps.

Sample IHS-SF336-2B1-13 **Lab Number** 2007-00082- 13 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	texture/joint compound	100	white	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF336-2B2-14 **Lab Number** 2007-00082- 14 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	texture/joint compound	100	white	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF336-2B3-15 **Lab Number** 2007-00082- 15 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	texture/joint compound	100	white	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF336-3A1-16 **Lab Number** 2007-00082- 16 **Sampled:** 12/20/2006 **Condition:** acceptable

Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Acoustic Tile Fibrous Mat

Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3

Non-Fibrous Components (in approx. decreasing order): binder, glass,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	acoustical tile	100	Gray	3	60-70%	-	-	-	-	-
	Total %	100		**Average %**	60-70%	-	-	-	-	-
				Fiber Identification:	glass fiber					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	glass fiber	CL	D	Y									
2													
3													
4													
5													
6													

Sample Analytical Note

Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF336-3A2-17 **Lab Number** 2007-00082- 17 **Sampled:** 12/20/2006 **Condition:** acceptable

Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Acoustic Tile Fibrous Mat

Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3

Non-Fibrous Components (in approx. decreasing order): binder, glass,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	acoustical tile	100	Gray	3	60-70%	-	-	-	-	-
	Total %	100		**Average %**	60-70%	-	-	-	-	-
				Fiber Identification:	glass fiber					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	glass fiber	CL	D	Y									
2													
3													
4													
5													
6													

Sample Analytical Note

Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF336-3A3-18 **Lab Number** 2007-00082- 18 **Sampled:** 12/20/2006 **Condition:** acceptable

Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Acoustic Tile Fibrous Mat

Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3

Non-Fibrous Components (in approx. decreasing order): binder, glass,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	acoustical tile	100	Gray	3	60-70%	-	-	-	-	-
	Total %	100		**Average %**	60-70%	-	-	-	-	-
				Fiber Identification:	glass fiber					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	glass fiber	CL	D	Y									
2													
3													
4													
5													
6													

Sample Analytical Note

Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF336-3B1-19 **Lab Number** 2007-00082- 19 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Acoustic Tile Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): perlite, glass, binder

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	acoustical tile	100	Off-white	3	40-50%	20-30%	-	-	-	-
	Total %	100		**Average %**	40-50%	20-30%	-	-	-	-
				Fiber Identification:	glass fiber	cellulose fiber				

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	glass fiber	CL	D	Y									
2	cellulose fiber	W	F	N	N	H	+	U					
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF336-3B2-20 **Lab Number** 2007-00082- 20 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Acoustic Tile Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): perlite, glass, binder

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	acoustical tile	100	Off-white	3	40-50%	20-30%	-	-	-	-
	Total %	100		**Average %**	40-50%	20-30%	-	-	-	-
				Fiber Identification:	glass fiber	cellulose fiber				

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	glass fiber	CL	D	Y									
2	cellulose fiber	W	F	N	N	H	+	U					
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF336-3B3-21 **Lab Number** 2007-00082- 21 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Acoustic Tile Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): perlite, glass, binder

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	acoustical tile	100	Off-white	3	40-50%	20-30%	-	-	-	-
	Total %	100		**Average %**	40-50%	20-30%	-	-	-	-
				Fiber Identification:	glass fiber	cellulose fiber				

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	glass fiber	CL	D	Y									
2	cellulose fiber	W	F	N	N	H	+	U					
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF336-7A1-22 **Lab Number** 2007-00082- 22 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	100	Brown	1	<=1%	-	-	-	-	-
	Total %	100		**Average %**	<=1%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF336-7A2-23 **Lab Number** 2007-00082- 23 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	100	Brown	1	<=1%	-	-	-	-	-
	Total %	100		**Average %**	<=1%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF336-7A3-24 **Lab Number** 2007-00082- 24 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 6
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	40	Brown	1	<=1%	n.d.	-	-	-	-
2	mastic	60	Yellow	1	<=1%	<=1%	-	-	-	-
	Total %	100		**Average %**	<=1%	<=1%	-	-	-	-
				Fiber Identification:	cellulose fiber	glass fiber				

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2	glass fiber	CL	D	Y									
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF336-7B1-25 **Lab Number** 2007-00082- 25 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	white	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF336-7B2-26 **Lab Number** 2007-00082- 26 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	white	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF336-7B3-27 **Lab Number** 2007-00082- 27 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	white	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF336-7C1-28 **Lab Number** 2007-00082- 28 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Black	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF336-7C2-29 **Lab Number** 2007-00082- 29 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Black	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF336-7C3-30 **Lab Number** 2007-00082- 30 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Black	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF336-7D1-31 **Lab Number** 2007-00082- 31 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Insulation Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): binder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	insulation	100	Tan	3	90-100%	-	-	-	-	-
	Total %	100		**Average %**	90-100%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps.

Sample IHS-SF336-7D2-32 **Lab Number** 2007-00082- 32 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Insulation Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): binder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	insulation	100	Tan	3	90-100%	-	-	-	-	-
	Total %	100		**Average %**	90-100%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps.

Sample IHS-SF336-7D3-33 **Lab Number** 2007-00082- 33 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By RAM 1/23/2007 **An?** OK **Apparent Smp Type** Insulation Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): binder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	insulation	100	Tan	3	90-100%	-	-	-	-	-
	Total %	100		**Average %**	90-100%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps.

Fr=Friability: 1=very non-friable; 2= non-friable; 3=friable; 4=highly friable
Colors: B=black;BL=blue;BR=brown;CL=clear;G=Green;GY=gray;OR=orange;OW=off-white;PN=pink;PU=purple;R=red;TN=tan;W=white;Y=yellow;V=various
Fiber Morphology: A=fine fibers/bundles, white, sinewy, flexible; B=fine fibers/bundles, w-br, straight, broomed ends; C=fine fibers/bundles, blue, straight, broomed ends; D=fine to coarse fibers, CL-B, brittle; E=coarse fibers,CL or dyed, striated; F=coarse fibers or splinters, W-BR, ribbon-like; G=lath-like or shards, low aspect ratio, may taper
Iso=isotropism - may be yes or no; Pleo=pleochroism - may be yes or no; Bi=birefringence - may be None, Low, Medium or High
Elg=sign of elongation - may be + or -; Ext=extinction - may be Parallel, Oblique, None or Undulating; Oil=medium used to for dispersion staining
Col Par=dispersion staining colors parallel to the fiber (fiber/halo): b/w=black/white; dg/py=dark gray/pale yellow; vg/y=violet gray/yellow; db/ly=dark blue/lemon yellow; vb/g= vivid blue/gold; sb/o=sky blue/orange; pb/r=pale blue/red; gb/dr=gray blue/dark red; w/b=white/black. Col Perp=same only perpendicular to fiber.
RI Par=refractive index parallel to fiber; RI Perp=refractive index perpendicular to fiber

Analyst: ROBERT A. McCORMICK

Printed: 23-Jan-07

Original Print Date: 23-Jan-07

Larry S. Pierce, Approved Accreditation Signatory

FIBERQUANT ANALYTICAL SERVICES

Polarized Light Microscope (PLM) Analysis for Asbestos

JobNumber:	**200700202**

Client: **IHI ENVIRONMENTAL**

(PHOENIX OFFICE)
4527 N 16th ST STE 105
PHOENIX, AZ 85016-0000
Office Phone: (602) 776-0300
FAX: (602) 776-0301

Samples: 186 PLM **Rec:** 1/9/2007 **Method:** Interim (EPA/600/M4-82-020) PLM analysis for asbestos in bulk smp
Client Job: 07P-3031A / IHS-SF2000 **PO Number:** 07P-3031A
Report Date: 1/29/2007 **Date Analyzed:** 1/27/2007 **Routing Number:** -

Method and Analysis Information: **Fiberquant Internal SOP:** PLMi

Each bulk sample is first dissected under a 7-30x magnification stereo-microscope. This examination is used to determine the general type of sample, how many and what type of layers it has, and initial estimates of fiber types and quantities. Second, liquid media mounts are made of each layer - such mounts may be of selected fibers (used solely for identification purposes) or may be representative of the layer as a whole (used for quantitation purposes). The mounts may be made in a synthetic Canadian balsam, one of several solvents, or in refractive index oils (media of known refractive index). Generally, a variety of different mounts are made: some optimized for fiber visibility, some optimized for fiber identification, and some optimized for fiber quantitation. The mounted slides are then examined at 50-400x magnification on a Nikon Labphot-pol microscope. Optical characteristics are used to identify each observed fiber type; the optical data are contained for each sample on its detail analysis sheet, attached.

Current EPA, NESHAP and OSHA regulations designate a result of <=1 % asbestos as "negative" and >1 % asbestos as "positive". Samples containing layers that have been determined to be "positive" may have to be handled differently during a renovation or demolition than samples whose layers have been determined to be "negative."

The method of fiber analysis and identification is the EPA Interim Method (EPA Method 600/M4-82-020). The method of fiber quantitation is an estimation technique in which the analysts quantitation is routinely calibrated by reference quantitation standards, and which has been shown to be equivalent in precision and accuracy to point counting. Friability is estimated for the purposes of deciding when to point count. Friabilities determined in the field take precedence over those determined in the laboratory. Those sample layers which are friable and estimated by the analyst to contain <= 1% asbestos are point counted using 400 points, and the results listed on the detail analysis sheet. Such point counting is required by NESHAP (National Emission Standards for Hazardous Air Polutants, Nov. 1990) in order to rely on analytical results that are <= 1%. The coefficient of variation for the estimation quantitation technique is 100% in the range 0-5%. This means that PLM analysis is not capable of conclusively determining whether a layer containing close to 1% asbestos is actually "positive" or "negative". For this reason, Fiberquant refers to results where asbestos was detected but <= 1% as "borderline negative", and results where asbestos was >1 % but <= 2% as "borderline positive" to indicate the uncertainty in assigning a "positive" or "negative" label. In the sample summary, a "ND" means that no asbestos was detected during the analysis. A "Tr" or "Trace" of asbestos is defined for our purposes as the detection of several asbestos asbestos fibers during the analysis; an amount essentially at the limit of detection for the method. The limit of detection (the smallest % of asbestos that can be detected) varies greatly depending on the matrix in which the asbestos is found. As little as 0.001% asbestos can be detected in favorable samples, while detection in unfavorable samples may approach the detection limit of 1% stated in the method. During the analysis, the analyst, for Fiberquant identification purposes only, determines the "apparent sample type" and "apparent layer types." It must be emphasized that these types are only what is apparent. Often, different materials appear similar or identical after sampling, so the analyst may assign a type other than what was sampled.

Floor tiles present a special problem for PLM asbestos analysis. Floor tile can contain chrysotile fibers so thin that they cannot be resolved by optical methods. In such a case, we may observe a percentage of asbestos which is lower than the actual percentage, or not observe asbestos at all when some is present. For this reason, floor tiles reported as negative should be confirmed to be negative using transmission electron microscope (TEM) analysis. Likewise, vermiculite insulation materials containing traces of asbestiform asbestos present a problem for routine PLM analysis - the amphiboles are sometimes present in trace amounts inhomogeneously distributed. We recommend a hydro-separation technique for such samples.

Vermiculite-containing samples may contain trace amounts of asbestiform amphibole that may or may not be detected during routine PLM analysis. For this reason, loose vermiculite samples reported as negative should be confirmed to contain no amphibole using hydroseparation techniques.

The samples were analyzed under the following ongoing quality assurance program: Blank samples are routinely analyzed to maintain contamination-free materials. Each analyst is a degreed geologist or mineralogist, and has also completed extensive training specific to asbestos analysis for 1-3 months before being allowed to analyze client samples. Qualitative reference samples are routinely analyzed to assure that analysts can identify asbestos and asbestos-look-alike fibers. Quantitative reference samples are routinely analyzed to calibrate and characterize the estimation procedure. Microscope alignment is checked each day. Refractive index oils are calibrated at least monthly. At least 10% of client samples are re-analyzed from scratch by a different analyst than the original, and any discrepancies are resolved for the sample and similar sample types before the results are reported. All quality checks performed for these samples were in control except as detailed in the "Analytical Notes" below. All analysts participate in interlab round robins and proficiency testing to assure competence. Fiberquant is accredited by NVLAP (#101031) for the analysis of bulk samples for asbestos using PLM. Accreditation does not imply endorsement by the EPA, any other United States governmental agency or any private agency or association. Each lab analysis refers only to the sample tested, and may not, due to the sampling process, be

representative of the material sampled. This report may not be reproduced except in full, without the approval of Fiberquant Analytical Services.

The "Analysis Results" below are the averages for all layers of each sample listed. The "+ Layer" column indicates whether any of the layers in the sample were >1% asbestos. Please refer to each detailed analysis sheet to determine which layer(s) of a sample was(were) positive.

Some results may have been calculated using client supplied data, such as volume or area sampled, for which Fiberquant assumes no liability for accuracy.

Job Analysis Notes:

Single layer sample analysis as per client request. Any material or layer other than that indicated on the chain of custody was not analyzed, even if a suspect material.

PLM Analysis Summary: **Job Number: 200700202** 07P-3031A / IHS-SF2000

Sample Number / Layer	Color	Apparent Layer Type *	Lab Number	Apparent Sample Type * / *Asbestos Results*	Positive Layer Yes or No
Sample IHS-SF2000-1A1-1			2007-00202- 1	Flooring	Positive Layer? Yes
Layer # 1	off-white	floor tile		*no asbestos detected*	
Layer # 2	Yellow	mastic		*no asbestos detected*	
Layer # 3	Black	mastic		*2-5% chrysotile asbestos*	
Sample IHS-SF2000-1A2-2			2007-00202- 2	Flooring	Positive Layer? No
Layer # 1	off-white	floor tile		*no asbestos detected*	
Layer # 2	Clear	mastic		*no asbestos detected*	
Sample IHS-SF2000-1A3-3			2007-00202- 3	Flooring	Positive Layer? No
Layer # 1	off-white	floor tile		*no asbestos detected*	
Layer # 2	Yellow	mastic		*no asbestos detected*	
Sample IHS-SF2000-1B1-4			2007-00202- 4	Adhesive/caulk	Positive Layer? No
Layer # 1	Yellow	mastic		*no asbestos detected*	
Sample IHS-SF2000-1B2-5			2007-00202- 5	Adhesive/caulk	Positive Layer? No
Layer # 1	Green	mastic		*no asbestos detected*	
Sample IHS-SF2000-1B3-6			2007-00202- 6	Adhesive/caulk	Positive Layer? No
Layer # 1	Tan	mastic		*no asbestos detected*	
Sample IHS-SF2000-1C1-7			2007-00202- 7	Surfacing	Positive Layer? No
Layer # 1	Gray	grout		*no asbestos detected*	
Sample IHS-SF2000-1C2-8			2007-00202- 8	Surfacing	Positive Layer? No
Layer # 1	Gray	grout		*no asbestos detected*	
Sample IHS-SF2000-1C3-9			2007-00202- 9	Surfacing	Positive Layer? No
Layer # 1	Gray	grout		*no asbestos detected*	
Sample IHS-SF2000-1D1-10			2007-00202- 10	Flooring	Positive Layer? Yes
Layer # 1	Yellow	mastic		*no asbestos detected*	
Layer # 2	Off-Whit	floor tile		*5-10% chrysotile asbestos*	
Layer # 3	Black	mastic		*2-5% chrysotile asbestos*	
Sample IHS-SF2000-1D2-11			2007-00202- 11	Flooring	Positive Layer? Yes
Layer # 1	Yellow	mastic		*no asbestos detected*	
Layer # 2	Off-Whit	floor tile		*5-10% chrysotile asbestos*	
Layer # 3	Black	mastic		*10-20% chrysotile asbestos*	
Sample IHS-SF2000-1D3-12			2007-00202- 12	Flooring	Positive Layer? Yes
Layer # 1	Off-whit	mastic		*no asbestos detected*	
Layer # 2	Off-Whit	floor tile		*5-10% chrysotile asbestos*	
Layer # 3	Black	mastic		*5-10% chrysotile asbestos*	
Sample IHS-SF2000-1E1-13			2007-00202- 13	Flooring	Positive Layer? No
Layer # 1	off-white	floor tile		*no asbestos detected*	
Layer # 2	Off-whit	mastic		*no asbestos detected*	
Sample IHS-SF2000-1E2-14			2007-00202- 14	Flooring	Positive Layer? No
Layer # 1	off-white	floor tile		*no asbestos detected*	
Layer # 2	Off-whit	mastic		*no asbestos detected*	
Sample IHS-SF2000-1E3-15			2007-00202- 15	Flooring	Positive Layer? No
Layer # 1	off-white	floor tile		*no asbestos detected*	
Layer # 2	Off-whit	mastic		*no asbestos detected*	
Sample IHS-SF2000-1F1-16			2007-00202- 16	Surfacing	Positive Layer? No
Layer # 1	Gray	grout		*no asbestos detected*	
Sample IHS-SF2000-1F2-17			2007-00202- 17	Surfacing	Positive Layer? No
Layer # 1	Gray	grout		*no asbestos detected*	
Sample IHS-SF2000-1F3-18			2007-00202- 18	Surfacing	Positive Layer? No
Layer # 1	Gray	grout		*no asbestos detected*	
Sample IHS-SF2000-1G1-19			2007-00202- 19	Flooring	Positive Layer? No
Layer # 1	off-white	floor tile		*no asbestos detected*	
Layer # 2	Tan	mastic		*no asbestos detected*	
Sample IHS-SF2000-1G2-20			2007-00202- 20	Flooring	Positive Layer? No
Layer # 1	off-white	floor tile		*no asbestos detected*	
Layer # 2	Yellow	mastic		*no asbestos detected*	
Sample IHS-SF2000-1G3-21			2007-00202- 21	Flooring	Positive Layer? No
Layer # 1	off-white	floor tile		*no asbestos detected*	
Layer # 2	Yellow	mastic		*no asbestos detected*	

Sample	**IHS-SF2000-1H1-22**	2007-00202- 22	Flooring	Positive Layer?	No
	Layer # 1 Orange floor tile		*no asbestos detected*		
	Layer # 2 Clear mastic		*no asbestos detected*		
Sample	**IHS-SF2000-1H2-23**	2007-00202- 23	Flooring	Positive Layer?	No
	Layer # 1 Orange floor tile		*no asbestos detected*		
	Layer # 2 Clear mastic		*no asbestos detected*		
Sample	**IHS-SF2000-1H3-24**	2007-00202- 24	Flooring	Positive Layer?	No
	Layer # 1 Orange floor tile		*no asbestos detected*		
	Layer # 2 Clear mastic		*no asbestos detected*		
Sample	**IHS-SF2000-1I1-25**	2007-00202- 25	Adhesive/caulk	Positive Layer?	No
	Layer # 1 Yellow mastic		*no asbestos detected*		
Sample	**IHS-SF2000-1I2-26**	2007-00202- 26	Adhesive/caulk	Positive Layer?	No
	Layer # 1 Yellow mastic		*no asbestos detected*		
Sample	**IHS-SF2000-1I3-27**	2007-00202- 27	Adhesive/caulk	Positive Layer?	No
	Layer # 1 Yellow mastic		*no asbestos detected*		
Sample	**IHS-SF2000-1J1-28**	2007-00202- 28	Adhesive/caulk	Positive Layer?	Yes
	Layer # 1 Black mastic		*5-10% chrysotile asbestos*		
Sample	**IHS-SF2000-1J2-29**	2007-00202- 29	Adhesive/caulk	Positive Layer?	Yes
	Layer # 1 Black mastic		*5-10% chrysotile asbestos*		
Sample	**IHS-SF2000-1J3-30**	2007-00202- 30	Adhesive/caulk	Positive Layer?	Yes
	Layer # 1 Black mastic		*5-10% chrysotile asbestos*		
Sample	**IHS-SF2000-1K1-31**	2007-00202- 31	Flooring	Positive Layer?	No
	Layer # 1 Gray floor tile		*no asbestos detected*		
	Layer # 2 Yellow mastic		*no asbestos detected*		
Sample	**IHS-SF2000-1K2-32**	2007-00202- 32	Flooring	Positive Layer?	No
	Layer # 1 Gray floor tile		*no asbestos detected*		
	Layer # 2 Yellow mastic		*no asbestos detected*		
Sample	**IHS-SF2000-1K3-33**	2007-00202- 33	Flooring	Positive Layer?	No
	Layer # 1 Gray floor tile		*no asbestos detected*		
	Layer # 2 Yellow mastic		*no asbestos detected*		
Sample	**IHS-SF2000-1L1-34**	2007-00202- 34	Flooring	Positive Layer?	Yes
	Layer # 1 Gray floor tile		*no asbestos detected*		
	Layer # 2 Yellow mastic		*no asbestos detected*		
	Layer # 3 Black mastic		*2-5% chrysotile asbestos*		
Sample	**IHS-SF2000-1L2-35**	2007-00202- 35	Flooring	Positive Layer?	No
	Layer # 1 Gray floor tile		*no asbestos detected*		
	Layer # 2 Yellow mastic		*no asbestos detected*		
Sample	**IHS-SF2000-1L3-36**	2007-00202- 36	Flooring	Positive Layer?	No
	Layer # 1 Gray floor tile		*no asbestos detected*		
	Layer # 2 Yellow mastic		*no asbestos detected*		
Sample	**IHS-SF2000-1M1-37**	2007-00202- 37	Flooring	Positive Layer?	No
	Layer # 1 Black floor tile		*no asbestos detected*		
	Layer # 2 Yellow mastic		*no asbestos detected*		
Sample	**IHS-SF2000-1M2-38**	2007-00202- 38	Flooring	Positive Layer?	No
	Layer # 1 Black floor tile		*no asbestos detected*		
Sample	**IHS-SF2000-1M3-39**	2007-00202- 39	Flooring	Positive Layer?	No
	Layer # 1 Black floor tile		*no asbestos detected*		
Sample	**IHS-SF2000-2A1-40**	2007-00202- 40	Wall System	Positive Layer?	No
	Layer # 1 white drywall core		*no asbestos detected*		
Sample	**IHS-SF2000-2A2-41**	2007-00202- 41	Wall System	Positive Layer?	No
	Layer # 1 white drywall core		*no asbestos detected*		
Sample	**IHS-SF2000-2A3-42**	2007-00202- 42	Wall System	Positive Layer?	No
	Layer # 1 white drywall core		*no asbestos detected*		
Sample	**IHS-SF2000-2B1-43**	2007-00202- 43	Wall System	Positive Layer?	No
	Layer # 1 white texture/joint compound		*no asbestos detected*		
Sample	**IHS-SF2000-2B2-44**	2007-00202- 44	Wall System	Positive Layer?	No
	Layer # 1 white texture/joint compound		*no asbestos detected*		
Sample	**IHS-SF2000-2B3-45**	2007-00202- 45	Wall System	Positive Layer?	No
	Layer # 1 white texture/joint compound		*no asbestos detected*		
Sample	**IHS-SF2000-2C1-46**	2007-00202- 46	Wall System	Positive Layer?	No
	Layer # 1 white texture/joint compound		*no asbestos detected*		
Sample	**IHS-SF2000-2C2-47**	2007-00202- 47	Wall System	Positive Layer?	No
	Layer # 1 white texture/joint compound		*no asbestos detected*		
Sample	**IHS-SF2000-2C3-48**	2007-00202- 48	Wall System	Positive Layer?	No
	Layer # 1 white texture/joint compound		*no asbestos detected*		
Sample	**IHS-SF2000-2C4-49**	2007-00202- 49	Wall System	Positive Layer?	No
	Layer # 1 white texture/joint compound		*no asbestos detected*		
Sample	**IHS-SF2000-2C5-50**	2007-00202- 50	Wall System	Positive Layer?	No
	Layer # 1 white texture/joint compound		*no asbestos detected*		
Sample	**IHS-SF2000-2C6-51**	2007-00202- 51	Wall System	Positive Layer?	No
	Layer # 1 white texture/joint compound		*no asbestos detected*		
Sample	**IHS-SF2000-2C7-52**	2007-00202- 52	Wall System	Positive Layer?	No
	Layer # 1 white texture/joint compound		*no asbestos detected*		
Sample	**IHS-SF2000-2D1-53**	2007-00202- 53	Wall System	Positive Layer?	No
	Layer # 1 White wall covering		*no asbestos detected*		

Sample	**IHS-SF2000-2D2-54**	2007-00202- 54	Wall System		Positive Layer? No
	Layer # 1 White wall covering		*no asbestos detected*		
Sample	**IHS-SF2000-2D3-55**	2007-00202- 55	Wall System		Positive Layer? No
	Layer # 1 White wall covering		*no asbestos detected*		
Sample	**IHS-SF2000-2E1-56**	2007-00202- 56	Wall System		Positive Layer? No
	Layer # 1 white texture/joint compound		*no asbestos detected*		
Sample	**IHS-SF2000-2E2-57**	2007-00202- 57	Wall System		Positive Layer? No
	Layer # 1 white texture/joint compound		*no asbestos detected*		
Sample	**IHS-SF2000-2E3-58**	2007-00202- 58	Wall System		Positive Layer? No
	Layer # 1 white texture/joint compound		*no asbestos detected*		
Sample	**IHS-SF2000-2E4-59**	2007-00202- 59	Wall System		Positive Layer? No
	Layer # 1 white texture/joint compound		*no asbestos detected*		
Sample	**IHS-SF2000-2E5-60**	2007-00202- 60	Wall System		Positive Layer? No
	Layer # 1 white texture/joint compound		*no asbestos detected*		
Sample	**IHS-SF2000-2F1-61**	2007-00202- 61	Wall System		Positive Layer? No
	Layer # 1 white texture/joint compound		*no asbestos detected*		
Sample	**IHS-SF2000-2F2-62**	2007-00202- 62	Wall System		Positive Layer? No
	Layer # 1 white texture/joint compound		*no asbestos detected*		
Sample	**IHS-SF2000-2F3-63**	2007-00202- 63	Wall System		Positive Layer? No
	Layer # 1 white texture/joint compound		*no asbestos detected*		
Sample	**IHS-SF2000-2F4-64**	2007-00202- 64	Wall System		Positive Layer? No
	Layer # 1 white texture/joint compound		*no asbestos detected*		
Sample	**IHS-SF2000-2F5-65**	2007-00202- 65	Wall System		Positive Layer? No
	Layer # 1 white texture/joint compound		*no asbestos detected*		
Sample	**IHS-SF2000-2G1-66**	2007-00202- 66	Wall System		Positive Layer? No
	Layer # 1 white texture/joint compound		*no asbestos detected*		
Sample	**IHS-SF2000-2G2-67**	2007-00202- 67	Wall System		Positive Layer? No
	Layer # 1 white texture/joint compound		*no asbestos detected*		
Sample	**IHS-SF2000-2G3-68**	2007-00202- 68	Wall System		Positive Layer? No
	Layer # 1 white texture/joint compound		*no asbestos detected*		
Sample	**IHS-SF2000-2H1-69**	2007-00202- 69	Wall System		Positive Layer? No
	Layer # 1 white texture/joint compound		*no asbestos detected*		
Sample	**IHS-SF2000-2H2-70**	2007-00202- 70	Wall System		Positive Layer? No
	Layer # 1 white texture/joint compound		*no asbestos detected*		
Sample	**IHS-SF2000-2H3-71**	2007-00202- 71	Wall System		Positive Layer? No
	Layer # 1 white texture/joint compound		*no asbestos detected*		
Sample	**IHS-SF2000-2I1-72**	2007-00202- 72	Surfacing		Positive Layer? No
	Layer # 1 White grout		*no asbestos detected*		
Sample	**IHS-SF2000-2I2-73**	2007-00202- 73	Surfacing		Positive Layer? No
	Layer # 1 White grout		*no asbestos detected*		
Sample	**IHS-SF2000-2I3-74**	2007-00202- 74	Surfacing		Positive Layer? No
	Layer # 1 White grout		*no asbestos detected*		
Sample	**IHS-SF2000-2J1-75**	2007-00202- 75	Surfacing		Positive Layer? No
	Layer # 1 White grout		*no asbestos detected*		
Sample	**IHS-SF2000-2J2-76**	2007-00202- 76	Surfacing		Positive Layer? No
	Layer # 1 White grout		*no asbestos detected*		
Sample	**IHS-SF2000-2J3-77**	2007-00202- 77	Surfacing		Positive Layer? No
	Layer # 1 White grout		*no asbestos detected*		
Sample	**IHS-SF2000-2K1-78**	2007-00202- 78	Acoustic Tile		Positive Layer? No
	Layer # 1 Off-whit acoustical tile		*no asbestos detected*		
Sample	**IHS-SF2000-2K2-79**	2007-00202- 79	Acoustic Tile		Positive Layer? No
	Layer # 1 Off-whit acoustical tile		*no asbestos detected*		
Sample	**IHS-SF2000-2K3-80**	2007-00202- 80	Acoustic Tile		Positive Layer? No
	Layer # 1 Off-whit acoustical tile		*no asbestos detected*		
Sample	**IHS-SF2000-2L1-81**	2007-00202- 81	Wall System		Positive Layer? No
	Layer # 1 Gray block		*no asbestos detected*		
Sample	**IHS-SF2000-2L2-82**	2007-00202- 82	Wall System		Positive Layer? No
	Layer # 1 Gray block		*no asbestos detected*		
Sample	**IHS-SF2000-2L3-83**	2007-00202- 83	Wall System		Positive Layer? No
	Layer # 1 Gray block		*no asbestos detected*		
Sample	**IHS-SF2000-2M1-84**	2007-00202- 84	Wall System		Positive Layer? No
	Layer # 1 Gray mortar		*no asbestos detected*		
Sample	**IHS-SF2000-2M2-85**	2007-00202- 85	Wall System		Positive Layer? No
	Layer # 1 Gray mortar		*no asbestos detected*		
Sample	**IHS-SF2000-2M3-86**	2007-00202- 86	Wall System		Positive Layer? No
	Layer # 1 Gray mortar		*no asbestos detected*		
Sample	**IHS-SF2000-2N1-87**	2007-00202- 87	Miscellaneous		Positive Layer? Yes
	Layer # 1 Gray cem/asb board		*10-20% chrysotile asbestos*		
Sample	**IHS-SF2000-2N2-88**	2007-00202- 88	Miscellaneous		Positive Layer? Yes
	Layer # 1 Gray cem/asb board		*10-20% chrysotile asbestos*		
Sample	**IHS-SF2000-2N3-89**	2007-00202- 89	Miscellaneous		Positive Layer? Yes
	Layer # 1 Gray cem/asb board		*10-20% chrysotile asbestos*		
Sample	**IHS-SF2000-2O1-90**	2007-00202- 90	Wall System		Positive Layer? No
	Layer # 1 Brown paint, textured		*no asbestos detected*		

Sample	**IHS-SF2000-2O2-91**	2007-00202- 91	Wall System		Positive Layer? No
	Layer # 1 Gray stucco		*no asbestos detected*		
Sample	**IHS-SF2000-2O3-92**	2007-00202- 92	Wall System		Positive Layer? No
	Layer # 1 Brown paint, textured		*no asbestos detected*		
Sample	**IHS-SF2000-2O4-93**	2007-00202- 93	Wall System		Positive Layer? No
	Layer # 1 Brown paint, textured		*no asbestos detected*		
Sample	**IHS-SF2000-2O5-94**	2007-00202- 94	Wall System		Positive Layer? No
	Layer # 1 Gray stucco		*no asbestos detected*		
Sample	**IHS-SF2000-2O6-95**	2007-00202- 95	Wall System		Positive Layer? No
	Layer # 1 Gray stucco		*no asbestos detected*		
Sample	**IHS-SF2000-2O7-96**	2007-00202- 96	Wall System		Positive Layer? No
	Layer # 1 Brown paint, textured		*no asbestos detected*		
Sample	**IHS-SF2000-3A1-97**	2007-00202- 97	Acoustic Tile		Positive Layer? No
	Layer # 1 Off-whit acoustical tile		*no asbestos detected*		
Sample	**IHS-SF2000-3A2-98**	2007-00202- 98	Acoustic Tile		Positive Layer? No
	Layer # 1 Off-whit acoustical tile		*no asbestos detected*		
Sample	**IHS-SF2000-3A3-99**	2007-00202- 99	Acoustic Tile		Positive Layer? No
	Layer # 1 Off-whit acoustical tile		*no asbestos detected*		
Sample	**IHS-SF2000-3B1-100**	2007-00202- 100	Acoustic Tile		Positive Layer? No
	Layer # 1 Off-whit acoustical tile		*no asbestos detected*		
Sample	**IHS-SF2000-3B2-101**	2007-00202- 101	Acoustic Tile		Positive Layer? No
	Layer # 1 Off-whit acoustical tile		*no asbestos detected*		
Sample	**IHS-SF2000-3B3-102**	2007-00202- 102	Acoustic Tile		Positive Layer? No
	Layer # 1 Off-whit acoustical tile		*no asbestos detected*		
Sample	**IHS-SF2000-3C1-103**	2007-00202- 103	Acoustic Tile		Positive Layer? No
	Layer # 1 Off-whit acoustical tile		*no asbestos detected*		
Sample	**IHS-SF2000-3C2-104**	2007-00202- 104	Acoustic Tile		Positive Layer? No
	Layer # 1 Off-whit acoustical tile		*no asbestos detected*		
Sample	**IHS-SF2000-3C3-105**	2007-00202- 105	Acoustic Tile		Positive Layer? No
	Layer # 1 Off-whit acoustical tile		*no asbestos detected*		
Sample	**IHS-SF2000-3D1-106**	2007-00202- 106	Acoustic Tile		Positive Layer? No
	Layer # 1 Off-whit acoustical tile		*no asbestos detected*		
Sample	**IHS-SF2000-3D2-107**	2007-00202- 107	Acoustic Tile		Positive Layer? No
	Layer # 1 Off-whit acoustical tile		*no asbestos detected*		
Sample	**IHS-SF2000-3D3-108**	2007-00202- 108	Acoustic Tile		Positive Layer? No
	Layer # 1 Off-whit acoustical tile		*no asbestos detected*		
Sample	**IHS-SF2000-3E1-109**	2007-00202- 109	Wall System		Positive Layer? No
	Layer # 1 Tan texture/joint compound		*no asbestos detected*		
	Layer # 2 White texture/joint compound		*no asbestos detected*		
Sample	**IHS-SF2000-3E2-110**	2007-00202- 110	Wall System		Positive Layer? No
	Layer # 1 Tan texture/joint compound		*no asbestos detected*		
	Layer # 2 White texture/joint compound		*no asbestos detected*		
Sample	**IHS-SF2000-3E3-111**	2007-00202- 111	Wall System		Positive Layer? No
	Layer # 1 Tan texture/joint compound		*no asbestos detected*		
	Layer # 2 White texture/joint compound		*no asbestos detected*		
Sample	**IHS-SF2000-3E4-112**	2007-00202- 112	Wall System		Positive Layer? No
	Layer # 1 Tan texture/joint compound		*no asbestos detected*		
	Layer # 2 White texture/joint compound		*no asbestos detected*		
Sample	**IHS-SF2000-3E5-113**	2007-00202- 113	Wall System		Positive Layer? No
	Layer # 1 Tan texture/joint compound		*no asbestos detected*		
	Layer # 2 White texture/joint compound		*no asbestos detected*		
Sample	**IHS-SF2000-5A1-114**	2007-00202- 114	Insulation		Positive Layer? No
	Layer # 1 White insulation		*no asbestos detected*		
Sample	**IHS-SF2000-5A2-115**	2007-00202- 115	Insulation		Positive Layer? No
	Layer # 1 White insulation		*no asbestos detected*		
Sample	**IHS-SF2000-5A3-116**	2007-00202- 116	Insulation		Positive Layer? No
	Layer # 1 White insulation		*no asbestos detected*		
Sample	**IHS-SF2000-5B1-117**	2007-00202- 117	Miscellaneous		Positive Layer? No
	Layer # 1 Green gasket		*no asbestos detected*		
Sample	**IHS-SF2000-5B2-118**	2007-00202- 118	Miscellaneous		Positive Layer? No
	Layer # 1 Green gasket		*no asbestos detected*		
Sample	**IHS-SF2000-5B3-119**	2007-00202- 119	Miscellaneous		Positive Layer? No
	Layer # 1 Green gasket		*no asbestos detected*		
Sample	**IHS-SF2000-5C1-120**	2007-00202- 120	Insulation		Positive Layer? No
	Layer # 1 Off-whit insulation		*no asbestos detected*		
Sample	**IHS-SF2000-5C2-121**	2007-00202- 121	Insulation		Positive Layer? No
	Layer # 1 Off-whit insulation		*no asbestos detected*		
Sample	**IHS-SF2000-5C3-122**	2007-00202- 122	Insulation		Positive Layer? No
	Layer # 1 Off-whit insulation		*no asbestos detected*		
Sample	**IHS-SF2000-5D1-123**	2007-00202- 123	Miscellaneous		Positive Layer? No
	Layer # 1 Off-whit gasket		*no asbestos detected*		
Sample	**IHS-SF2000-5D2-124**	2007-00202- 124	Miscellaneous		Positive Layer? No
	Layer # 1 Off-whit gasket		*no asbestos detected*		
Sample	**IHS-SF2000-5D3-125**	2007-00202- 125	Miscellaneous		Positive Layer? No
	Layer # 1 Off-whit gasket		*no asbestos detected*		

Sample	**IHS-SF2000-5E1-126**	2007-00202- 126	Insulation		Positive Layer? No
	Layer # 1	Tan	insulation wrap	*no asbestos detected*	
	Layer # 2	Tan	insulation	*no asbestos detected*	
Sample	**IHS-SF2000-5E2-127**	2007-00202- 127	Insulation		Positive Layer? No
	Layer # 1	Tan	insulation wrap	*no asbestos detected*	
	Layer # 2	Tan	insulation	*no asbestos detected*	
Sample	**IHS-SF2000-5E3-128**	2007-00202- 128	Insulation		Positive Layer? No
	Layer # 1	Tan	insulation wrap	*no asbestos detected*	
	Layer # 2	Tan	insulation	*no asbestos detected*	
Sample	**IHS-SF2000-6A1-129**	2007-00202- 129	Insulation		Positive Layer? No
	Layer # 1	Off-whit	fireproofing	*no asbestos detected*	
Sample	**IHS-SF2000-6A2-130**	2007-00202- 130	Insulation		Positive Layer? No
	Layer # 1	Off-whit	fireproofing	*no asbestos detected*	
Sample	**IHS-SF2000-6A3-131**	2007-00202- 131	Insulation		Positive Layer? No
	Layer # 1	Off-whit	fireproofing	*no asbestos detected*	
Sample	**IHS-SF2000-6A4-132**	2007-00202- 132	Insulation		Positive Layer? No
	Layer # 1	Off-whit	fireproofing	*no asbestos detected*	
Sample	**IHS-SF2000-6A5-133**	2007-00202- 133	Insulation		Positive Layer? No
	Layer # 1	Off-whit	fireproofing	*no asbestos detected*	
Sample	**IHS-SF2000-6A6-134**	2007-00202- 134	Insulation		Positive Layer? No
	Layer # 1	Off-whit	fireproofing	*no asbestos detected*	
Sample	**IHS-SF2000-6A7-135**	2007-00202- 135	Insulation		Positive Layer? No
	Layer # 1	Off-whit	fireproofing	*no asbestos detected*	
Sample	**IHS-SF2000-7A1-136**	2007-00202- 136	Miscellaneous		Positive Layer? No
	Layer # 1	Black	coating	*<=1% chrysotile asbestos*	
Sample	**IHS-SF2000-7A2-137**	2007-00202- 137	Miscellaneous		Positive Layer? No
	Layer # 1	Black	coating	*<=1% chrysotile asbestos*	
Sample	**IHS-SF2000-7A3-138**	2007-00202- 138	Miscellaneous		Positive Layer? No
	Layer # 1	Black	coating	*<=1% chrysotile asbestos*	
Sample	**IHS-SF2000-7B1-139**	2007-00202- 139	Miscellaneous		Positive Layer? Yes
	Layer # 1	Purple	coating	*>1-2% chrysotile asbestos*	
Sample	**IHS-SF2000-7B2-140**	2007-00202- 140	Miscellaneous		Positive Layer? Yes
	Layer # 1	Purple	coating	*>1-2% chrysotile asbestos*	
Sample	**IHS-SF2000-7B3-141**	2007-00202- 141	Miscellaneous		Positive Layer? Yes
	Layer # 1	Purple	coating	*>1-2% chrysotile asbestos*	
Sample	**IHS-SF2000-7C1-142**	2007-00202- 142	Miscellaneous		Positive Layer? No
	Layer # 1	Brown	coating	*no asbestos detected*	
Sample	**IHS-SF2000-7C2-143**	2007-00202- 143	Miscellaneous		Positive Layer? No
	Layer # 1	Brown	coating	*no asbestos detected*	
Sample	**IHS-SF2000-7C3-144**	2007-00202- 144	Miscellaneous		Positive Layer? Yes
	Layer # 1	Black	coating	*10-20% chrysotile asbestos*	
Sample	**IHS-SF2000-7D1-145**	2007-00202- 145	Adhesive/caulk		Positive Layer? No
	Layer # 1	Tan	mastic	*no asbestos detected*	
Sample	**IHS-SF2000-7D2-146**	2007-00202- 146	Adhesive/caulk		Positive Layer? No
	Layer # 1	Tan	mastic	*no asbestos detected*	
Sample	**IHS-SF2000-7D3-147**	2007-00202- 147	Adhesive/caulk		Positive Layer? No
	Layer # 1	Tan	mastic	*no asbestos detected*	
Sample	**IHS-SF2000-7E1-148**	2007-00202- 148	Adhesive/caulk		Positive Layer? No
	Layer # 1	Black	caulk	*no asbestos detected*	
Sample	**IHS-SF2000-7E2-149**	2007-00202- 149	Adhesive/caulk		Positive Layer? No
	Layer # 1	Brown	caulk	*no asbestos detected*	
Sample	**IHS-SF2000-7E3-150**	2007-00202- 150	Adhesive/caulk		Positive Layer? No
	Layer # 1	Black	caulk	*no asbestos detected*	
Sample	**IHS-SF2000-7F1-151**	2007-00202- 151	Adhesive/caulk		Positive Layer? Yes
	Layer # 1	white	caulk	*5-10% chrysotile asbestos*	
Sample	**IHS-SF2000-7F2-152**	2007-00202- 152	Adhesive/caulk		Positive Layer? Yes
	Layer # 1	white	caulk	*5-10% chrysotile asbestos*	
Sample	**IHS-SF2000-7F3-153**	2007-00202- 153	Adhesive/caulk		Positive Layer? Yes
	Layer # 1	white	caulk	*5-10% chrysotile asbestos*	
Sample	**IHS-SF2000-7G1-154**	2007-00202- 154	Adhesive/caulk		Positive Layer? Yes
	Layer # 1	Gray	caulk	*5-10% chrysotile asbestos*	
Sample	**IHS-SF2000-7G2-155**	2007-00202- 155	Adhesive/caulk		Positive Layer? No
	Layer # 1	White	caulk	*no asbestos detected*	
Sample	**IHS-SF2000-7G3-156**	2007-00202- 156	Adhesive/caulk		Positive Layer? No
	Layer # 1	White	caulk	*no asbestos detected*	
Sample	**IHS-SF2000-7H1-157**	2007-00202- 157	Adhesive/caulk		Positive Layer? No
	Layer # 1	Gray	caulk	*<=1% chrysotile asbestos*	
Sample	**IHS-SF2000-7H2-158**	2007-00202- 158	Adhesive/caulk		Positive Layer? No
	Layer # 1	Gray	caulk	*<=1% chrysotile asbestos*	
Sample	**IHS-SF2000-7H3-159**	2007-00202- 159	Adhesive/caulk		Positive Layer? No
	Layer # 1	Gray	caulk	*<=1% chrysotile asbestos*	
Sample	**IHS-SF2000-7I1-160**	2007-00202- 160	Insulation		Positive Layer? No
	Layer # 1	Yellow	insulation	*no asbestos detected*	
Sample	**IHS-SF2000-7I2-161**	2007-00202- 161	Insulation		Positive Layer? No
	Layer # 1	Yellow	insulation	*no asbestos detected*	

Sample	**IHS-SF2000-7I3-162**	2007-00202- 162	Insulation	Positive Layer?	No
	Layer # 1 Yellow insulation		*no asbestos detected*		
Sample	**IHS-SF2000-7J1-163**	2007-00202- 163	Adhesive/caulk	Positive Layer?	No
	Layer # 1 Black mastic		*no asbestos detected*		
Sample	**IHS-SF2000-7J2-164**	2007-00202- 164	Adhesive/caulk	Positive Layer?	No
	Layer # 1 Black mastic		*no asbestos detected*		
Sample	**IHS-SF2000-7J3-165**	2007-00202- 165	Adhesive/caulk	Positive Layer?	No
	Layer # 1 Black mastic		*no asbestos detected*		
Sample	**IHS-SF2000-7K1-166**	2007-00202- 166	Insulation	Positive Layer?	No
	Layer # 1 Yellow foam		*no asbestos detected*		
Sample	**IHS-SF2000-7K2-167**	2007-00202- 167	Insulation	Positive Layer?	No
	Layer # 1 Yellow foam		*no asbestos detected*		
Sample	**IHS-SF2000-7K3-168**	2007-00202- 168	Insulation	Positive Layer?	No
	Layer # 1 Yellow foam		*no asbestos detected*		
Sample	**IHS-SF2000-7L1-169**	2007-00202- 169	Adhesive/caulk	Positive Layer?	No
	Layer # 1 Tan caulk		*no asbestos detected*		
Sample	**IHS-SF2000-7L2-170**	2007-00202- 170	Adhesive/caulk	Positive Layer?	No
	Layer # 1 Tan caulk		*no asbestos detected*		
Sample	**IHS-SF2000-7L3-171**	2007-00202- 171	Adhesive/caulk	Positive Layer?	No
	Layer # 1 Tan caulk		*no asbestos detected*		
Sample	**IHS-SF2000-7M1-172**	2007-00202- 172	Adhesive/caulk	Positive Layer?	No
	Layer # 1 white caulk		*no asbestos detected*		
Sample	**IHS-SF2000-7M2-173**	2007-00202- 173	Adhesive/caulk	Positive Layer?	No
	Layer # 1 white caulk		*no asbestos detected*		
Sample	**IHS-SF2000-7M3-174**	2007-00202- 174	Adhesive/caulk	Positive Layer?	No
	Layer # 1 white caulk		*no asbestos detected*		
Sample	**IHS-SF2000-7N1-175**	2007-00202- 175	Adhesive/caulk	Positive Layer?	Yes
	Layer # 1 Gray caulk		*5-10% chrysotile asbestos*		
Sample	**IHS-SF2000-7N2-176**	2007-00202- 176	Adhesive/caulk	Positive Layer?	Yes
	Layer # 1 Gray caulk		*5-10% chrysotile asbestos*		
Sample	**IHS-SF2000-7N3-177**	2007-00202- 177	Adhesive/caulk	Positive Layer?	Yes
	Layer # 1 Gray caulk		*5-10% chrysotile asbestos*		
Sample	**IHS-SF2000-7O1-178**	2007-00202- 178	Adhesive/caulk	Positive Layer?	No
	Layer # 1 Black caulk		*no asbestos detected*		
Sample	**IHS-SF2000-7O2-179**	2007-00202- 179	Adhesive/caulk	Positive Layer?	No
	Layer # 1 Black caulk		*no asbestos detected*		
Sample	**IHS-SF2000-7O3-180**	2007-00202- 180	Adhesive/caulk	Positive Layer?	No
	Layer # 1 Black caulk		*no asbestos detected*		
Sample	**IHS-SF2000-8A1-181**	2007-00202- 181	Roofing	Positive Layer?	Yes
	Layer # 1 Black roof ply/bitumen		*30-40% chrysotile asbestos*		
Sample	**IHS-SF2000-8A2-182**	2007-00202- 182	Roofing	Positive Layer?	Yes
	Layer # 1 Black roof ply/bitumen		*30-40% chrysotile asbestos*		
Sample	**IHS-SF2000-8A3-183**	2007-00202- 183	Roofing	Positive Layer?	Yes
	Layer # 1 Black roof ply/bitumen		*30-40% chrysotile asbestos*		
Sample	**IHS-SF2000-8B1-184**	2007-00202- 184	Roofing	Positive Layer?	Yes
	Layer # 1 Black caulk		*5-10% chrysotile asbestos*		
Sample	**IHS-SF2000-8B2-185**	2007-00202- 185	Roofing	Positive Layer?	Yes
	Layer # 1 Black caulk		*5-10% chrysotile asbestos*		
Sample	**IHS-SF2000-8B3-186**	2007-00202- 186	Roofing	Positive Layer?	Yes
	Layer # 1 Black caulk		*5-10% chrysotile asbestos*		

* Apparent Sample Types and Apparent Layer Types are as they appeared to the analyst. Since many types of materials appear similar after sampling damage, the apparent type of material may not be the actual type of material.

Sample IHS-SF2000-1A1-1 **Lab Number** 2007-00202- 1 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid
Homogeneous No **# Layers** 3 **Pos Layer?** Yes **# Sub-Samples** 7
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	floor tile	97	off-white	1	n.d.	-	-	-	-	-
2	mastic	2	Yellow	1	n.d.	-	-	-	-	-
3	mastic	1	Black	1	2-5%	-	-	-	-	-
	Total %	100		**Average %**	<=1%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-1A2-2 **Lab Number** 2007-00202- 2 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 6
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	floor tile	98	off-white	1	n.d.	-	-	-	-	-
2	mastic	2	Clear	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-1A3-3 **Lab Number** 2007-00202- 3 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 6
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	floor tile	98	off-white	1	n.d.	-	-	-	-	-
2	mastic	2	Yellow	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-1B1-4 **Lab Number** 2007-00202- 4 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Rubbery
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	100	Yellow	1	2-5%	-	-	-	-	-
	Total %	100		**Average %**	2-5%	-	-	-	-	-
				Fiber Identification:	synthetic fiber (extr					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	synthetic fiber (extruded)	W	E	N	N	H	+	P					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-1B2-5 **Lab Number** 2007-00202- 5 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Rubbery
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	100	Green	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-1B3-6 **Lab Number** 2007-00202- 6 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Rubbery
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	100	Tan	1	5-10%	-	-	-	-	-
	Total %	100		**Average %**	5-10%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-1C1-7 **Lab Number** 2007-00202- 7 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Surfacing Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	grout	100	Gray	2	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-1C2-8 **Lab Number** 2007-00202- 8 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Surfacing Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	grout	100	Gray	2	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-1C3-9 **Lab Number** 2007-00202- 9 **Sampled:** 12/13/2006 **Condition:** acceptable

Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Surfacing Non-fibrous Solid

Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3

Non-Fibrous Components (in approx. decreasing order): powder, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	grout	100	Gray	2	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note

Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-1D1-10 **Lab Number** 2007-00202- 10 **Sampled:** 12/13/2006 **Condition:** acceptable

Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid

Homogeneous No **# Layers** 3 **Pos Layer?** Yes **# Sub-Samples** 8

Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	2	Yellow	1	n.d.	-	-	-	-	-
2	floor tile	93	Off-White	1	5-10%	-	-	-	-	-
3	mastic	5	Black	1	2-5%	-	-	-	-	-
	Total %	100		**Average %**	5-10%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2													
3													
4													
5													
6													

Sample Analytical Note

Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-1D2-11 **Lab Number** 2007-00202- 11 **Sampled:** 12/13/2006 **Condition:** acceptable

Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid

Homogeneous No **# Layers** 3 **Pos Layer?** Yes **# Sub-Samples** 8

Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	7	Yellow	1	n.d.	-	-	-	-	-
2	floor tile	90	Off-White	1	5-10%	-	-	-	-	-
3	mastic	3	Black	1	10-20%	-	-	-	-	-
	Total %	100		**Average %**	5-10%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2													
3													
4													
5													
6													

Sample Analytical Note

Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-1D3-12 **Lab Number** 2007-00202- 12 **Sampled:** 12/13/2006 **Condition:** acceptable

Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid

Homogeneous No **# Layers** 3 **Pos Layer?** Yes **# Sub-Samples** 8

Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	3	Off-white	1	n.d.	-	-	-	-	-
2	floor tile	92	Off-White	1	5-10%	-	-	-	-	-
3	mastic	5	Black	1	5-10%	-	-	-	-	-
	Total %	100		**Average %**	5-10%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2													
3													
4													
5													
6													

Sample Analytical Note

Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-1E1-13 **Lab Number** 2007-00202- 13 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 6
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	floor tile	95	off-white	1	n.d.	-	-	-	-	-
2	mastic	5	Off-white	1	2-5%	-	-	-	-	-
	Total %	100		**Average %**	<=1%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-1E2-14 **Lab Number** 2007-00202- 14 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 6
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	floor tile	95	off-white	1	n.d.	-	-	-	-	-
2	mastic	5	Off-white	1	2-5%	-	-	-	-	-
	Total %	100		**Average %**	<=1%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-1E3-15 **Lab Number** 2007-00202- 15 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 6
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	floor tile	97	off-white	1	n.d.	-	-	-	-	-
2	mastic	2	Off-white	1	>1-2%	-	-	-	-	-
	Total %	99		**Average %**	<=1%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-1F1-16 **Lab Number** 2007-00202- 16 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Surfacing Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	grout	100	Gray	2	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-1F2-17 **Lab Number** 2007-00202- 17 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Surfacing Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	grout	100	Gray	2	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-1F3-18 **Lab Number** 2007-00202- 18 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Surfacing Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	grout	100	Gray	2	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-1G1-19 **Lab Number** 2007-00202- 19 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 6
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	floor tile	95	off-white	1	n.d.	-	-	-	-	-
2	mastic	5	Tan	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-1G2-20 **Lab Number** 2007-00202- 20 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 6
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	floor tile	95	off-white	1	n.d.	-	-	-	-	-
2	mastic	5	Yellow	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-1G3-21 **Lab Number** 2007-00202- 21 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 6
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	floor tile	95	off-white	1	n.d.	-	-	-	-	-
2	mastic	5	Yellow	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-1H1-22 **Lab Number** 2007-00202- 22 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 5
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	floor tile	98	Orange	1	n.d.	-	-	-	-	-
2	mastic	2	Clear	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-1H2-23 **Lab Number** 2007-00202- 23 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 5
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	floor tile	96	Orange	1	n.d.	-	-	-	-	-
2	mastic	4	Clear	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-1H3-24 **Lab Number** 2007-00202- 24 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 5
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	floor tile	97	Orange	1	n.d.	-	-	-	-	-
2	mastic	3	Clear	1	<=1%	-	-	-	-	-
	Total %	100		**Average %**	<=1%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-1I1-25 **Lab Number** 2007-00202- 25 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	100	Yellow	1	2-5%	-	-	-	-	-
	Total %	100		**Average %**	2-5%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-1I2-26 **Lab Number** 2007-00202- 26 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	100	Yellow	1	<=1%	-	-	-	-	-
	Total %	100		**Average %**	<=1%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-1I3-27 **Lab Number** 2007-00202- 27 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Rubbery
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	100	Yellow	1	>1-2%	-	-	-	-	-
	Total %	100		**Average %**	>1-2%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-1J1-28 **Lab Number** 2007-00202- 28 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Sticky
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	100	Black	1	5-10%	-	-	-	-	-
	Total %	100		**Average %**	5-10%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-1J2-29 **Lab Number** 2007-00202- 29 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Sticky
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	100	Black	1	5-10%	-	-	-	-	-
	Total %	100		**Average %**	5-10%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-1J3-30 **Lab Number** 2007-00202- 30 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Sticky
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	100	Black	1	5-10%	-	-	-	-	-
	Total %	100		**Average %**	5-10%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-1K1-31 **Lab Number** 2007-00202- 31 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 6
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	floor tile	95	Gray	1	n.d.	-	-	-	-	-
2	mastic	5	Yellow	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-1K2-32 **Lab Number** 2007-00202- 32 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 6
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	floor tile	95	Gray	1	n.d.	-	-	-	-	-
2	mastic	5	Yellow	1	<=1%	-	-	-	-	-
	Total %	100		**Average %**	<=1%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-1K3-33 **Lab Number** 2007-00202- 33 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 6
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	floor tile	95	Gray	1	n.d.	-	-	-	-	-
2	mastic	5	Yellow	1	<=1%	-	-	-	-	-
	Total %	100		**Average %**	<=1%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-1L1-34 **Lab Number** 2007-00202- 34 **Sampled:** 12/13/2006 **Condition:** acceptable

Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid

Homogeneous No **# Layers** 3 **Pos Layer?** Yes **# Sub-Samples** 7

Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	floor tile	95	Gray	1	n.d.	n.d.	-	-	-	-
2	mastic	4	Yellow	1	<=1%	n.d.	-	-	-	-
3	mastic	1	Black	1	n.d.	2-5%	-	-	-	-
	Total %	100		**Average %**	<=1%	<=1%	-	-	-	-
				Fiber Identification:	cellulose fiber	chrysotile asbestos				

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
3													
4													
5													
6													

Sample Analytical Note

Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-1L2-35 **Lab Number** 2007-00202- 35 **Sampled:** 12/13/2006 **Condition:** acceptable

Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid

Homogeneous No **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 6

Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	floor tile	95	Gray	1	n.d.	-	-	-	-	-
2	mastic	5	Yellow	1	<=1%	-	-	-	-	-
	Total %	100		**Average %**	<=1%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note

Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-1L3-36 **Lab Number** 2007-00202- 36 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 6
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	floor tile	95	Gray	1	n.d.	-	-	-	-	-
2	mastic	5	Yellow	1	<=1%	-	-	-	-	-
	Total %	100		**Average %**	<=1%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-1M1-37 **Lab Number** 2007-00202- 37 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 5
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	floor tile	92	Black	1	n.d.	-	-	-	-	-
2	mastic	8	Yellow	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-1M2-38 **Lab Number** 2007-00202- 38 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	floor tile	100	Black	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-1M3-39 **Lab Number** 2007-00202- 39 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	floor tile	100	Black	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-2A1-40 **Lab Number** 2007-00202- 40 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	drywall core	100	white	3	<=1%	<=1%	-	-	-	-
	Total %	100		**Average %**	<=1%	<=1%	-	-	-	-
				Fiber Identification:	cellulose	glass fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose	W	F	N	N	H	+	U					
2	glass fiber	CL	D	Y									
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps.

Sample IHS-SF2000-2A2-41 **Lab Number** 2007-00202- 41 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	drywall core	100	white	3	<=1%	<=1%	-	-	-	-
	Total %	100		**Average %**	<=1%	<=1%	-	-	-	-
				Fiber Identification:	cellulose	glass fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose	W	F	N	N	H	+	U					
2	glass fiber	CL	D	Y									
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps.

Sample IHS-SF2000-2A3-42 **Lab Number** 2007-00202- 42 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	drywall core	100	white	3	<=1%	-	-	-	-	-
	Total %	100		**Average %**	<=1%	-	-	-	-	-
				Fiber Identification:	cellulose					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps.

Sample IHS-SF2000-2B1-43 **Lab Number** 2007-00202- 43 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	texture/joint compound	100	white	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-2B2-44 **Lab Number** 2007-00202- 44 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	texture/joint compound	100	white	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-2B3-45 **Lab Number** 2007-00202- 45 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	texture/joint compound	100	white	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-2C1-46 **Lab Number** 2007-00202- 46 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	texture/joint compound	100	white	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid. Note: sample size was too small - analysis may not be representative of whole.

Sample IHS-SF2000-2C2-47 **Lab Number** 2007-00202- 47 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	texture/joint compound	100	white	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid. Note: sample size was too small - analysis may not be representative of whole.

Sample IHS-SF2000-2C3-48 **Lab Number** 2007-00202- 48 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	texture/joint compound	100	white	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid. Note: sample size was too small - analysis may not be representative of whole.

Sample IHS-SF2000-2C4-49 **Lab Number** 2007-00202- 49 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	texture/joint compound	100	white	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid. Note: sample size was too small - analysis may not be representative of whole.

Sample IHS-SF2000-2C5-50 **Lab Number** 2007-00202- 50 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	texture/joint compound	100	white	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid. Note: sample size was too small - analysis may not be representative of whole.

Sample IHS-SF2000-2C6-51 **Lab Number** 2007-00202- 51 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	texture/joint compound	100	white	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid. Note: sample size was too small - analysis may not be representative of whole.

Sample IHS-SF2000-2C7-52 **Lab Number** 2007-00202- 52 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	texture/joint compound	100	white	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid. Note: sample size was too small - analysis may not be representative of whole.

Sample IHS-SF2000-2D1-53 **Lab Number** 2007-00202- 53 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): binder, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	wall covering	100	White	1	30-40%	-	-	-	-	-
	Total %	100		**Average %**	30-40%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-2D2-54 **Lab Number** 2007-00202- 54 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): binder, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	wall covering	100	White	1	30-40%	-	-	-	-	-
	Total %	100		**Average %**	30-40%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-2D3-55 **Lab Number** 2007-00202- 55 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): binder, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	wall covering	100	White	1	30-40%	-	-	-	-	-
	Total %	100		**Average %**	30-40%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-2E1-56 **Lab Number** 2007-00202- 56 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	texture/joint compound	100	white	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-2E2-57 **Lab Number** 2007-00202- 57 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	texture/joint compound	100	white	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-2E3-58 **Lab Number** 2007-00202- 58 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	texture/joint compound	100	white	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-2E4-59 **Lab Number** 2007-00202- 59 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	texture/joint compound	100	white	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-2E5-60 **Lab Number** 2007-00202- 60 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	texture/joint compound	100	white	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-2F1-61 **Lab Number** 2007-00202- 61 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	texture/joint compound	100	white	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-2F2-62 **Lab Number** 2007-00202- 62 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	texture/joint compound	100	white	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-2F3-63 **Lab Number** 2007-00202- 63 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	texture/joint compound	100	white	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-2F4-64 **Lab Number** 2007-00202- 64 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	texture/joint compound	100	white	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-2F5-65 **Lab Number** 2007-00202- 65 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	texture/joint compound	100	white	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-2G1-66 **Lab Number** 2007-00202- 66 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	texture/joint compound	100	white	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-2G2-67 **Lab Number** 2007-00202- 67 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	texture/joint compound	100	white	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-2G3-68 **Lab Number** 2007-00202- 68 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	texture/joint compound	100	white	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-2H1-69 **Lab Number** 2007-00202- 69 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	texture/joint compound	100	white	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-2H2-70 **Lab Number** 2007-00202- 70 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	texture/joint compound	100	white	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-2H3-71 **Lab Number** 2007-00202- 71 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	texture/joint compound	100	white	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-2I1-72 **Lab Number** 2007-00202- 72 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Surfacing Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	grout	100	White	2	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-2I2-73 **Lab Number** 2007-00202- 73 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Surfacing Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	grout	100	White	2	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-2I3-74 **Lab Number** 2007-00202- 74 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Surfacing Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	grout	100	White	2	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-2J1-75 **Lab Number** 2007-00202- 75 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Surfacing Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	grout	100	White	2	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-2J2-76 **Lab Number** 2007-00202- 76 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Surfacing Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	grout	100	White	2	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-2J3-77 **Lab Number** 2007-00202- 77 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Surfacing Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	grout	100	White	2	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-2K1-78 **Lab Number** 2007-00202- 78 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Acoustic Tile Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): perlite, glass, binder

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	acoustical tile	100	Off-white	3	40-50%	10-20%	-	-	-	-
	Total %	100		**Average %**	40-50%	10-20%	-	-	-	-
				Fiber Identification:	cellulose fiber	glass fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2	glass fiber	CL	D	Y									
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-2K2-79 **Lab Number** 2007-00202- 79 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Acoustic Tile Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): perlite, glass, binder

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	acoustical tile	100	Off-white	3	40-50%	5-10%	-	-	-	-
	Total %	100		**Average %**	40-50%	5-10%	-	-	-	-
				Fiber Identification:	cellulose fiber	glass fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2	glass fiber	CL	D	Y									
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-2K3-80 **Lab Number** 2007-00202- 80 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Acoustic Tile Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): perlite, glass, binder

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	acoustical tile	100	Off-white	3	40-50%	5-10%	-	-	-	-
	Total %	100		**Average %**	40-50%	5-10%	-	-	-	-
				Fiber Identification:	cellulose fiber	glass fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2	glass fiber	CL	D	Y									
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-2L1-81 **Lab Number** 2007-00202- 81 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, rock,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	block	100	Gray	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-2L2-82 **Lab Number** 2007-00202- 82 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, rock,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	block	100	Gray	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-2L3-83 **Lab Number** 2007-00202- 83 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, rock,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	block	100	Gray	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-2M1-84 **Lab Number** 2007-00202- 84 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, rock,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mortar	100	Gray	2	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-2M2-85 **Lab Number** 2007-00202- 85 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, rock,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mortar	100	Gray	2	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-2M3-86 **Lab Number** 2007-00202- 86 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, rock,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mortar	100	Gray	2	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-2N1-87 **Lab Number** 2007-00202- 87 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Miscellaneous Fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	cem/asb board	100	Gray	2	10-20%	-	-	-	-	-
	Total %	100		**Average %**	10-20%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-2N2-88 **Lab Number** 2007-00202- 88 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Miscellaneous Fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	cem/asb board	100	Gray	2	10-20%	-	-	-	-	-
	Total %	100		**Average %**	10-20%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-2N3-89 **Lab Number** 2007-00202- 89 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Miscellaneous Fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	cem/asb board	100	Gray	2	10-20%	-	-	-	-	-
	Total %	100		**Average %**	10-20%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-2O1-90 **Lab Number** 2007-00202- 90 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): binder, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	paint, textured	100	Brown	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent. No cementicious type stucco was observed.

Sample IHS-SF2000-2O2-91 **Lab Number** 2007-00202- 91 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, rock,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	stucco	100	Gray	2	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: dissolution of matrix using dilute HCl acid. Procedure: tweased apart using forceps.

Sample IHS-SF2000-2O3-92 **Lab Number** 2007-00202- 92 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): binder, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	paint, textured	100	Brown	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent. No cementicious type stucco was observed.

Sample IHS-SF2000-2O4-93 **Lab Number** 2007-00202- 93 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): binder, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	paint, textured	100	Brown	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent. No cementicious type stucco was observed.

Sample IHS-SF2000-2O5-94 **Lab Number** 2007-00202- 94 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, rock,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	stucco	100	Gray	2	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: dissolution of matrix using dilute HCl acid. Procedure: tweased apart using forceps.

Sample IHS-SF2000-2O6-95 **Lab Number** 2007-00202- 95 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, rock,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	stucco	100	Gray	2	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: dissolution of matrix using dilute HCl acid. Procedure: tweased apart using forceps.

Sample IHS-SF2000-2O7-96 **Lab Number** 2007-00202- 96 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): binder, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	paint, textured	100	Brown	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent. No cementicious type stucco was observed.

Sample IHS-SF2000-3A1-97 **Lab Number** 2007-00202- 97 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Acoustic Tile Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): perlite, glass, binder

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	acoustical tile	100	Off-white	3	40-50%	20-30%	-	-	-	-
	Total %	100		**Average %**	40-50%	20-30%	-	-	-	-
				Fiber Identification:	cellulose fiber	glass fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2	glass fiber	CL	D	Y									
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-3A2-98 **Lab Number** 2007-00202- 98 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Acoustic Tile Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): perlite, glass, binder

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	acoustical tile	100	Off-white	3	40-50%	20-30%	-	-	-	-
	Total %	100		**Average %**	40-50%	20-30%	-	-	-	-
				Fiber Identification:	cellulose fiber	glass fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2	glass fiber	CL	D	Y									
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-3A3-99 **Lab Number** 2007-00202- 99 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Acoustic Tile Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): perlite, glass, binder

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	acoustical tile	100	Off-white	3	40-50%	20-30%	-	-	-	-
	Total %	100		**Average %**	40-50%	20-30%	-	-	-	-
				Fiber Identification:	cellulose fiber	glass fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2	glass fiber	CL	D	Y									
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-3B1-100 **Lab Number** 2007-00202- 100 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Acoustic Tile Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): perlite, glass, binder

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	acoustical tile	100	Off-white	3	30-40%	20-30%	-	-	-	-
	Total %	100		**Average %**	30-40%	20-30%	-	-	-	-
				Fiber Identification:	glass fiber	cellulose fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	glass fiber	CL	D	Y									
2	cellulose fiber	W	F	N	N	H	+	U					
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-3B2-101 **Lab Number** 2007-00202- 101 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Acoustic Tile Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): perlite, glass, binder

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	acoustical tile	100	Off-white	3	40-50%	5-10%	-	-	-	-
	Total %	100		**Average %**	40-50%	5-10%	-	-	-	-
				Fiber Identification:	cellulose fiber	glass fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2	glass fiber	CL	D	Y									
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-3B3-102 **Lab Number** 2007-00202- 102 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Acoustic Tile Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): perlite, glass, binder

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	acoustical tile	100	Off-white	3	40-50%	5-10%	-	-	-	-
	Total %	100		**Average %**	40-50%	5-10%	-	-	-	-
				Fiber Identification:	cellulose fiber	glass fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2	glass fiber	CL	D	Y									
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-3C1-103 **Lab Number** 2007-00202- 103 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Acoustic Tile Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): perlite, glass, binder

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	acoustical tile	100	Off-white	3	30-40%	20-30%	-	-	-	-
	Total %	100		**Average %**	30-40%	20-30%	-	-	-	-
				Fiber Identification:	glass fiber	cellulose fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	glass fiber	CL	D	Y									
2	cellulose fiber	W	F	N	N	H	+	U					
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-3C2-104 **Lab Number** 2007-00202- 104 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Acoustic Tile Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): perlite, glass, binder

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	acoustical tile	100	Off-white	3	30-40%	30-40%	-	-	-	-
	Total %	100		**Average %**	30-40%	30-40%	-	-	-	-
				Fiber Identification:	glass fiber	cellulose fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	glass fiber	CL	D	Y									
2	cellulose fiber	W	F	N	N	H	+	U					
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-3C3-105 **Lab Number** 2007-00202- 105 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Acoustic Tile Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): perlite, glass, binder

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	acoustical tile	100	Off-white	3	30-40%	30-40%	-	-	-	-
	Total %	100		**Average %**	30-40%	30-40%	-	-	-	-
				Fiber Identification:	glass fiber	cellulose fiber				

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	glass fiber	CL	D	Y									
2	cellulose fiber	W	F	N	N	H	+	U					
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-3D1-106 **Lab Number** 2007-00202- 106 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Acoustic Tile Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): perlite, glass, binder

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	acoustical tile	100	Off-white	3	30-40%	20-30%	-	-	-	-
	Total %	100		**Average %**	30-40%	20-30%	-	-	-	-
				Fiber Identification:	glass fiber	cellulose fiber				

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	glass fiber	CL	D	Y									
2	cellulose fiber	W	F	N	N	H	+	U					
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-3D2-107 **Lab Number** 2007-00202- 107 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Acoustic Tile Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): perlite, glass, binder

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	acoustical tile	100	Off-white	3	30-40%	20-30%	-	-	-	-
	Total %	100		**Average %**	30-40%	20-30%	-	-	-	-
				Fiber Identification:	glass fiber	cellulose fiber				

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	glass fiber	CL	D	Y									
2	cellulose fiber	W	F	N	N	H	+	U					
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-3D3-108 **Lab Number** 2007-00202- 108 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Acoustic Tile Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): perlite, glass, binder

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	acoustical tile	100	Off-white	3	30-40%	20-30%	-	-	-	-
	Total %	100		**Average %**	30-40%	20-30%	-	-	-	-
				Fiber Identification:	glass fiber	cellulose fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	glass fiber	CL	D	Y									
2	cellulose fiber	W	F	N	N	H	+	U					
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-3E1-109 **Lab Number** 2007-00202- 109 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 6
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	texture/joint compound	80	Tan	3	n.d.	-	-	-	-	-
2	texture/joint compound	20	White	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid. Textures were seperated by a paint layer.

Sample IHS-SF2000-3E2-110 **Lab Number** 2007-00202- 110 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 6
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	texture/joint compound	15	Tan	3	n.d.	-	-	-	-	-
2	texture/joint compound	85	White	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid. Textures were seperated by a paint layer.

Sample IHS-SF2000-3E3-111 **Lab Number** 2007-00202- 111 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 6
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	texture/joint compound	40	Tan	3	n.d.	-	-	-	-	-
2	texture/joint compound	60	White	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid. Textures were seperated by a paint layer.

Sample IHS-SF2000-3E4-112 **Lab Number** 2007-00202- 112 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 6
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	texture/joint compound	40	Tan	3	n.d.	-	-	-	-	-
2	texture/joint compound	60	White	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid. Textures were seperated by a paint layer.

Sample IHS-SF2000-3E5-113 **Lab Number** 2007-00202- 113 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 6
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	texture/joint compound	60	Tan	3	n.d.	-	-	-	-	-
2	texture/joint compound	40	White	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid. Textures were seperated by a paint layer.

Sample IHS-SF2000-5A1-114 **Lab Number** 2007-00202- 114 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Insulation Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	insulation	100	White	3	>1-2%	<=1%	-	-	-	-
	Total %	100		**Average %**	>1-2%	<=1%	-	-	-	-
				Fiber Identification:	synthetic fiber (extr	glass fiber				

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	synthetic fiber (extruded)	W	E	N	N	H	+	P					
2	glass fiber	CL	D	Y									
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-5A2-115 **Lab Number** 2007-00202- 115 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Insulation Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	insulation	100	White	3	2-5%	>1-2%	-	-	-	-
	Total %	100		**Average %**	2-5%	>1-2%	-	-	-	-
				Fiber Identification:	synthetic fiber (extr	glass fiber				

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	synthetic fiber (extruded)	W	E	N	N	H	+	P					
2	glass fiber	CL	D	Y									
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-5A3-116 **Lab Number** 2007-00202- 116 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Insulation Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	insulation	100	White	3	2-5%	>1-2%	-	-	-	-
	Total %	100		**Average %**	2-5%	>1-2%	-	-	-	-
				Fiber Identification:	synthetic fiber (extr	glass fiber				

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	synthetic fiber (extruded)	W	E	N	N	H	+	P					
2	glass fiber	CL	D	Y									
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-5B1-117 **Lab Number** 2007-00202- 117 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Miscellaneous Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): binder, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	gasket	100	Green	1	40-50%	2-5%	>1-2%	-	-	-
	Total %	100		**Average %**	40-50%	2-5%	>1-2%	-	-	-
				Fiber Identification:	wollastonite	cellulose fiber	glass fiber			

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	wollastonite	W	G	N	N	M	B	P					
2	cellulose fiber	W	F	N	N	H	+	U					
3	glass fiber	CL	D	Y									
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-5B2-118 **Lab Number** 2007-00202- 118 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Miscellaneous Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): binder, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	gasket	100	Green	1	40-50%	2-5%	>1-2%	-	-	-
	Total %	100		**Average %**	40-50%	2-5%	>1-2%	-	-	-
				Fiber Identification:	wollastonite	cellulose fiber	glass fiber			

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	wollastonite	W	G	N	N	M	B	P					
2	cellulose fiber	W	F	N	N	H	+	U					
3	glass fiber	CL	D	Y									
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-5B3-119 **Lab Number** 2007-00202- 119 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Miscellaneous Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): binder, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	gasket	100	Green	1	40-50%	2-5%	>1-2%	-	-	-
	Total %	100		**Average %**	40-50%	2-5%	>1-2%	-	-	-
				Fiber Identification:	wollastonite	cellulose fiber	glass fiber			

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	wollastonite	W	G	N	N	M	B	P					
2	cellulose fiber	W	F	N	N	H	+	U					
3	glass fiber	CL	D	Y									
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-5C1-120 **Lab Number** 2007-00202- 120 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Insulation Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	insulation	100	Off-white	3	<=1%	-	-	-	-	-
	Total %	100		**Average %**	<=1%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-5C2-121 **Lab Number** 2007-00202- 121 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Insulation Powder
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	insulation	100	Off-white	3	10-20%	2-5%	-	-	-	-
	Total %	100		**Average %**	10-20%	2-5%	-	-	-	-
				Fiber Identification:	glass fiber	cellulose fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	glass fiber	CL	D	Y									
2	cellulose fiber	W	F	N	N	H	+	U					
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-5C3-122 **Lab Number** 2007-00202- 122 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Insulation Powder
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	insulation	100	Off-white	3	10-20%	2-5%	-	-	-	-
	Total %	100		**Average %**	10-20%	2-5%	-	-	-	-
				Fiber Identification:	glass fiber	cellulose fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	glass fiber	CL	D	Y									
2	cellulose fiber	W	F	N	N	H	+	U					
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-5D1-123 **Lab Number** 2007-00202- 123 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Miscellaneous Sticky
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): binder, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	gasket	100	Off-white	1	70-80%	-	-	-	-	-
	Total %	100		**Average %**	70-80%	-	-	-	-	-
				Fiber Identification:	glass fiber					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	glass fiber	CL	D	Y									
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-5D2-124 **Lab Number** 2007-00202- 124 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Miscellaneous Sticky
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): binder, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	gasket	100	Off-white	1	70-80%	-	-	-	-	-
	Total %	100		**Average %**	70-80%	-	-	-	-	-
				Fiber Identification:	glass fiber					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	glass fiber	CL	D	Y									
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-5D3-125 **Lab Number** 2007-00202- 125 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Miscellaneous Sticky
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): binder, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	gasket	100	Off-white	1	70-80%	-	-	-	-	-
	Total %	100		**Average %**	70-80%	-	-	-	-	-
				Fiber Identification:	glass fiber					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	glass fiber	CL	D	Y									
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-5E1-126 **Lab Number** 2007-00202- 126 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Insulation Fibrous Mat
Homogeneous No **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 6
Non-Fibrous Components (in approx. decreasing order): binder, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	insulation wrap	25	Tan	2	80-90%	n.d.	-	-	-	-
2	insulation	75	Tan	3	n.d.	50-60%	-	-	-	-
	Total %	100		**Average %**	20-30%	40-50%	-	-	-	-
				Fiber Identification:	cellulose fiber	glass fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2	glass fiber	CL	D	Y									
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-5E2-127 **Lab Number** 2007-00202- 127 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Insulation Fibrous Mat
Homogeneous No **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 6
Non-Fibrous Components (in approx. decreasing order): binder, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	insulation wrap	25	Tan	2	80-90%	n.d.	-	-	-	-
2	insulation	75	Tan	3	n.d.	50-60%	-	-	-	-
	Total %	100		**Average %**	20-30%	40-50%	-	-	-	-
				Fiber Identification:	cellulose fiber	glass fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2	glass fiber	CL	D	Y									
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-5E3-128 **Lab Number** 2007-00202- 128 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Insulation Fibrous Mat
Homogeneous No **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 6
Non-Fibrous Components (in approx. decreasing order): binder, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	insulation wrap	15	Tan	2	80-90%	n.d.	-	-	-	-
2	insulation	85	Tan	3	n.d.	50-60%	-	-	-	-
	Total %	100		**Average %**	10-20%	40-50%	-	-	-	-
				Fiber Identification:	cellulose fiber	glass fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2	glass fiber	CL	D	Y									
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-6A1-129 **Lab Number** 2007-00202- 129 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Insulation Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, mica/vermiculite,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	fireproofing	100	Off-white	3	5-10%	-	-	-	-	-
	Total %	100		**Average %**	5-10%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-6A2-130 **Lab Number** 2007-00202- 130 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Insulation Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, mica/vermiculite,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	fireproofing	100	Off-white	3	5-10%	-	-	-	-	-
	Total %	100		**Average %**	5-10%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-6A3-131 **Lab Number** 2007-00202- 131 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Insulation Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, mica/vermiculite,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	fireproofing	100	Off-white	3	5-10%	-	-	-	-	-
	Total %	100		**Average %**	5-10%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-6A4-132 **Lab Number** 2007-00202- 132 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Insulation Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, mica/vermiculite,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	fireproofing	100	Off-white	3	5-10%	-	-	-	-	-
	Total %	100		**Average %**	5-10%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-6A5-133 **Lab Number** 2007-00202- 133 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Insulation Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, mica/vermiculite,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	fireproofing	100	Off-white	3	5-10%	-	-	-	-	-
	Total %	100		**Average %**	5-10%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-6A6-134 **Lab Number** 2007-00202- 134 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Insulation Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, mica/vermiculite,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	fireproofing	100	Off-white	3	5-10%	-	-	-	-	-
	Total %	100		**Average %**	5-10%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-6A7-135 **Lab Number** 2007-00202- 135 **Sampled:** 12/13/2006 **Condition:** acceptable

Analyzed By RAM 1/29/2007 **An?** OK **Apparent Smp Type** Insulation Fibrous Mat

Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3

Non-Fibrous Components (in approx. decreasing order): powder, mica/vermiculite,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	fireproofing	100	Off-white	3	5-10%	-	-	-	-	-
	Total %	100		**Average %**	5-10%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note

Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-7A1-136 **Lab Number** 2007-00202- 136 **Sampled:** 12/13/2006 **Condition:** acceptable

Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Miscellaneous Non-fibrous Solid

Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3

Non-Fibrous Components (in approx. decreasing order): filler, bitumen,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	coating	100	Black	1	<=1%	-	-	-	-	-
	Total %	100		**Average %**	<=1%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2													
3													
4													
5													
6													

Sample Analytical Note

Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent. This sample is very close to 1%. It is recommended that this sample be analyzed by TEM.

Sample IHS-SF2000-7A2-137 **Lab Number** 2007-00202- 137 **Sampled:** 12/13/2006 **Condition:** acceptable

Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Miscellaneous Non-fibrous Solid

Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3

Non-Fibrous Components (in approx. decreasing order): filler, bitumen,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	coating	100	Black	1	<=1%	-	-	-	-	-
	Total %	100		**Average %**	<=1%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2													
3													
4													
5													
6													

Sample Analytical Note

Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent. This sample is very close to 1%. It is recommended that this sample be analyzed by TEM.

Sample IHS-SF2000-7A3-138 **Lab Number** 2007-00202- 138 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Miscellaneous Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, bitumen,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	coating	100	Black	1	<=1%	-	-	-	-	-
	Total %	100		**Average %**	<=1%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent. This sample is very close to 1%. It is recommended that this sample be analyzed by TEM.

Sample IHS-SF2000-7B1-139 **Lab Number** 2007-00202- 139 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Miscellaneous Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	coating	100	Purple	1	>1-2%	-	-	-	-	-
	Total %	100		**Average %**	>1-2%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-7B2-140 **Lab Number** 2007-00202- 140 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Miscellaneous Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	coating	100	Purple	1	>1-2%	-	-	-	-	-
	Total %	100		**Average %**	>1-2%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-7B3-141 **Lab Number** 2007-00202- 141 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Miscellaneous Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	coating	100	Purple	1	>1-2%	-	-	-	-	-
	Total %	100		**Average %**	>1-2%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-7C1-142 **Lab Number** 2007-00202- 142 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Miscellaneous Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	coating	100	Brown	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-7C2-143 **Lab Number** 2007-00202- 143 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Miscellaneous Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	coating	100	Brown	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-7C3-144 **Lab Number** 2007-00202- 144 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Miscellaneous Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	coating	100	Black	1	10-20%	-	-	-	-	-
	Total %	100		**Average %**	10-20%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-7D1-145 **Lab Number** 2007-00202- 145 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	100	Tan	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-7D2-146 **Lab Number** 2007-00202- 146 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	100	Tan	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-7D3-147 **Lab Number** 2007-00202- 147 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	100	Tan	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-7E1-148 **Lab Number** 2007-00202- 148 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Black	1	>1-2%	-	-	-	-	-
	Total %	100		**Average %**	>1-2%	-	-	-	-	-
				Fiber Identification:	synthetic fiber (extr					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	synthetic fiber (extruded)	W	E	N	N	H	+	P					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-7E2-149 **Lab Number** 2007-00202- 149 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Brown	1	>1-2%	-	-	-	-	-
	Total %	100		**Average %**	>1-2%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-7E3-150 **Lab Number** 2007-00202- 150 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Black	1	>1-2%	-	-	-	-	-
	Total %	100		**Average %**	>1-2%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-7F1-151 **Lab Number** 2007-00202- 151 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Sticky
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	white	1	5-10%	-	-	-	-	-
	Total %	100		**Average %**	5-10%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-7F2-152 **Lab Number** 2007-00202- 152 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Sticky
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	white	1	5-10%	-	-	-	-	-
	Total %	100		**Average %**	5-10%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-7F3-153 **Lab Number** 2007-00202- 153 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Sticky
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	white	1	5-10%	-	-	-	-	-
	Total %	100		**Average %**	5-10%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-7G1-154 **Lab Number** 2007-00202- 154 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Sticky
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Gray	1	5-10%	-	-	-	-	-
	Total %	100		**Average %**	5-10%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-7G2-155 **Lab Number** 2007-00202- 155 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	White	1	5-10%	-	-	-	-	-
	Total %	100		**Average %**	5-10%	-	-	-	-	-
				Fiber Identification:	synthetic fiber (shr					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	synthetic fiber (shredded)	W	A	N	N	H	+	P					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-7G3-156 **Lab Number** 2007-00202- 156 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	White	1	5-10%	-	-	-	-	-
	Total %	100		**Average %**	5-10%	-	-	-	-	-
				Fiber Identification:	synthetic fiber (shr					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	synthetic fiber (shredded)	W	A	N	N	H	+	P					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-7H1-157 **Lab Number** 2007-00202- 157 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Rubbery
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Gray	1	<=1%	-	-	-	-	-
	Total %	100		**Average %**	<=1%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	vb/g	pb/r	1.556	1.549
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-7H2-158 **Lab Number** 2007-00202- 158 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Rubbery
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Gray	1	<=1%	-	-	-	-	-
	Total %	100		**Average %**	<=1%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	vb/g	pb/r	1.556	1.549
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-7H3-159 **Lab Number** 2007-00202- 159 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Rubbery
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Gray	1	<=1%	-	-	-	-	-
	Total %	100		**Average %**	<=1%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	vb/g	pb/r	1.556	1.549
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-7I1-160 **Lab Number** 2007-00202- 160 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Insulation Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): binder, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	insulation	100	Yellow	3	90-100%	-	-	-	-	-
	Total %	100		**Average %**	90-100%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-7I2-161 **Lab Number** 2007-00202- 161 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Insulation Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): binder, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	insulation	100	Yellow	3	90-100%	-	-	-	-	-
	Total %	100		**Average %**	90-100%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-7I3-162 **Lab Number** 2007-00202- 162 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Insulation Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): binder, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	insulation	100	Yellow	3	90-100%	-	-	-	-	-
	Total %	100		**Average %**	90-100%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid.

Sample IHS-SF2000-7J1-163 **Lab Number** 2007-00202- 163 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): bitumen, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	100	Black	1	2-5%	-	-	-	-	-
	Total %	100		**Average %**	2-5%	-	-	-	-	-
				Fiber Identification:	glass fiber					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	glass fiber	CL	D	Y									
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-7J2-164 **Lab Number** 2007-00202- 164 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): bitumen, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	100	Black	1	2-5%	-	-	-	-	-
	Total %	100		**Average %**	2-5%	-	-	-	-	-
				Fiber Identification:	glass fiber					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	glass fiber	CL	D	Y									
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-7J3-165 **Lab Number** 2007-00202- 165 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): bitumen, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	100	Black	1	2-5%	-	-	-	-	-
	Total %	100		**Average %**	2-5%	-	-	-	-	-
				Fiber Identification:	glass fiber					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	glass fiber	CL	D	Y									
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-7K1-166 **Lab Number** 2007-00202- 166 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Insulation Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): polymer foam, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	foam	100	Yellow	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps.

Sample IHS-SF2000-7K2-167 **Lab Number** 2007-00202- 167 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Insulation Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): polymer foam, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	foam	100	Yellow	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps.

Sample IHS-SF2000-7K3-168 **Lab Number** 2007-00202- 168 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Insulation Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): polymer foam, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	foam	100	Yellow	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps.

Sample IHS-SF2000-7L1-169 **Lab Number** 2007-00202- 169 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Rubbery
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Tan	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-7L2-170 **Lab Number** 2007-00202- 170 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Rubbery
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Tan	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-7L3-171 **Lab Number** 2007-00202- 171 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Rubbery
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Tan	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-7M1-172 **Lab Number** 2007-00202- 172 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	white	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-7M2-173 **Lab Number** 2007-00202- 173 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	white	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-7M3-174 **Lab Number** 2007-00202- 174 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	white	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-7N1-175 **Lab Number** 2007-00202- 175 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Sticky
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Gray	1	5-10%	-	-	-	-	-
	Total %	100		**Average %**	5-10%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-7N2-176 **Lab Number** 2007-00202- 176 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Sticky
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Gray	1	5-10%	-	-	-	-	-
	Total %	100		**Average %**	5-10%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-7N3-177 **Lab Number** 2007-00202- 177 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Sticky
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Gray	1	5-10%	-	-	-	-	-
	Total %	100		**Average %**	5-10%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-7O1-178 **Lab Number** 2007-00202- 178 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Sticky
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Black	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-7O2-179 **Lab Number** 2007-00202- 179 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Sticky
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Black	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-7O3-180 **Lab Number** 2007-00202- 180 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Sticky
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Black	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-8A1-181 **Lab Number** 2007-00202- 181 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Roofing Fibrous Mat
Homogeneous No **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): bitumen, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	roof ply/bitumen	100	Black	1	30-40%	30-40%	-	-	-	-
	Total %	100		**Average %**	30-40%	30-40%	-	-	-	-
				Fiber Identification:	chrysotile asbestos	cellulose fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2	cellulose fiber	W	F	N	N	H	+	U					
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-8A2-182 **Lab Number** 2007-00202- 182 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Roofing Fibrous Mat
Homogeneous No **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): bitumen, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	roof ply/bitumen	100	Black	1	30-40%	30-40%	2-5%	-	-	-
	Total %	100		**Average %**	30-40%	30-40%	2-5%	-	-	-
				Fiber Identification:	chrysotile asbestos	cellulose fiber	glass fiber			

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2	cellulose fiber	W	F	N	N	H	+	U					
3	glass fiber	CL	D	Y									
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-8A3-183 **Lab Number** 2007-00202- 183 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Roofing Fibrous Mat
Homogeneous No **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): bitumen, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	roof ply/bitumen	100	Black	1	30-40%	30-40%	>1-2%	-	-	-
	Total %	100		**Average %**	30-40%	30-40%	>1-2%	-	-	-
				Fiber Identification:	chrysotile asbestos	cellulose fiber	glass fiber			

Fibers									Refractive Index Determinations				
		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2	cellulose fiber	W	F	N	N	H	+	U					
3	glass fiber	CL	D	Y									
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-8B1-184 **Lab Number** 2007-00202- 184 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Roofing Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): bitumen, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Black	1	5-10%	-	-	-	-	-
	Total %	100		**Average %**	5-10%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers									Refractive Index Determinations				
		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-8B2-185 **Lab Number** 2007-00202- 185 **Sampled:** 12/13/2006 **Condition:** acceptable
Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Roofing Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): bitumen, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Black	1	5-10%	-	-	-	-	-
	Total %	100		**Average %**	5-10%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers									Refractive Index Determinations				
		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2000-8B3-186 **Lab Number** 2007-00202- 186 **Sampled:** 12/13/2006 **Condition:** acceptable

Analyzed By RAM 1/27/2007 **An?** OK **Apparent Smp Type** Roofing Non-fibrous Solid

Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3

Non-Fibrous Components (in approx. decreasing order): bitumen, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Black	1	5-10%	-	-	-	-	-
	Total %	100		**Average %**	5-10%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	db/ly	sb/o	1.561	1.553
2													
3													
4													
5													
6													

Sample Analytical Note

Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Fr=Friability: 1=very non-friable; 2= non-friable; 3=friable; 4=highly friable
Colors: B=black;BL=blue;BR=brown;CL=clear;G=Green;GY=gray;OR=orange;OW=off-white;PN=pink;PU=purple;R=red;TN=tan;W=white;Y=yellow;V=various
Fiber Morphology: A=fine fibers/bundles, white, sinewy, flexible; B=fine fibers/bundles, w-br, straight, broomed ends; C=fine fibers/bundles, blue, straight, broomed ends; D=fine to coarse fibers, CL-B, brittle; E=coarse fibers,CL or dyed, striated; F=coarse fibers or splinters, W-BR, ribbon-like; G=lath-like or shards, low aspect ratio, may taper
Iso=isotropism - may be yes or no; Pleo=pleochroism - may be yes or no; Bi=birefringence - may be None, Low, Medium or High
Elg=sign of elongation - may be + or -; Ext=extinction - may be Parallel, Oblique, None or Undulating; Oil=medium used to for dispersion staining
Col Par=dispersion staining colors parallel to the fiber (fiber/halo): b/w=black/white; dg/py=dark gray/pale yellow; vg/y=violet gray/yellow; db/ly=dark blue/lemon yellow; vb/g= vivid blue/gold; sb/o=sky blue/orange; pb/r=pale blue/red; gb/dr=gray blue/dark red; w/b=white/black. Col Perp=same only perpendicular to fiber.
RI Par=refractive index parallel to fiber; RI Perp=refractive index perpendicular to fiber

Analyst: ROBERT A. McCORMICK

Printed: 29-Jan-07

Original Print Date: 29-Jan-07

Larry S. Pierce, Approved Accreditation Signatory

FIBERQUANT ANALYTICAL SERVICES

Polarized Light Microscope (PLM) Analysis for Asbestos

JobNumber:	**200700080**

Client: **IHI ENVIRONMENTAL**

(PHOENIX OFFICE)
4527 N 16th ST STE 105
PHOENIX, AZ 85016-0000
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Samples: 50 PLM **Rec:** 1/4/2007 **Method:** Interim (EPA/600/M4-82-020) PLM analysis for asbestos in bulk smp
Client Job: 07P-3031A / IHS-SF2001 **PO Number:** 07P-3031A
Report Date: 1/23/2007 **Date Analyzed:** 1/23/2007 **Routing Number:** -

Method and Analysis Information: **Fiberquant Internal SOP:** PLMi

Each bulk sample is first dissected under a 7-30x magnification stereo-microscope. This examination is used to determine the general type of sample, how many and what type of layers it has, and initial estimates of fiber types and quantities. Second, liquid media mounts are made of each layer - such mounts may be of selected fibers (used solely for identification purposes) or may be representative of the layer as a whole (used for quantitation purposes). The mounts may be made in a synthetic Canadian balsam, one of several solvents, or in refractive index oils (media of known refractive index). Generally, a variety of different mounts are made: some optimized for fiber visibility, some optimized for fiber identification, and some optimized for fiber quantitation. The mounted slides are then examined at 50-400x magnification on a Nikon Labphot-pol microscope. Optical characteristics are used to identify each observed fiber type; the optical data are contained for each sample on its detail analysis sheet, attached.

Current EPA, NESHAP and OSHA regulations designate a result of <=1 % asbestos as "negative" and >1 % asbestos as "positive". Samples containing layers that have been determined to be "positive" may have to be handled differently during a renovation or demolition than samples whose layers have been determined to be "negative."

The method of fiber analysis and identification is the EPA Interim Method (EPA Method 600/M4-82-020). The method of fiber quantitation is an estimation technique in which the analysts quantitation is routinely calibrated by reference quantitation standards, and which has been shown to be equivalent in precision and accuracy to point counting. Friability is estimated for the purposes of deciding when to point count. Friabilities determined in the field take precedence over those determined in the laboratory. Those sample layers which are friable and estimated by the analyst to contain <= 1% asbestos are point counted using 400 points, and the results listed on the detail analysis sheet. Such point counting is required by NESHAP (National Emission Standards for Hazardous Air Polutants, Nov. 1990) in order to rely on analytical results that are <= 1%. The coefficient of variation for the estimation quantitation technique is 100% in the range 0-5%. This means that PLM analysis is not capable of conclusively determining whether a layer containing close to 1% asbestos is actually "positive" or "negative". For this reason, Fiberquant refers to results where asbestos was detected but <= 1% as "borderline negative", and results where asbestos was >1 % but <= 2% as "borderline positive" to indicate the uncertainty in assigning a "positive" or "negative" label. In the sample summary, a "ND" means that no asbestos was detected during the analysis. A "Tr" or "Trace" of asbestos is defined for our purposes as the detection of several asbestos asbestos fibers during the analysis; an amount essentially at the limit of detection for the method. The limit of detection (the smallest % of asbestos that can be detected) varies greatly depending on the matrix in which the asbestos is found. As little as 0.001% asbestos can be detected in favorable samples, while detection in unfavorable samples may approach the detection limit of 1% stated in the method. During the analysis, the analyst, for Fiberquant identification purposes only, determines the "apparent sample type" and "apparent layer types." It must be emphasized that these types are only what is apparent. Often, different materials appear similar or identical after sampling, so the analyst may assign a type other than what was sampled.

Floor tiles present a special problem for PLM asbestos analysis. Floor tile can contain chrysotile fibers so thin that they cannot be resolved by optical methods. In such a case, we may observe a percentage of asbestos which is lower than the actual percentage, or not observe asbestos at all when some is present. For this reason, floor tiles reported as negative should be confirmed to be negative using transmission electron microscope (TEM) analysis. Likewise, vermiculite insulation materials containing traces of asbestiform asbestos present a problem for routine PLM analysis - the amphiboles are sometimes present in trace amounts inhomogeneously distributed. We recommend a hydro-separation technique for such samples.

Vermiculite-containing samples may contain trace amounts of asbestiform amphibole that may or may not be detected during routine PLM analysis. For this reason, loose vermiculite samples reported as negative should be confirmed to contain no amphibole using hydroseparation techniques.

The samples were analyzed under the following ongoing quality assurance program: Blank samples are routinely analyzed to maintain contamination-free materials. Each analyst is a degreed geologist or mineralogist, and has also completed extensive training specific to asbestos analysis for 1-3 months before being allowed to analyze client samples. Qualitative reference samples are routinely analyzed to assure that analysts can identify asbestos and asbestos-look-alike fibers. Quantitative reference samples are routinely analyzed to calibrate and characterize the estimation procedure. Microscope alignment is checked each day. Refractive index oils are calibrated at least monthly. At least 10% of client samples are re-analyzed from scratch by a different analyst than the original, and any discrepancies are resolved for the sample and similar sample types before the results are reported. All quality checks performed for these samples were in control except as detailed in the "Analytical Notes" below. All analysts participate in interlab round robins and proficiency testing to assure competence. Fiberquant is accredited by NVLAP (#101031) for the analysis of bulk samples for asbestos using PLM. Accreditation does not imply endorsement by the EPA, any other United States governmental agency or any private agency or association. Each lab analysis refers only to the sample tested, and may not, due to the sampling process, be

representative of the material sampled. This report may not be reproduced except in full, without the approval of Fiberquant Analytical Services.

The "Analysis Results" below are the averages for all layers of each sample listed. The "+ Layer" column indicates whether any of the layers in the sample were >1% asbestos. Please refer to each detailed analysis sheet to determine which layer(s) of a sample was(were) positive.

Some results may have been calculated using client supplied data, such as volume or area sampled, for which Fiberquant assumes no liability for accuracy.

Job Analysis Notes:

Single layer sample analysis as per client request. Any material or layer other than that indicated on the chain of custody was not analyzed, even if a suspect material.

PLM Analysis Summary: **Job Number: 200700080** 07P-3031A / IHS-SF2001

Sample Number / Layer	Color	Lab Number / Apparent Layer Type *	Apparent Sample Type * / *Asbestos Results*	Positive Layer Yes or No
Sample **IHS-SF2001-1A1-1**		2007-00080- 1	Flooring	Positive Layer? No
Layer # 1	off-white	floor tile	*no asbestos detected*	
Layer # 2	Yellow	mastic	*no asbestos detected*	
Sample **IHS-SF2001-1A2-2**		2007-00080- 2	Flooring	Positive Layer? No
Layer # 1	off-white	floor tile	*no asbestos detected*	
Layer # 2	Yellow	mastic	*no asbestos detected*	
Sample **IHS-SF2001-1A3-3**		2007-00080- 3	Flooring	Positive Layer? No
Layer # 1	off-white	floor tile	*no asbestos detected*	
Layer # 2	Yellow	mastic	*no asbestos detected*	
Sample **IHS-SF2001-1B1-4**		2007-00080- 4	Adhesive/caulk	Positive Layer? No
Layer # 1	Green	mastic	*no asbestos detected*	
Sample **IHS-SF2001-1B2-5**		2007-00080- 5	Adhesive/caulk	Positive Layer? No
Layer # 1	Green	mastic	*no asbestos detected*	
Sample **IHS-SF2001-1B3-6**		2007-00080- 6	Adhesive/caulk	Positive Layer? No
Layer # 1	Green	mastic	*no asbestos detected*	
Sample **IHS-SF2001-1C1-7**		2007-00080- 7	Adhesive/caulk	Positive Layer? No
Layer # 1	Tan	mastic	*no asbestos detected*	
Sample **IHS-SF2001-1C2-8**		2007-00080- 8	Adhesive/caulk	Positive Layer? No
Layer # 1	Tan	mastic	*no asbestos detected*	
Sample **IHS-SF2001-1C3-9**		2007-00080- 9	Adhesive/caulk	Positive Layer? No
Layer # 1	Tan	mastic	*no asbestos detected*	
Sample **IHS-SF2001-1D1-10**		2007-00080- 10	Flooring	Positive Layer? No
Layer # 1	off-white	floor tile	*2-5% chrysotile asbestos*	
Layer # 2	Yellow	mastic	*no asbestos detected*	
Sample **IHS-SF2001-1D2-11**		2007-00080- 11	Flooring	Positive Layer? No
Layer # 1	off-white	floor tile	*2-5% chrysotile asbestos*	
Layer # 2	Yellow	mastic	*no asbestos detected*	
Sample **IHS-SF2001-1D3-12**		2007-00080- 12	Flooring	Positive Layer? No
Layer # 1	off-white	floor tile	*2-5% chrysotile asbestos*	
Layer # 2	Yellow	mastic	*no asbestos detected*	
Sample **IHS-SF2001-1E1-13**		2007-00080- 13	Flooring	Positive Layer? No
Layer # 1	Blue	sheet flooring	*no asbestos detected*	
Sample **IHS-SF2001-1E2-14**		2007-00080- 14	Flooring	Positive Layer? No
Layer # 1	Blue	sheet flooring	*no asbestos detected*	
Layer # 2	Yellow	mastic	*no asbestos detected*	
Sample **IHS-SF2001-1E3-15**		2007-00080- 15	Flooring	Positive Layer? No
Layer # 1	Blue	sheet flooring	*no asbestos detected*	
Layer # 2	Yellow	mastic	*no asbestos detected*	
Sample **IHS-SF2001-2A1-16**		2007-00080- 16	Wall System	Positive Layer? No
Layer # 1	White	powder	*no asbestos detected*	
Sample **IHS-SF2001-2A2-17**		2007-00080- 17	Wall System	Positive Layer? No
Layer # 1	White	powder	*no asbestos detected*	
Sample **IHS-SF2001-2A3-18**		2007-00080- 18	Wall System	Positive Layer? No
Layer # 1	White	powder	*no asbestos detected*	
Sample **IHS-SF2001-2B1-19**		2007-00080- 19	Wall System	Positive Layer? No
Layer # 1	white	texture/joint compound	*no asbestos detected*	
Sample **IHS-SF2001-2B2-20**		2007-00080- 20	Wall System	Positive Layer? No
Layer # 1	white	texture/joint compound	*no asbestos detected*	
Sample **IHS-SF2001-2B3-21**		2007-00080- 21	Wall System	Positive Layer? No
Layer # 1	white	texture/joint compound	*no asbestos detected*	
Sample **IHS-SF2001-2C1-22**		2007-00080- 22	Wall System	Positive Layer? No
Layer # 1	Off-white	paint	*no asbestos detected*	
Sample **IHS-SF2001-2C2-23**		2007-00080- 23	Wall System	Positive Layer? No
Layer # 1	white	texture/joint compound	*no asbestos detected*	
Sample **IHS-SF2001-2C3-24**		2007-00080- 24	Wall System	Positive Layer? No
Layer # 1	white	texture/joint compound	*no asbestos detected*	
Sample **IHS-SF2001-2C4-25**		2007-00080- 25	Wall System	Positive Layer? No
Layer # 1	Off-white	paint	*no asbestos detected*	

Sample	**IHS-SF2001-2C5-26**	2007-00080- 26	Wall System	Positive Layer?	No
	Layer # 1 white texture/joint compound		*no asbestos detected*		
Sample	**IHS-SF2001-2D1-27**	2007-00080- 27	Wall System	Positive Layer?	No
	Layer # 1 white texture/joint compound		*no asbestos detected*		
Sample	**IHS-SF2001-2D2-28**	2007-00080- 28	Wall System	Positive Layer?	No
	Layer # 1 white texture/joint compound		*no asbestos detected*		
Sample	**IHS-SF2001-2D3-29**	2007-00080- 29	Wall System	Positive Layer?	No
	Layer # 1 white texture/joint compound		*no asbestos detected*		
Sample	**IHS-SF2001-3A1-30**	2007-00080- 30	Acoustic Tile	Positive Layer?	No
	Layer # 1 Tan acoustical tile		*no asbestos detected*		
Sample	**IHS-SF2001-3A2-31**	2007-00080- 31	Acoustic Tile	Positive Layer?	No
	Layer # 1 Tan acoustical tile		*no asbestos detected*		
Sample	**IHS-SF2001-3A3-32**	2007-00080- 32	Acoustic Tile	Positive Layer?	No
	Layer # 1 Tan acoustical tile		*no asbestos detected*		
Sample	**IHS-SF2001-3B1-33**	2007-00080- 33	Adhesive/caulk	Positive Layer?	No
	Layer # 1 Brown mastic		*no asbestos detected*		
Sample	**IHS-SF2001-3B2-34**	2007-00080- 34	Adhesive/caulk	Positive Layer?	No
	Layer # 1 Brown mastic		*no asbestos detected*		
Sample	**IHS-SF2001-3B3-35**	2007-00080- 35	Adhesive/caulk	Positive Layer?	No
	Layer # 1 Brown mastic		*no asbestos detected*		
Sample	**IHS-SF2001-3C1-36**	2007-00080- 36	Acoustic Tile	Positive Layer?	No
	Layer # 1 Yellow acoustical tile		*no asbestos detected*		
Sample	**IHS-SF2001-3C2-37**	2007-00080- 37	Acoustic Tile	Positive Layer?	No
	Layer # 1 Yellow acoustical tile		*no asbestos detected*		
Sample	**IHS-SF2001-3C3-38**	2007-00080- 38	Acoustic Tile	Positive Layer?	No
	Layer # 1 Yellow acoustical tile		*no asbestos detected*		
Sample	**IHS-SF2001-7A1-39**	2007-00080- 39	Adhesive/caulk	Positive Layer?	No
	Layer # 1 Yellow mastic		*no asbestos detected*		
Sample	**IHS-SF2001-7A2-40**	2007-00080- 40	Adhesive/caulk	Positive Layer?	No
	Layer # 1 Yellow mastic		*no asbestos detected*		
Sample	**IHS-SF2001-7A3-41**	2007-00080- 41	Adhesive/caulk	Positive Layer?	No
	Layer # 1 Yellow mastic		*no asbestos detected*		
Sample	**IHS-SF2001-7B1-42**	2007-00080- 42	Adhesive/caulk	Positive Layer?	No
	Layer # 1 white caulk		*no asbestos detected*		
Sample	**IHS-SF2001-7B2-43**	2007-00080- 43	Adhesive/caulk	Positive Layer?	No
	Layer # 1 white caulk		*no asbestos detected*		
Sample	**IHS-SF2001-7B3-44**	2007-00080- 44	Adhesive/caulk	Positive Layer?	No
	Layer # 1 white caulk		*no asbestos detected*		
Sample	**IHS-SF2001-7C1-45**	2007-00080- 45	Roofing	Positive Layer?	No
	Layer # 1 Black caulk		*no asbestos detected*		
Sample	**IHS-SF2001-7C2-46**	2007-00080- 46	Roofing	Positive Layer?	No
	Layer # 1 Black caulk		*no asbestos detected*		
Sample	**IHS-SF2001-7C3-47**	2007-00080- 47	Roofing	Positive Layer?	No
	Layer # 1 Black caulk		*no asbestos detected*		
Sample	**IHS-SF2001-8A1-48**	2007-00080- 48	Roofing	Positive Layer?	Yes
	Layer # 1 Black caulk		*5-10% chrysotile asbestos*		
Sample	**IHS-SF2001-8A2-49**	2007-00080- 49	Roofing	Positive Layer?	Yes
	Layer # 1 Black caulk		*5-10% chrysotile asbestos*		
Sample	**IHS-SF2001-8A3-50**	2007-00080- 50	Roofing	Positive Layer?	Yes
	Layer # 1 Black caulk		*5-10% chrysotile asbestos*		

* Apparent Sample Types and Apparent Layer Types are as they appeared to the analyst. Since many types of materials appear similar after sampling damage, the apparent type of material may not be the actual type of material.

Sample IHS-SF2001-1A1-1 **Lab Number** 2007-00080- 1 **Sampled:** 12/15/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 6
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	floor tile	98	off-white	1	n.d.	-	-	-	-	-
2	mastic	2	Yellow	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2001-1A2-2 **Lab Number** 2007-00080- 2 **Sampled:** 12/15/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 6
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	floor tile	98	off-white	1	n.d.	-	-	-	-	-
2	mastic	2	Yellow	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2001-1A3-3 **Lab Number** 2007-00080- 3 **Sampled:** 12/15/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 6
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	floor tile	99	off-white	1	n.d.	-	-	-	-	-
2	mastic	1	Yellow	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2001-1B1-4 **Lab Number** 2007-00080- 4 **Sampled:** 12/15/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Sticky
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	100	Green	1	<=1%	<=1%	-	-	-	-
	Total %	100		**Average %**	<=1%	<=1%	-	-	-	-
				Fiber Identification:	synthetic fiber (extr	cellulose fiber				

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	synthetic fiber (extruded)	W	E	N	N	H	+	P					
2	cellulose fiber	W	F	N	N	H	+	U					
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2001-1B2-5 **Lab Number** 2007-00080- 5 **Sampled:** 12/15/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Sticky
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	100	Green	1	<=1%	<=1%	-	-	-	-
	Total %	100		**Average %**	<=1%	<=1%	-	-	-	-
				Fiber Identification:	synthetic fiber (extr	cellulose fiber				

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	synthetic fiber (extruded)	W	E	N	N	H	+	P					
2	cellulose fiber	W	F	N	N	H	+	U					
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2001-1B3-6 **Lab Number** 2007-00080- 6 **Sampled:** 12/15/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Sticky
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	100	Green	1	<=1%	<=1%	-	-	-	-
	Total %	100		**Average %**	<=1%	<=1%	-	-	-	-
				Fiber Identification:	synthetic fiber (extr	cellulose fiber				

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	synthetic fiber (extruded)	W	E	N	N	H	+	P					
2	cellulose fiber	W	F	N	N	H	+	U					
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2001-1C1-7 **Lab Number** 2007-00080- 7 **Sampled:** 12/15/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Sticky
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	100	Tan	1	<=1%	-	-	-	-	-
	Total %	100		**Average %**	<=1%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2001-1C2-8 **Lab Number** 2007-00080- 8 **Sampled:** 12/15/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Sticky
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	100	Tan	1	<=1%	<=1%	-	-	-	-
	Total %	100		**Average %**	<=1%	<=1%	-	-	-	-
				Fiber Identification:	cellulose fiber	synthetic fiber (extr				

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2	synthetic fiber (extruded)	W	E	N	N	H	+	P					
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2001-1C3-9 **Lab Number** 2007-00080- 9 **Sampled:** 12/15/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Sticky
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	100	Tan	1	<=1%	-	-	-	-	-
	Total %	100		**Average %**	<=1%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2001-1D1-10 **Lab Number** 2007-00080- 10 **Sampled:** 12/15/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid
Homogeneous Yes **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 5
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	floor tile	99	off-white	1	2-5%	n.d.	-	-	-	-
2	mastic	1	Yellow	1	n.d.	2-5%	-	-	-	-
	Total %	100		**Average %**	2-5%	<=1%	-	-	-	-
				Fiber Identification:	chrysotile asbestos	cellulose fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	vb/g	sb/o	1.556	1.553
2	cellulose fiber	W	F	N	N	H	+	U					
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2001-1D2-11 **Lab Number** 2007-00080- 11 **Sampled:** 12/15/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid
Homogeneous Yes **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 5
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	floor tile	99	off-white	1	2-5%	n.d.	-	-	-	-
2	mastic	1	Yellow	1	n.d.	2-5%	-	-	-	-
	Total %	100		**Average %**	2-5%	<=1%	-	-	-	-
				Fiber Identification:	chrysotile asbestos	cellulose fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	vb/g	sb/o	1.556	1.553
2	cellulose fiber	W	F	N	N	H	+	U					
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2001-1D3-12 **Lab Number** 2007-00080- 12 **Sampled:** 12/15/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Flooring Non-fibrous Solid
Homogeneous Yes **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 5
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	floor tile	99	off-white	1	2-5%	n.d.	-	-	-	-
2	mastic	1	Yellow	1	n.d.	2-5%	-	-	-	-
	Total %	100		**Average %**	2-5%	<=1%	-	-	-	-
				Fiber Identification:	chrysotile asbestos	cellulose fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	vb/g	sb/o	1.556	1.553
2	cellulose fiber	W	F	N	N	H	+	U					
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2001-1E1-13 **Lab Number** 2007-00080- 13 **Sampled:** 12/15/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Flooring Fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	sheet flooring	100	Blue	2	10-20%	2-5%	-	-	-	-
	Total %	100		**Average %**	10-20%	2-5%	-	-	-	-
				Fiber Identification:	cellulose fiber	glass fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2	glass fiber	CL	D	Y									
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2001-1E2-14 **Lab Number** 2007-00080- 14 **Sampled:** 12/15/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Flooring Fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 6
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	sheet flooring	98	Blue	2	10-20%	2-5%	-	-	-	-
2	mastic	2	Yellow	1	2-5%	n.d.	-	-	-	-
	Total %	100		**Average %**	10-20%	2-5%	-	-	-	-
				Fiber Identification:	cellulose fiber	glass fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2	glass fiber	CL	D	Y									
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2001-1E3-15 **Lab Number** 2007-00080- 15 **Sampled:** 12/15/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Flooring Fibrous Solid
Homogeneous No **# Layers** 2 **Pos Layer?** No **# Sub-Samples** 6
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	sheet flooring	97	Blue	2	10-20%	2-5%	-	-	-	-
2	mastic	3	Yellow	1	2-5%	n.d.	-	-	-	-
	Total %	100		**Average %**	10-20%	2-5%	-	-	-	-
				Fiber Identification:	cellulose fiber	glass fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2	glass fiber	CL	D	Y									
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2001-2A1-16 **Lab Number** 2007-00080- 16 **Sampled:** 12/15/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Powder
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	powder	100	White	4	>1-2%	-	-	-	-	-
	Total %	100		**Average %**	>1-2%	-	-	-	-	-
				Fiber Identification:	cellulose					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2001-2A2-17 **Lab Number** 2007-00080- 17 **Sampled:** 12/15/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Powder
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	powder	100	White	4	>1-2%	-	-	-	-	-
	Total %	100		**Average %**	>1-2%	-	-	-	-	-
				Fiber Identification:	cellulose					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2001-2A3-18 **Lab Number** 2007-00080- 18 **Sampled:** 12/15/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Powder
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	powder	100	White	4	>1-2%	-	-	-	-	-
	Total %	100		**Average %**	>1-2%	-	-	-	-	-
				Fiber Identification:	cellulose					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2001-2B1-19 **Lab Number** 2007-00080- 19 **Sampled:** 12/15/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	texture/joint compound	100	white	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2001-2B2-20 **Lab Number** 2007-00080- 20 **Sampled:** 12/15/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	texture/joint compound	100	white	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2001-2B3-21 **Lab Number** 2007-00080- 21 **Sampled:** 12/15/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	texture/joint compound	100	white	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2001-2C1-22 **Lab Number** 2007-00080- 22 **Sampled:** 12/15/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	paint	100	Off-white	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent. The sample contained only paint, there was no texture/joint compound.

Sample IHS-SF2001-2C2-23 **Lab Number** 2007-00080- 23 **Sampled:** 12/15/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	texture/joint compound	100	white	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2001-2C3-24 **Lab Number** 2007-00080- 24 **Sampled:** 12/15/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	texture/joint compound	100	white	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2001-2C4-25 **Lab Number** 2007-00080- 25 **Sampled:** 12/15/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	paint	100	Off-white	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent. The sample contained only paint, there was no texture/joint compound.

Sample IHS-SF2001-2C5-26 **Lab Number** 2007-00080- 26 **Sampled:** 12/15/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	texture/joint compound	100	white	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2001-2D1-27 **Lab Number** 2007-00080- 27 **Sampled:** 12/15/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	texture/joint compound	100	white	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2001-2D2-28 **Lab Number** 2007-00080- 28 **Sampled:** 12/15/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	texture/joint compound	100	white	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2001-2D3-29 **Lab Number** 2007-00080- 29 **Sampled:** 12/15/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	texture/joint compound	100	white	3	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using dilute HCl acid. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2001-3A1-30 **Lab Number** 2007-00080- 30 **Sampled:** 12/15/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Acoustic Tile Fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): glass, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	acoustical tile	100	Tan	3	20-30%	40-50%	-	-	-	-
	Total %	100		**Average %**	20-30%	40-50%	-	-	-	-
				Fiber Identification:	cellulose fiber	glass fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2	glass fiber	CL	D	Y									
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2001-3A2-31 **Lab Number** 2007-00080- 31 **Sampled:** 12/15/2006 **Condition:** acceptable

Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Acoustic Tile Fibrous Solid

Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3

Non-Fibrous Components (in approx. decreasing order): glass, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	acoustical tile	100	Tan	3	20-30%	40-50%	-	-	-	-
	Total %	100		**Average %**	20-30%	40-50%	-	-	-	-
				Fiber Identification:	cellulose fiber	glass fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2	glass fiber	CL	D	Y									
3													
4													
5													
6													

Sample Analytical Note

Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2001-3A3-32 **Lab Number** 2007-00080- 32 **Sampled:** 12/15/2006 **Condition:** acceptable

Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Acoustic Tile Fibrous Solid

Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3

Non-Fibrous Components (in approx. decreasing order): glass, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	acoustical tile	100	Tan	3	20-30%	40-50%	-	-	-	-
	Total %	100		**Average %**	20-30%	40-50%	-	-	-	-
				Fiber Identification:	cellulose fiber	glass fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2	glass fiber	CL	D	Y									
3													
4													
5													
6													

Sample Analytical Note

Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2001-3B1-33 **Lab Number** 2007-00080- 33 **Sampled:** 12/15/2006 **Condition:** acceptable

Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid

Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3

Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	100	Brown	2	<=1%	<=1%	-	-	-	-
	Total %	100		**Average %**	<=1%	<=1%	-	-	-	-
				Fiber Identification:	talc and transitional	non-fibrous tremolit				

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	talc and transitional talc fiber	W	B	N	N	H	+	P	1.605	pb/r	w/b	1.603	<1.60
2	non-fibrous tremolite/actinolite	W	G	N	N	M	+	O	1.605	db/ly	sb/o	1.613	1.607
3													
4													
5													
6													

Sample Analytical Note

Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent. Note: sample contained transitional talc and/or non-fibrous tremolite/actinolite. Transitional talc is an intimate mixture of anthophyllite and talc. Even though it may have been asbestos once, the EPA has ruled not to regulate 'fibers of mixed mineral assemblege'. However, these fibers may be regulated when analyzed by TEM, since some are indistinguishable from anthophyllite asbestos. Likewise, non-fibrous tremolite/actinolite, while not regulated by EPA, may be counted (and thus regulated) when analyzed by TEM.

Sample IHS-SF2001-3B2-34 **Lab Number** 2007-00080- 34 **Sampled:** 12/15/2006 **Condition:** acceptable

Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid

Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3

Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	100	Brown	2	<=1%	<=1%	-	-	-	-
	Total %	100		**Average %**	<=1%	<=1%	-	-	-	-
				Fiber Identification:	talc and transitional	non-fibrous tremolit				

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	talc and transitional talc fiber	W	B	N	N	H	+	P	1.605	pb/r	w/b	1.603	<1.60
2	non-fibrous tremolite/actinolite	W	G	N	N	M	+	O	1.605	db/ly	sb/o	1.613	1.607
3													
4													
5													
6													

Sample Analytical Note

Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent. Note: sample contained transitional talc and/or non-fibrous tremolite/actinolite. Transitional talc is an intimate mixture of anthophyllite and talc. Even though it may have been asbestos once, the EPA has ruled not to regulate 'fibers of mixed mineral assemblege'. However, these fibers may be regulated when analyzed by TEM, since some are indistinguishable from anthophyllite asbestos. Likewise, non-fibrous tremolite/actinolite, while not regulated by EPA, may be counted (and thus regulated) when analyzed by TEM.

Sample IHS-SF2001-3B3-35 **Lab Number** 2007-00080- 35 **Sampled:** 12/15/2006 **Condition:** acceptable

Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid

Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3

Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	100	Brown	2	<=1%	<=1%	-	-	-	-
	Total %	100		**Average %**	<=1%	<=1%	-	-	-	-
				Fiber Identification:	talc and transitional	non-fibrous tremolit				

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	talc and transitional talc fiber	W	B	N	N	H	+	P	1.605	pb/r	w/b	1.603	<1.60
2	non-fibrous tremolite/actinolite	W	G	N	N	M	+	O	1.605	db/ly	sb/o	1.613	1.607
3													
4													
5													
6													

Sample Analytical Note

Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent. Note: sample contained transitional talc and/or non-fibrous tremolite/actinolite. Transitional talc is an intimate mixture of anthophyllite and talc. Even though it may have been asbestos once, the EPA has ruled not to regulate 'fibers of mixed mineral assemblege'. However, these fibers may be regulated when analyzed by TEM, since some are indistinguishable from anthophyllite asbestos. Likewise, non-fibrous tremolite/actinolite, while not regulated by EPA, may be counted (and thus regulated) when analyzed by TEM.

Sample IHS-SF2001-3C1-36 **Lab Number** 2007-00080- 36 **Sampled:** 12/15/2006 **Condition:** acceptable

Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Acoustic Tile Fibrous Solid

Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3

Non-Fibrous Components (in approx. decreasing order): binder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	acoustical tile	100	Yellow	3	90-100%	-	-	-	-	-
	Total %	100		**Average %**	90-100%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note

Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2001-3C2-37 **Lab Number** 2007-00080- 37 **Sampled:** 12/15/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Acoustic Tile Fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): binder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	acoustical tile	100	Yellow	3	90-100%	-	-	-	-	-
	Total %	100		**Average %**	90-100%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2001-3C3-38 **Lab Number** 2007-00080- 38 **Sampled:** 12/15/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Acoustic Tile Fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): binder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	acoustical tile	100	Yellow	3	90-100%	-	-	-	-	-
	Total %	100		**Average %**	90-100%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2001-7A1-39 **Lab Number** 2007-00080- 39 **Sampled:** 12/15/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	100	Yellow	1	<=1%	-	-	-	-	-
	Total %	100		**Average %**	<=1%	-	-	-	-	-
				Fiber Identification:	glass fiber					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	glass fiber	CL	D	Y									
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2001-7A2-40 **Lab Number** 2007-00080- 40 **Sampled:** 12/15/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	100	Yellow	1	<=1%	-	-	-	-	-
	Total %	100		**Average %**	<=1%	-	-	-	-	-
				Fiber Identification:	glass fiber					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	glass fiber	CL	D	Y									
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2001-7A3-41 **Lab Number** 2007-00080- 41 **Sampled:** 12/15/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	100	Yellow	1	<=1%	-	-	-	-	-
	Total %	100		**Average %**	<=1%	-	-	-	-	-
				Fiber Identification:	glass fiber					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	glass fiber	CL	D	Y									
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2001-7B1-42 **Lab Number** 2007-00080- 42 **Sampled:** 12/15/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Rubbery
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	white	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2001-7B2-43 **Lab Number** 2007-00080- 43 **Sampled:** 12/15/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Rubbery
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	white	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2001-7B3-44 **Lab Number** 2007-00080- 44 **Sampled:** 12/15/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Rubbery
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	white	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2001-7C1-45 **Lab Number** 2007-00080- 45 **Sampled:** 12/15/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Roofing Sticky
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): bitumen, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Black	1	5-10%	-	-	-	-	-
	Total %	100		**Average %**	5-10%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2001-7C2-46 **Lab Number** 2007-00080- 46 **Sampled:** 12/15/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Roofing Sticky
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): bitumen, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Black	1	5-10%	-	-	-	-	-
	Total %	100		**Average %**	5-10%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2001-7C3-47 **Lab Number** 2007-00080- 47 **Sampled:** 12/15/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Roofing Sticky
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): bitumen, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Black	1	5-10%	-	-	-	-	-
	Total %	100		**Average %**	5-10%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2001-8A1-48 **Lab Number** 2007-00080- 48 **Sampled:** 12/15/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Roofing Sticky
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): bitumen, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Black	1	5-10%	-	-	-	-	-
	Total %	100		**Average %**	5-10%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	vb/g	sb/o	1.557	1.553
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2001-8A2-49 **Lab Number** 2007-00080- 49 **Sampled:** 12/15/2006 **Condition:** acceptable

Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Roofing Sticky

Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3

Non-Fibrous Components (in approx. decreasing order): bitumen, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Black	1	5-10%	-	-	-	-	-
	Total %	100		**Average %**	5-10%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	vb/g	sb/o	1.557	1.553
2													
3													
4													
5													
6													

Sample Analytical Note

Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2001-8A3-50 **Lab Number** 2007-00080- 50 **Sampled:** 12/15/2006 **Condition:** acceptable

Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Roofing Sticky

Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3

Non-Fibrous Components (in approx. decreasing order): bitumen, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Black	1	5-10%	-	-	-	-	-
	Total %	100		**Average %**	5-10%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	vb/g	sb/o	1.557	1.553
2													
3													
4													
5													
6													

Sample Analytical Note

Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Fr=Friability: 1=very non-friable; 2= non-friable; 3=friable; 4=highly friable
Colors: B=black;BL=blue;BR=brown;CL=clear;G=Green;GY=gray;OR=orange;OW=off-white;PN=pink;PU=purple;R=red;TN=tan;W=white;Y=yellow;V=various
Fiber Morphology: A=fine fibers/bundles, white, sinewy, flexible; B=fine fibers/bundles, w-br, straight, broomed ends; C=fine fibers/bundles, blue, straight, broomed ends; D=fine to coarse fibers, CL-B, brittle; E=coarse fibers,CL or dyed, striated; F=coarse fibers or splinters, W-BR, ribbon-like; G=lath-like or shards, low aspect ratio, may taper
Iso=isotropism - may be yes or no; Pleo=pleochroism - may be yes or no; Bi=birefringence - may be None, Low, Medium or High
Elg=sign of elongation - may be + or -; Ext=extinction - may be Parallel, Oblique, None or Undulating; Oil=medium used to for dispersion staining
Col Par=dispersion staining colors parallel to the fiber (fiber/halo): b/w=black/white; dg/py=dark gray/pale yellow; vg/y=violet gray/yellow; db/ly=dark blue/lemon yellow; vb/g= vivid blue/gold; sb/o=sky blue/orange; pb/r=pale blue/red; gb/dr=gray blue/dark red; w/b=white/black. Col Perp=same only perpendicular to fiber.
RI Par=refractive index parallel to fiber; RI Perp=refractive index perpendicular to fiber

Analyst: UWE .. STEIMLE

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Larry S. Pierce, Approved Accreditation Signatory

FIBERQUANT ANALYTICAL SERVICES

Polarized Light Microscope (PLM) Analysis for Asbestos

JobNumber:	**200700079**

Client: **IHI ENVIRONMENTAL**

(PHOENIX OFFICE)
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PHOENIX, AZ 85016-0000
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Samples: 12 PLM **Rec:** 1/4/2007 **Method:** Interim (EPA/600/M4-82-020) PLM analysis for asbestos in bulk smp
Client Job: 07P-3031A / IHS-SF2002 **PO Number:** 07P-3031A
Report Date: 1/8/2007 **Date Analyzed:** 1/8/2007 **Routing Number:** -

Method and Analysis Information: **Fiberquant Internal SOP:** PLMi

Each bulk sample is first dissected under a 7-30x magnification stereo-microscope. This examination is used to determine the general type of sample, how many and what type of layers it has, and initial estimates of fiber types and quantities. Second, liquid media mounts are made of each layer - such mounts may be of selected fibers (used solely for identification purposes) or may be representative of the layer as a whole (used for quantitation purposes). The mounts may be made in a synthetic Canadian balsam, one of several solvents, or in refractive index oils (media of known refractive index). Generally, a variety of different mounts are made: some optimized for fiber visibility, some optimized for fiber identification, and some optimized for fiber quantitation. The mounted slides are then examined at 50-400x magnification on a Nikon Labphot-pol microscope. Optical characteristics are used to identify each observed fiber type; the optical data are contained for each sample on its detail analysis sheet, attached.

Current EPA, NESHAP and OSHA regulations designate a result of <=1 % asbestos as "negative" and >1 % asbestos as "positive". Samples containing layers that have been determined to be "positive" may have to be handled differently during a renovation or demolition than samples whose layers have been determined to be "negative."

The method of fiber analysis and identification is the EPA Interim Method (EPA Method 600/M4-82-020). The method of fiber quantitation is an estimation technique in which the analysts quantitation is routinely calibrated by reference quantitation standards, and which has been shown to be equivalent in precision and accuracy to point counting. Friability is estimated for the purposes of deciding when to point count. Friabilities determined in the field take precedence over those determined in the laboratory. Those sample layers which are friable and estimated by the analyst to contain <= 1% asbestos are point counted using 400 points, and the results listed on the detail analysis sheet. Such point counting is required by NESHAP (National Emission Standards for Hazardous Air Polutants, Nov. 1990) in order to rely on analytical results that are <= 1%. The coefficient of variation for the estimation quantitation technique is 100% in the range 0-5%. This means that PLM analysis is not capable of conclusively determining whether a layer containing close to 1% asbestos is actually "positive" or "negative". For this reason, Fiberquant refers to results where asbestos was detected but <= 1% as "borderline negative", and results where asbestos was >1 % but <= 2% as "borderline positive" to indicate the uncertainty in assigning a "positive" or "negative" label. In the sample summary, a "ND" means that no asbestos was detected during the analysis. A "Tr" or "Trace" of asbestos is defined for our purposes as the detection of several asbestos asbestos fibers during the analysis; an amount essentially at the limit of detection for the method. The limit of detection (the smallest % of asbestos that can be detected) varies greatly depending on the matrix in which the asbestos is found. As little as 0.001% asbestos can be detected in favorable samples, while detection in unfavorable samples may approach the detection limit of 1% stated in the method. During the analysis, the analyst, for Fiberquant identification purposes only, determines the "apparent sample type" and "apparent layer types." It must be emphasized that these types are only what is apparent. Often, different materials appear similar or identical after sampling, so the analyst may assign a type other than what was sampled.

Floor tiles present a special problem for PLM asbestos analysis. Floor tile can contain chrysotile fibers so thin that they cannot be resolved by optical methods. In such a case, we may observe a percentage of asbestos which is lower than the actual percentage, or not observe asbestos at all when some is present. For this reason, floor tiles reported as negative should be confirmed to be negative using transmission electron microscope (TEM) analysis. Likewise, vermiculite insulation materials containing traces of asbestiform asbestos present a problem for routine PLM analysis - the amphiboles are sometimes present in trace amounts inhomogeneously distributed. We recommend a hydro-separation technique for such samples.

Vermiculite-containing samples may contain trace amounts of asbestiform amphibole that may or may not be detected during routine PLM analysis. For this reason, loose vermiculite samples reported as negative should be confirmed to contain no amphibole using hydroseparation techniques.

The samples were analyzed under the following ongoing quality assurance program: Blank samples are routinely analyzed to maintain contamination-free materials. Each analyst is a degreed geologist or mineralogist, and has also completed extensive training specific to asbestos analysis for 1-3 months before being allowed to analyze client samples. Qualitative reference samples are routinely analyzed to assure that analysts can identify asbestos and asbestos-look-alike fibers. Quantitative reference samples are routinely analyzed to calibrate and characterize the estimation procedure. Microscope alignment is checked each day. Refractive index oils are calibrated at least monthly. At least 10% of client samples are re-analyzed from scratch by a different analyst than the original, and any discrepancies are resolved for the sample and similar sample types before the results are reported. All quality checks performed for these samples were in control except as detailed in the "Analytical Notes" below. All analysts participate in interlab round robins and proficiency testing to assure competence. Fiberquant is accredited by NVLAP (#101031) for the analysis of bulk samples for asbestos using PLM. Accreditation does not imply endorsement by the EPA, any other United States governmental agency or any private agency or association. Each lab analysis refers only to the sample tested, and may not, due to the sampling process, be

representative of the material sampled. This report may not be reproduced except in full, without the approval of Fiberquant Analytical Services.

The "Analysis Results" below are the averages for all layers of each sample listed. The "+ Layer" column indicates whether any of the layers in the sample were >1% asbestos. Please refer to each detailed analysis sheet to determine which layer(s) of a sample was(were) positive.

Some results may have been calculated using client supplied data, such as volume or area sampled, for which Fiberquant assumes no liability for accuracy.

Job Analysis Notes:

Single layer sample analysis as per client request. Any material or layer other than that indicated on the chain of custody was not analyzed, even if a suspect material.

PLM Analysis Summary: **Job Number: 200700079** 07P-3031A / IHS-SF2002

Sample Number / Layer	Color	Lab Number / Apparent Layer Type *	Apparent Sample Type * / *Asbestos Results*	Positive Layer Yes or No
Sample **IHS-SF2002-7A1-1** Layer # 1	Yellow insulation	2007-00079- 1	Insulation *no asbestos detected*	Positive Layer? No
Sample **IHS-SF2002-7A2-2** Layer # 1	Yellow insulation	2007-00079- 2	Insulation *no asbestos detected*	Positive Layer? No
Sample **IHS-SF2002-7A3-3** Layer # 1	Yellow insulation	2007-00079- 3	Insulation *no asbestos detected*	Positive Layer? No
Sample **IHS-SF2002-7B1-4** Layer # 1	White caulk	2007-00079- 4	Adhesive/caulk *2-5% chrysotile asbestos*	Positive Layer? Yes
Sample **IHS-SF2002-7B2-5** Layer # 1	White caulk	2007-00079- 5	Adhesive/caulk *2-5% chrysotile asbestos*	Positive Layer? Yes
Sample **IHS-SF2002-7B3-6** Layer # 1	White caulk	2007-00079- 6	Adhesive/caulk *2-5% chrysotile asbestos*	Positive Layer? Yes
Sample **IHS-SF2002-8A1-7** Layer # 1	Black caulk	2007-00079- 7	Adhesive/caulk *10-20% chrysotile asbestos*	Positive Layer? Yes
Sample **IHS-SF2002-8A2-8** Layer # 1	Black caulk	2007-00079- 8	Adhesive/caulk *10-20% chrysotile asbestos*	Positive Layer? Yes
Sample **IHS-SF2002-8A3-9** Layer # 1	Black caulk	2007-00079- 9	Adhesive/caulk *10-20% chrysotile asbestos*	Positive Layer? Yes
Sample **IHS-SF2002-8B1-10** Layer # 1	Gray caulk	2007-00079- 10	Adhesive/caulk *>1-2% chrysotile asbestos*	Positive Layer? Yes
Sample **IHS-SF2002-8B2-11** Layer # 1	Gray caulk	2007-00079- 11	Adhesive/caulk *>1-2% chrysotile asbestos*	Positive Layer? Yes
Sample **IHS-SF2002-8B3-12** Layer # 1	Gray caulk	2007-00079- 12	Adhesive/caulk *>1-2% chrysotile asbestos*	Positive Layer? Yes

* Apparent Sample Types and Apparent Layer Types are as they appeared to the analyst. Since many types of materials appear similar after sampling damage, the apparent type of material may not be the actual type of material.

Sample IHS-SF2002-7A1-1 **Lab Number** 2007-00079- 1 **Sampled:** 12/14/2006 **Condition:** acceptable
Analyzed By DMS 1/8/2007 **An?** OK **Apparent Smp Type** Insulation Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): binder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	insulation	100	Yellow	3	90-100%	-	-	-	-	-
	Total %	100		**Average %**	90-100%	-	-	-	-	-
				Fiber Identification:	glass fiber					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	glass fiber	Y	D	Y									
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps.

Sample IHS-SF2002-7A2-2 **Lab Number** 2007-00079- 2 **Sampled:** 12/14/2006 **Condition:** acceptable
Analyzed By DMS 1/8/2007 **An?** OK **Apparent Smp Type** Insulation Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): binder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	insulation	100	Yellow	3	90-100%	-	-	-	-	-
	Total %	100		**Average %**	90-100%	-	-	-	-	-
				Fiber Identification:	glass fiber					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	glass fiber	Y	D	Y									
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps.

Sample IHS-SF2002-7A3-3 **Lab Number** 2007-00079- 3 **Sampled:** 12/14/2006 **Condition:** acceptable
Analyzed By DMS 1/8/2007 **An?** OK **Apparent Smp Type** Insulation Fibrous Mat
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): binder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	insulation	100	Yellow	3	90-100%	-	-	-	-	-
	Total %	100		**Average %**	90-100%	-	-	-	-	-
				Fiber Identification:	glass fiber					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	glass fiber	Y	D	Y									
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps.

Sample IHS-SF2002-7B1-4 **Lab Number** 2007-00079- 4 **Sampled:** 12/14/2006 **Condition:** acceptable
Analyzed By DMS 1/8/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Sticky
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): binder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	White	1	2-5%	-	-	-	-	-
	Total %	100		**Average %**	2-5%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	vb/g	pb/r	1.556	1.549
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps.

Sample IHS-SF2002-7B2-5 **Lab Number** 2007-00079- 5 **Sampled:** 12/14/2006 **Condition:** acceptable
Analyzed By DMS 1/8/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Sticky
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): binder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	White	1	2-5%	-	-	-	-	-
	Total %	100		**Average %**	2-5%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	vb/g	pb/r	1.556	1.549
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps.

Sample IHS-SF2002-7B3-6 **Lab Number** 2007-00079- 6 **Sampled:** 12/14/2006 **Condition:** acceptable
Analyzed By DMS 1/8/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Sticky
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): binder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	White	1	2-5%	-	-	-	-	-
	Total %	100		**Average %**	2-5%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	vb/g	pb/r	1.556	1.549
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps.

Sample IHS-SF2002-8A1-7 **Lab Number** 2007-00079- 7 **Sampled:** 12/14/2006 **Condition:** acceptable
Analyzed By DMS 1/8/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Sticky
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): bitumen, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Black	1	10-20%	-	-	-	-	-
	Total %	100		**Average %**	10-20%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	vb/g	pb/r	1.556	1.549
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps.

Sample IHS-SF2002-8A2-8 **Lab Number** 2007-00079- 8 **Sampled:** 12/14/2006 **Condition:** acceptable
Analyzed By DMS 1/8/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Sticky
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): bitumen, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Black	1	10-20%	-	-	-	-	-
	Total %	100		**Average %**	10-20%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	vb/g	pb/r	1.556	1.549
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps.

Sample IHS-SF2002-8A3-9 **Lab Number** 2007-00079- 9 **Sampled:** 12/14/2006 **Condition:** acceptable
Analyzed By DMS 1/8/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Sticky
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): bitumen, filler,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Black	1	10-20%	-	-	-	-	-
	Total %	100		**Average %**	10-20%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	vb/g	pb/r	1.556	1.549
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps.

Sample IHS-SF2002-8B1-10 **Lab Number** 2007-00079- 10 **Sampled:** 12/14/2006 **Condition:** acceptable
Analyzed By DMS 1/8/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Sticky
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): binder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Gray	1	>1-2%	-	-	-	-	-
	Total %	100		**Average %**	>1-2%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	vb/g	pb/r	1.556	1.549
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps.

Sample IHS-SF2002-8B2-11 **Lab Number** 2007-00079- 11 **Sampled:** 12/14/2006 **Condition:** acceptable
Analyzed By DMS 1/8/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Sticky
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): binder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Gray	1	>1-2%	-	-	-	-	-
	Total %	100		**Average %**	>1-2%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	vb/g	pb/r	1.556	1.549
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps.

Sample IHS-SF2002-8B3-12 **Lab Number** 2007-00079- 12 **Sampled:** 12/14/2006 **Condition:** acceptable
Analyzed By DMS 1/8/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Sticky
Homogeneous Yes **# Layers** 1 **Pos Layer?** Yes **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): binder, ,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Gray	1	>1-2%	-	-	-	-	-
	Total %	100		**Average %**	>1-2%	-	-	-	-	-
				Fiber Identification:	chrysotile asbestos					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations Oil	Col Par	Col Per	RI Par	RI Per
1	chrysotile asbestos	W	A	N	N	L	+	P	1.550	vb/g	pb/r	1.556	1.549
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps.

Fr=Friability: 1=very non-friable; 2= non-friable; 3=friable; 4=highly friable
Colors: B=black;BL=blue;BR=brown;CL=clear;G=Green;GY=gray;OR=orange;OW=off-white;PN=pink;PU=purple;R=red;TN=tan;W=white;Y=yellow;V=various
Fiber Morphology: A=fine fibers/bundles, white, sinewy, flexible; B=fine fibers/bundles, w-br, straight, broomed ends; C=fine fibers/bundles, blue, straight, broomed ends; D=fine to coarse fibers, CL-B, brittle; E=coarse fibers,CL or dyed, striated; F=coarse fibers or splinters, W-BR, ribbon-like; G=lath-like or shards, low aspect ratio, may taper
Iso=isotropism - may be yes or no; Pleo=pleochroism - may be yes or no; Bi=birefringence - may be None, Low, Medium or High
Elg=sign of elongation - may be + or -; Ext=extinction - may be Parallel, Oblique, None or Undulating; Oil=medium used to for dispersion staining
Col Par=dispersion staining colors parallel to the fiber (fiber/halo): b/w=black/white; dg/py=dark gray/pale yellow; vg/y=violet gray/yellow; db/ly=dark blue/lemon yellow; vb/g= vivid blue/gold; sb/o=sky blue/orange; pb/r=pale blue/red; gb/dr=gray blue/dark red; w/b=white/black. Col Perp=same only perpendicular to fiber.
RI Par=refractive index parallel to fiber; RI Perp=refractive index perpendicular to fiber

Analyst: DAVID M. SCHALLER

Printed: 08-Jan-07

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Larry S. Pierce, Approved Accreditation Signatory

FIBERQUANT ANALYTICAL SERVICES

Polarized Light Microscope (PLM) Analysis for Asbestos

JobNumber: 200700083

Client: IHI ENVIRONMENTAL

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PHOENIX, AZ 85016-0000
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Samples: 33 PLM **Rec:** 1/4/2007 **Method:** Interim (EPA/600/M4-82-020) PLM analysis for asbestos in bulk smp
Client Job: 07P-3031A / IHS-SF2003 **PO Number:** 07P-3031A
Report Date: 1/23/2007 **Date Analyzed:** 1/23/2007 **Routing Number:** -

Method and Analysis Information: **Fiberquant Internal SOP:** PLMi

Each bulk sample is first dissected under a 7-30x magnification stereo-microscope. This examination is used to determine the general type of sample, how many and what type of layers it has, and initial estimates of fiber types and quantities. Second, liquid media mounts are made of each layer - such mounts may be of selected fibers (used solely for identification purposes) or may be representative of the layer as a whole (used for quantitation purposes). The mounts may be made in a synthetic Canadian balsam, one of several solvents, or in refractive index oils (media of known refractive index). Generally, a variety of different mounts are made: some optimized for fiber visibility, some optimized for fiber identification, and some optimized for fiber quantitation. The mounted slides are then examined at 50-400x magnification on a Nikon Labphot-pol microscope. Optical characteristics are used to identify each observed fiber type; the optical data are contained for each sample on its detail analysis sheet, attached.

Current EPA, NESHAP and OSHA regulations designate a result of <=1 % asbestos as "negative" and >1 % asbestos as "positive". Samples containing layers that have been determined to be "positive" may have to be handled differently during a renovation or demolition than samples whose layers have been determined to be "negative."

The method of fiber analysis and identification is the EPA Interim Method (EPA Method 600/M4-82-020). The method of fiber quantitation is an estimation technique in which the analysts quantitation is routinely calibrated by reference quantitation standards, and which has been shown to be equivalent in precision and accuracy to point counting. Friability is estimated for the purposes of deciding when to point count. Friabilities determined in the field take precedence over those determined in the laboratory. Those sample layers which are friable and estimated by the analyst to contain <= 1% asbestos are point counted using 400 points, and the results listed on the detail analysis sheet. Such point counting is required by NESHAP (National Emission Standards for Hazardous Air Polutants, Nov. 1990) in order to rely on analytical results that are <= 1%. The coefficient of variation for the estimation quantitation technique is 100% in the range 0-5%. This means that PLM analysis is not capable of conclusively determining whether a layer containing close to 1% asbestos is actually "positive" or "negative". For this reason, Fiberquant refers to results where asbestos was detected but <= 1% as "borderline negative", and results where asbestos was >1 % but <= 2% as "borderline positive" to indicate the uncertainty in assigning a "positive" or "negative" label. In the sample summary, a "ND" means that no asbestos was detected during the analysis. A "Tr" or "Trace" of asbestos is defined for our purposes as the detection of several asbestos asbestos fibers during the analysis; an amount essentially at the limit of detection for the method. The limit of detection (the smallest % of asbestos that can be detected) varies greatly depending on the matrix in which the asbestos is found. As little as 0.001% asbestos can be detected in favorable samples, while detection in unfavorabie samples may approach the detection limit of 1% stated in the method. During the analysis, the analyst, for Fiberquant identification purposes only, determines the "apparent sample type" and "apparent layer types." It must be emphasized that these types are only what is apparent. Often, different materials appear similar or identical after sampling, so the analyst may assign a type other than what was sampled.

Floor tiles present a special problem for PLM asbestos analysis. Floor tile can contain chrysotile fibers so thin that they cannot be resolved by optical methods. In such a case, we may observe a percentage of asbestos which is lower than the actual percentage, or not observe asbestos at all when some is present. For this reason, floor tiles reported as negative should be confirmed to be negative using transmission electron microscope (TEM) analysis. Likewise, vermiculite insulation materials containing traces of asbestiform asbestos present a problem for routine PLM analysis - the amphiboles are sometimes present in trace amounts inhomogeneously distributed. We recommend a hydro-separation technique for such samples.

Vermiculite-containing samples may contain trace amounts of asbestiform amphibole that may or may not be detected during routine PLM analysis. For this reason, loose vermiculite samples reported as negative should be confirmed to contain no amphibole using hydroseparation techniques.

The samples were analyzed under the following ongoing quality assurance program: Blank samples are routinely analyzed to maintain contamination-free materials. Each analyst is a degreed geologist or mineralogist, and has also completed extensive training specific to asbestos analysis for 1-3 months before being allowed to analyze client samples. Qualitative reference samples are routinely analyzed to assure that analysts can identify asbestos and asbestos-look-alike fibers. Quantitative reference samples are routinely analyzed to calibrate and characterize the estimation procedure. Microscope alignment is checked each day. Refractive index oils are calibrated at least monthly. At least 10% of client samples are re-analyzed from scratch by a different analyst than the original, and any discrepancies are resolved for the sample and similar sample types before the results are reported. All quality checks performed for these samples were in control except as detailed in the "Analytical Notes" below. All analysts participate in interlab round robins and proficiency testing to assure competence. Fiberquant is accredited by NVLAP (#101031) for the analysis of bulk samples for asbestos using PLM. Accreditation does not imply endorsement by the EPA, any other United States governmental agency or any private agency or association. Each lab analysis refers only to the sample tested, and may not, due to the sampling process, be

representative of the material sampled. This report may not be reproduced except in full, without the approval of Fiberquant Analytical Services.

The "Analysis Results" below are the averages for all layers of each sample listed. The "+ Layer" column indicates whether any of the layers in the sample were >1% asbestos. Please refer to each detailed analysis sheet to determine which layer(s) of a sample was(were) positive.

Some results may have been calculated using client supplied data, such as volume or area sampled, for which Fiberquant assumes no liability for accuracy.

Job Analysis Notes:

Single layer sample analysis as per client request. Any material or layer other than that indicated on the chain of custody was not analyzed, even if a suspect material.

PLM Analysis Summary: **Job Number: 200700083** 07P-3031A / IHS-SF2003

Sample Number / Layer / Color	Lab Number / Apparent Layer Type *	Apparent Sample Type * / Asbestos Results	Positive Layer Yes or No
Sample **IHS-SF2003-1A1-1** Layer # 1 Pink sheet flooring	2007-00083- 1	Flooring *no asbestos detected*	Positive Layer? No
Sample **IHS-SF2003-1A2-2** Layer # 1 Pink sheet flooring	2007-00083- 2	Flooring *no asbestos detected*	Positive Layer? No
Sample **IHS-SF2003-1A3-3** Layer # 1 Pink sheet flooring	2007-00083- 3	Flooring *no asbestos detected*	Positive Layer? No
Sample **IHS-SF2003-1B1-4** Layer # 1 Off-whit mastic	2007-00083- 4	Adhesive/caulk *no asbestos detected*	Positive Layer? No
Sample **IHS-SF2003-1B2-5** Layer # 1 Off-whit mastic	2007-00083- 5	Adhesive/caulk *no asbestos detected*	Positive Layer? No
Sample **IHS-SF2003-1B3-6** Layer # 1 Off-whit mastic	2007-00083- 6	Adhesive/caulk *no asbestos detected*	Positive Layer? No
Sample **IHS-SF2003-1C1-7** Layer # 1 Yellow sheet flooring	2007-00083- 7	Flooring *no asbestos detected*	Positive Layer? No
Sample **IHS-SF2003-1C2-8** Layer # 1 Yellow sheet flooring	2007-00083- 8	Flooring *no asbestos detected*	Positive Layer? No
Sample **IHS-SF2003-1C3-9** Layer # 1 Yellow sheet flooring	2007-00083- 9	Flooring *no asbestos detected*	Positive Layer? No
Sample **IHS-SF2003-1D1-10** Layer # 1 Blue sheet flooring	2007-00083- 10	Flooring *no asbestos detected*	Positive Layer? No
Sample **IHS-SF2003-1D2-11** Layer # 1 Blue sheet flooring	2007-00083- 11	Flooring *no asbestos detected*	Positive Layer? No
Sample **IHS-SF2003-1D3-12** Layer # 1 Blue sheet flooring	2007-00083- 12	Flooring *no asbestos detected*	Positive Layer? No
Sample **IHS-SF2003-1E1-13** Layer # 1 Green mastic	2007-00083- 13	Adhesive/caulk *no asbestos detected*	Positive Layer? No
Sample **IHS-SF2003-1E2-14** Layer # 1 Green mastic	2007-00083- 14	Adhesive/caulk *no asbestos detected*	Positive Layer? No
Sample **IHS-SF2003-1E3-15** Layer # 1 Green mastic	2007-00083- 15	Adhesive/caulk *no asbestos detected*	Positive Layer? No
Sample **IHS-SF2003-2A1-16** Layer # 1 Off-whit powder	2007-00083- 16	Wall System *no asbestos detected*	Positive Layer? No
Sample **IHS-SF2003-2A2-17** Layer # 1 Off-whit drywall core	2007-00083- 17	Wall System *no asbestos detected*	Positive Layer? No
Sample **IHS-SF2003-2A3-18** Layer # 1 Off-whit drywall core	2007-00083- 18	Wall System *no asbestos detected*	Positive Layer? No
Sample **IHS-SF2003-3A1-19** Layer # 1 Tan acoustical tile	2007-00083- 19	Acoustic Tile *no asbestos detected*	Positive Layer? No
Sample **IHS-SF2003-3A2-20** Layer # 1 Tan acoustical tile	2007-00083- 20	Acoustic Tile *no asbestos detected*	Positive Layer? No
Sample **IHS-SF2003-3A3-21** Layer # 1 Tan acoustical tile	2007-00083- 21	Acoustic Tile *no asbestos detected*	Positive Layer? No
Sample **IHS-SF2003-7A1-22** Layer # 1 Off-whit caulk	2007-00083- 22	Adhesive/caulk *no asbestos detected*	Positive Layer? No
Sample **IHS-SF2003-7A2-23** Layer # 1 Off-whit caulk	2007-00083- 23	Adhesive/caulk *no asbestos detected*	Positive Layer? No
Sample **IHS-SF2003-7A3-24** Layer # 1 White caulk	2007-00083- 24	Adhesive/caulk *no asbestos detected*	Positive Layer? No
Sample **IHS-SF2003-7B1-25** Layer # 1 Yellow mastic	2007-00083- 25	Adhesive/caulk *no asbestos detected*	Positive Layer? No
Sample **IHS-SF2003-7B2-26** Layer # 1 Off-whit mastic	2007-00083- 26	Adhesive/caulk *no asbestos detected*	Positive Layer? No
Sample **IHS-SF2003-7B3-27** Layer # 1 Off-whit mastic	2007-00083- 27	Adhesive/caulk *no asbestos detected*	Positive Layer? No
Sample **IHS-SF2003-7C1-28** Layer # 1 Off-whit coating	2007-00083- 28	Sprayed Material *no asbestos detected*	Positive Layer? No
Sample **IHS-SF2003-7C2-29** Layer # 1 Off-whit coating	2007-00083- 29	Sprayed Material *no asbestos detected*	Positive Layer? No

Sample	**IHS-SF2003-7C3-30**	2007-00083- 30	Sprayed Material	Positive Layer? No
	Layer # 1 Off-whit coating		*no asbestos detected*	
Sample	**IHS-SF2003-7D1-31**	2007-00083- 31	Sprayed Material	Positive Layer? No
	Layer # 1 Gray coating		*no asbestos detected*	
Sample	**IHS-SF2003-7D2-32**	2007-00083- 32	Sprayed Material	Positive Layer? No
	Layer # 1 Gray coating		*no asbestos detected*	
Sample	**IHS-SF2003-7D3-33**	2007-00083- 33	Sprayed Material	Positive Layer? No
	Layer # 1 Gray coating		*no asbestos detected*	

* Apparent Sample Types and Apparent Layer Types are as they appeared to the analyst. Since many types of materials appear similar after sampling damage, the apparent type of material may not be the actual type of material.

Sample IHS-SF2003-1A1-1 **Lab Number** 2007-00083- 1 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Flooring Fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): polymer, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	sheet flooring	100	Pink	2	10-20%	2-5%	-	-	-	-
	Total %	100		**Average %**	10-20%	2-5%	-	-	-	-
				Fiber Identification:	cellulose fiber	glass fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2	glass fiber	CL	D	Y									
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2003-1A2-2 **Lab Number** 2007-00083- 2 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Flooring Fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): polymer, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	sheet flooring	100	Pink	2	10-20%	2-5%	-	-	-	-
	Total %	100		**Average %**	10-20%	2-5%	-	-	-	-
				Fiber Identification:	cellulose fiber	glass fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2	glass fiber	CL	D	Y									
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2003-1A3-3 **Lab Number** 2007-00083- 3 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Flooring Fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): polymer, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	sheet flooring	100	Pink	2	5-10%	<=1%	-	-	-	-
	Total %	100		**Average %**	5-10%	<=1%	-	-	-	-
				Fiber Identification:	cellulose fiber	glass fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2	glass fiber	CL	D	Y									
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2003-1B1-4 **Lab Number** 2007-00083- 4 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Sticky
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	100	Off-white	1	<=1%	<=1%	-	-	-	-
	Total %	100		**Average %**	<=1%	<=1%	-	-	-	-
				Fiber Identification:	synthetic fiber (extr	cellulose fiber				

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	synthetic fiber (extruded)	W	E	N	N	H	+	P					
2	cellulose fiber	W	F	N	N	H	+	U					
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2003-1B2-5 **Lab Number** 2007-00083- 5 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Sticky
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	100	Off-white	1	<=1%	<=1%	-	-	-	-
	Total %	100		**Average %**	<=1%	<=1%	-	-	-	-
				Fiber Identification:	synthetic fiber (extr	cellulose fiber				

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	synthetic fiber (extruded)	W	E	N	N	H	+	P					
2	cellulose fiber	W	F	N	N	H	+	U					
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2003-1B3-6 **Lab Number** 2007-00083- 6 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Sticky
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber: Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	100	Off-white	1	<=1%	<=1%	-	-	-	-
	Total %	100		**Average %**	<=1%	<=1%	-	-	-	-
				Fiber Identification:	synthetic fiber (extr	cellulose fiber				

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations: Oil	Col Par	Col Per	RI Par	RI Per
1	synthetic fiber (extruded)	W	E	N	N	H	+	P					
2	cellulose fiber	W	F	N	N	H	+	U					
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2003-1C1-7 **Lab Number** 2007-00083- 7 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Flooring Fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): polymer, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	sheet flooring	100	Yellow	2	20-30%	2-5%	-	-	-	-
	Total %	100		**Average %**	20-30%	2-5%	-	-	-	-
				Fiber Identification:	cellulose fiber	glass fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2	glass fiber	CL	D	Y									
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2003-1C2-8 **Lab Number** 2007-00083- 8 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Flooring Fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): polymer, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	sheet flooring	100	Yellow	2	20-30%	2-5%	-	-	-	-
	Total %	100		**Average %**	20-30%	2-5%	-	-	-	-
				Fiber Identification:	cellulose fiber	glass fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2	glass fiber	CL	D	Y									
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2003-1C3-9 **Lab Number** 2007-00083- 9 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Flooring Fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): polymer, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	sheet flooring	100	Yellow	2	20-30%	2-5%	-	-	-	-
	Total %	100		**Average %**	20-30%	2-5%	-	-	-	-
				Fiber Identification:	cellulose fiber	glass fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2	glass fiber	CL	D	Y									
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2003-1D1-10 **Lab Number** 2007-00083- 10 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Flooring Fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): polymer, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	sheet flooring	100	Blue	2	20-30%	2-5%	-	-	-	-
	Total %	100		**Average %**	20-30%	2-5%	-	-	-	-
				Fiber Identification:	cellulose fiber	glass fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2	glass fiber	CL	D	Y									
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2003-1D2-11 **Lab Number** 2007-00083- 11 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Flooring Fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): polymer, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	sheet flooring	100	Blue	2	20-30%	2-5%	-	-	-	-
	Total %	100		**Average %**	20-30%	2-5%	-	-	-	-
				Fiber Identification:	cellulose fiber	glass fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2	glass fiber	CL	D	Y									
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2003-1D3-12 **Lab Number** 2007-00083- 12 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Flooring Fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): polymer, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	sheet flooring	100	Blue	2	20-30%	2-5%	-	-	-	-
	Total %	100		**Average %**	20-30%	2-5%	-	-	-	-
				Fiber Identification:	cellulose fiber	glass fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2	glass fiber	CL	D	Y									
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2003-1E1-13 **Lab Number** 2007-00083- 13 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Sticky
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	100	Green	1	>1-2%	<=1%	-	-	-	-
	Total %	100		**Average %**	>1-2%	<=1%	-	-	-	-
				Fiber Identification:	cellulose fiber	synthetic fiber (extr				

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2	synthetic fiber (extruded)	W	E	N	N	H	+	P					
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2003-1E2-14 **Lab Number** 2007-00083- 14 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Sticky
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	100	Green	1	>1-2%	<=1%	-	-	-	-
	Total %	100		**Average %**	>1-2%	<=1%	-	-	-	-
				Fiber Identification:	cellulose fiber	synthetic fiber (extr				

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2	synthetic fiber (extruded)	W	E	N	N	H	+	P					
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2003-1E3-15 **Lab Number** 2007-00083- 15 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Sticky
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	mastic	100	Green	1	>1-2%	<=1%	-	-	-	-
	Total %	100		**Average %**	>1-2%	<=1%	-	-	-	-
				Fiber Identification:	cellulose fiber	synthetic fiber (extr				

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2	synthetic fiber (extruded)	W	E	N	N	H	+	P					
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2003-2A1-16 **Lab Number** 2007-00083- 16 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Powder
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): powder, ,

Layers

#	Layer Type	%	Color	Friability
1	powder	100	Off-white	4
	Total %	100		

Percents of Each Fiber	Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
	<=1%	<=1%	-	-	-	-
Average %	<=1%	<=1%	-	-	-	-
Fiber Identification:	cellulose	glass fiber				

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext
1	cellulose	W	F	N	N	H	+	U
2	glass fiber	CL	D	Y				
3								
4								
5								
6								

Refractive Index Determinations	Oil	Col Par	Col Per	RI Par	RI Per

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2003-2A2-17 **Lab Number** 2007-00083- 17 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability
1	drywall core	100	Off-white	3
	Total %	100		

Percents of Each Fiber	Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
	<=1%	<=1%	-	-	-	-
Average %	<=1%	<=1%	-	-	-	-
Fiber Identification:	cellulose	glass fiber				

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext
1	cellulose	W	F	N	N	H	+	U
2	glass fiber	CL	D	Y				
3								
4								
5								
6								

Refractive Index Determinations	Oil	Col Par	Col Per	RI Par	RI Per

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2003-2A3-18 **Lab Number** 2007-00083- 18 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Wall System Fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability
1	drywall core	100	Off-white	3
	Total %	100		

Percents of Each Fiber	Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
	<=1%	<=1%	-	-	-	-
Average %	<=1%	<=1%	-	-	-	-
Fiber Identification:	cellulose	glass fiber				

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext
1	cellulose	W	F	N	N	H	+	U
2	glass fiber	CL	D	Y				
3								
4								
5								
6								

Refractive Index Determinations	Oil	Col Par	Col Per	RI Par	RI Per

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2003-3A1-19 **Lab Number** 2007-00083- 19 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Acoustic Tile Fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): glass, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	acoustical tile	100	Tan	3	20-30%	40-50%	-	-	-	-
	Total %	100		**Average %**	20-30%	40-50%	-	-	-	-
				Fiber Identification:	cellulose fiber	glass fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2	glass fiber	CL	D	Y									
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2003-3A2-20 **Lab Number** 2007-00083- 20 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Acoustic Tile Fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): glass, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	acoustical tile	100	Tan	3	20-30%	40-50%	-	-	-	-
	Total %	100		**Average %**	20-30%	40-50%	-	-	-	-
				Fiber Identification:	cellulose fiber	glass fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2	glass fiber	CL	D	Y									
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2003-3A3-21 **Lab Number** 2007-00083- 21 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Acoustic Tile Fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): glass, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	acoustical tile	100	Tan	3	20-30%	40-50%	-	-	-	-
	Total %	100		**Average %**	20-30%	40-50%	-	-	-	-
				Fiber Identification:	cellulose fiber	glass fiber				

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2	glass fiber	CL	D	Y									
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2003-7A1-22 **Lab Number** 2007-00083- 22 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Rubbery
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Off-white	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2003-7A2-23 **Lab Number** 2007-00083- 23 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Rubbery
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	Off-white	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2003-7A3-24 **Lab Number** 2007-00083- 24 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Rubbery
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	caulk	100	White	1	n.d.	-	-	-	-	-
	Total %	100		**Average %**	n.d.	-	-	-	-	-
				Fiber Identification:	none					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	none												
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2003-7B1-25 **Lab Number** 2007-00083- 25 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Sticky
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability
1	mastic	100	Yellow	1
	Total %	100		

Percents of Each Fiber	Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
	2-5%	-	-	-	-	-
Average %	2-5%	-	-	-	-	-
Fiber Identification:	cellulose fiber					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext
1	cellulose fiber	W	F	N	N	H	+	U
2								
3								
4								
5								
6								

Refractive Index Determinations

Oil	Col Par	Col Per	RI Par	RI Per

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2003-7B2-26 **Lab Number** 2007-00083- 26 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability
1	mastic	100	Off-white	1
	Total %	100		

Percents of Each Fiber	Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
	n.d.	-	-	-	-	-
Average %	n.d.	-	-	-	-	-
Fiber Identification:	none					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext
1	none							
2								
3								
4								
5								
6								

Refractive Index Determinations

Oil	Col Par	Col Per	RI Par	RI Per

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2003-7B3-27 **Lab Number** 2007-00083- 27 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Adhesive/caulk Non-fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability
1	mastic	100	Off-white	1
	Total %	100		

Percents of Each Fiber	Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
	n.d.	-	-	-	-	-
Average %	n.d.	-	-	-	-	-
Fiber Identification:	none					

Fibers

		Color	Mrph	Iso	Pleo	Bi	Elg	Ext
1	none							
2								
3								
4								
5								
6								

Refractive Index Determinations

Oil	Col Par	Col Per	RI Par	RI Per

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2003-7C1-28 **Lab Number** 2007-00083- 28 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Sprayed Material Fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	coating	100	Off-white	1	10-20%	-	-	-	-	-
	Total %	100		**Average %**	10-20%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2003-7C2-29 **Lab Number** 2007-00083- 29 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Sprayed Material Fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	coating	100	Off-white	1	10-20%	-	-	-	-	-
	Total %	100		**Average %**	10-20%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2003-7C3-30 **Lab Number** 2007-00083- 30 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Sprayed Material Fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	coating	100	Off-white	1	10-20%	-	-	-	-	-
	Total %	100		**Average %**	10-20%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2003-7D1-31 **Lab Number** 2007-00083- 31 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Sprayed Material Fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	coating	100	Gray	1	10-20%	-	-	-	-	-
	Total %	100		**Average %**	10-20%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2003-7D2-32 **Lab Number** 2007-00083- 32 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Sprayed Material Fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	coating	100	Gray	1	10-20%	-	-	-	-	-
	Total %	100		**Average %**	10-20%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Sample IHS-SF2003-7D3-33 **Lab Number** 2007-00083- 33 **Sampled:** 12/20/2006 **Condition:** acceptable
Analyzed By US 1/23/2007 **An?** OK **Apparent Smp Type** Sprayed Material Fibrous Solid
Homogeneous Yes **# Layers** 1 **Pos Layer?** No **# Sub-Samples** 3
Non-Fibrous Components (in approx. decreasing order): filler, binder,

Layers

#	Layer Type	%	Color	Friability	Percents of Each Fiber					
					Fib 1	Fib 2	Fib 3	Fib 4	Fib 5	Fib 6
1	coating	100	Gray	1	10-20%	-	-	-	-	-
	Total %	100		**Average %**	10-20%	-	-	-	-	-
				Fiber Identification:	cellulose fiber					

Fibers		Color	Mrph	Iso	Pleo	Bi	Elg	Ext	Refractive Index Determinations				
									Oil	Col Par	Col Per	RI Par	RI Per
1	cellulose fiber	W	F	N	N	H	+	U					
2													
3													
4													
5													
6													

Sample Analytical Note
Procedure: tweased apart using forceps. Procedure: dissolution of matrix using solvent.

Fr=Friability: 1=very non-friable; 2= non-friable; 3=friable; 4=highly friable
Colors: B=black;BL=blue;BR=brown;CL=clear;G=Green;GY=gray;OR=orange;OW=off-white;PN=pink;PU=purple;R=red;TN=tan;W=white;Y=yellow;V=various
Fiber Morphology: A=fine fibers/bundles, white, sinewy, flexible; B=fine fibers/bundles, w-br, straight, broomed ends; C=fine fibers/bundles, blue, straight, broomed ends; D=fine to coarse fibers, CL-B, brittle; E=coarse fibers,CL or dyed, striated; F=coarse fibers or splinters, W-BR, ribbon-like; G=lath-like or shards, low aspect ratio, may taper
Iso=isotropism - may be yes or no; Pleo=pleochroism - may be yes or no; Bi=birefringence - may be None, Low, Medium or High
Elg=sign of elongation - may be + or -; Ext=extinction - may be Parallel, Oblique, None or Undulating; Oil=medium used to for dispersion staining
Col Par=dispersion staining colors parallel to the fiber (fiber/halo): b/w=black/white; dg/py=dark gray/pale yellow; vg/y=violet gray/yellow; db/ly=dark blue/lemon yellow; vb/g= vivid blue/gold; sb/o=sky blue/orange; pb/r=pale blue/red; gb/dr=gray blue/dark red; w/b=white/black. Col Perp=same only perpendicular to fiber.
RI Par=refractive index parallel to fiber; RI Perp=refractive index perpendicular to fiber

Uwe Steimle

Analyst: UWE .. STEIMLE

Printed: 23-Jan-07

Original Print Date: 23-Jan-07

Larry S. Pierce

Larry S. Pierce, Approved Accreditation Signatory

FIBERQUANT ANALYTICAL SERVICES

Fiberquant, Inc. 5025 S. 33rd St., Phoenix, Arizona 85040 602-276-6139 Fax 602-276-4558

Invoice: JobNumber: 200700083

Invoice Date: *24-Jan-07*
Printed: *24-Jan-07*

Submitted By:

IHI ENVIRONMENTAL
(PHOENIX OFFICE)
4527 N 16th ST STE 105
PHOENIX, AZ 85016-0000
Office Phone: (602) 776-0300
FAX: (602) 776-0301

Billed To:

IHI ENVIRONMENTAL
640 E WILMINGTON AVE
S L C, UT 84106-0000
Office Phone: (801) 466-2223
FAX: (801) 466-9616

Samples: 33 PLM Interim (EPA/600/M4-82-020) PLM analysis for asbestos in bulk smp
Client Job: 07P-3031A / IHS-SF2003 **PO Number:** 07P-3031A **Routing Number:** -
Submitted By: STEPHANIE JURY 1/4/2007

Charges:

Job #	Prod ID	Name	#	TAT	$/ea	SubTotal
200700083	PLM 1/10	PLM 1lyr 10-15 days	33	10-15	$6.00	$198.00

Total. Please Pay This Amount: $198.00

Note:

Thank You for Choosing Fiberquant

Fiberquant is pleased to provide complimentary shipping for each job totaling more than $200.00. Call for Details.

Terms are net 30 days unless other arrangements are made

For jobs awaiting payment from a third party, we appreciate the prompt payment of all applicable invoices as soon as the third party payment is received. The client understands and agrees that any credit granted shall be paid promptly in accordance with any terms and agreements, that Fiberquant may add one and one half percent (1.5%) per month to any overdue balance, and in event of default the client agrees to pay reasonable collection charges and/or attorney fees. Invoices over 90 days old may be turned over for collection unless other arrangements are made. Fiberquant is a member of the American Creditors Association.

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Fiberquant Analytical Services 5025 S. 33rd St.; Phoenix, AZ 85040; Phone: 602-276-6139; FAX: 602-276-4558; Fiberquant@abilnet.com

FIBERQUANT ANALYTICAL SERVICES

Chain-of-Custody Form

Submitted by (Company)	IHI ENVIRONMENTAL
Address	4527 N 16th St #105
City, State, Zip Code	Phoenix AZ 85016
Phone (602) 776-0300	FAX (602) 776-0301

Invoice to (Company)	IHI Env in SLC UT
Address	
City, State, Zip Code	
Phone	FAX

Contact (print)	Stephanie Jury
Sampled by (signature)	
Job Number or Project Name	07P-3031A
PO Number	IHS-SF2003

Sample Method Requested ONLY ONE METHOD per COC			Turn-around-time (circle one) Rush	Norm	Ext.
Asbestos by **PLM**	Improved	Interim	<6 hrs	1-3 days	5-10 days
	Analyze all samples?	Yes No			
	Analyze 'til positive found If so then by	(ATPF) Layer or Sample			
	Single Layer Protocol	Yes Np			
Fibers by **PCM**	7400(Area)	ORM (Personal)	<4 hrs	24 hrs	3-5 days
Asbestos by **TEM**	AIR: AHERA	Mod. AHERA	<6 hrs	24 hrs	3-5 days
	Water*: Water	Sludge	1-2 days	3-5 days	10 days
	Annex2 : Chatfield	Full			
	Vacuum Dust (ASTM)		3.5 days	5-10 days	N/A
Metals by **FLAA**	Analyte: Cd Cr Cu Ni Pb Zn		<6 hrs	2-3 days	N/A
	Matrix: Filter: MCE FG				
	Paint: by Area by Weight				
	Soil				
	Wipe				
	Initial here certifying wipes used are ASTM E1792 compliant				
Fungi	Air Sample: Zefon Other		<6 hrs	1-2 days	N/A
	ID/Count: Bulk Swab				
	Tape: Qualitative (%)				
	Tape:Quantitative (cm2)				
	Culturable Air Bulk/Dust Swab		7 days Only		
Dust	NIOSH 500		<4 hrs	24 hrs	N/A
Other			**Call**	**Call**	

Review of Analysis Request _____ Date_____

Sample Number	Description/Location (include agar type/maker/exp. Date)	Sample Date	Sample Time	Vol/Area
1) IHS-SF2003-1A1-1	Sheet Vinyl (Pink)	12/20/06		
2) 1A2-2	↓			
3) 1A3-3				
4) 1B1-4	Carpet Mastic (Yellow)			
5) 1B2-5	↓			
6) 1B3-6				
7) 1C1-7	Sheet Vinyl (Yellow)			
8) 1C2-8	↓			
9) 1C3-9				
10) 1D1-10	Sheet Vinyl (Blue)			
11) 1D2-11	↓			
12) 1D3-12				
13) 1E1-13	Carpet Mastic (Green)			
14) 1E2-14	↓			
15) 1E3-15				
16) 2A1-16	Gypsum Drywall			
17) 2A2-17	↓			
18) 2A3-18				
19)				
20)				

1)Relinquished by:	Date: 1/4/07	Time: 3:30p	3)Relinquished by:	Date:	Time:
2)Received by: Kathy	Date: 1-4-07	Time: 3:30	4)Received by:	Date:	Time:
* **TEM Water: Sampler's name Required by State of Arizona**	Print Name				

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Fiberquant Analytical Services 5025 S. 33rd St.; Phoenix, AZ 85040; Phone: 602-276-6139; FAX: 602-276-4558; Fiberquant@abilnet.com

FIBERQUANT ANALYTICAL SERVICES

Chain-of-Custody Form

Submitted by (Company)	IHI ENVIRONMENTAL
Address	4527 N 16th St #105
City, State, Zip Code	Phoenix AZ 85016
Phone (602) 776-0300	FAX (602) 776-0301

Invoice to (Company)	IHI Env in SLC UT
Address	
City, State, Zip Code	
Phone	FAX

Contact (print)	Stephanie Jury
Sampled by (signature)	
Job Number or Project Name	07P-3031A
PO Number	IHS-SF2003

Sample Method Requested ONLY ONE METHOD per COC			Turn-around-time (circle one) Rush	Norm	Ext.
Asbestos by **PLM**	Improved	Interim	<6 hrs	1-3 days	10-15 ~~30~~ days
	Analyze all samples?	Yes No			
	Analyze 'til positive found If so then by	(ATPF) Layer or Sample			
	Single Layer Protocol	Yes Np			
Fibers by **PCM**	7400(Area)	ORM (Personal)	<4 hrs	24 hrs	3-5 days
Asbestos by **TEM**	AIR: AHERA	Mod. AHERA	<6 hrs	24 hrs	3-5 days
	Water*: Water	Sludge	1-2 days	3-5 days	10 days
	Annex2 : Chatfield	Full			
	Vacuum Dust (ASTM)		3.5 days	5-10 days	N/A
Metals by **FLAA**	Analyte: Cd Cr Cu Ni Pb Zn		<6 hrs	2-3 days	N/A
	Matrix: Filter: MCE FG				
	Paint: by Area by Weight				
	Soil				
	Wipe				
	Initial here certifying wipes used are ASTM E1792 compliant _____				
Fungi	Air Sample: Zefon	Other	<6 hrs	1-2 days	N/A
	ID/Count: Bulk	Swab			
	Tape: Qualitative (%)				
	Tape:Quantitative (cm2)				
	Culturable Air Bulk/Dust Swab		7 days Only		
Dust	NIOSH 500		<4 hrs	24 hrs	N/A
Other			Call	Call	

Review of Analysis Request _____ Date_____

Sample Number	Description/Location (include agar type/maker/exp. Date)	Sample Date	Sample Time	Vol/Area
1) IHS-SF2003-3A1-19	2x4' Ceiling Tile	12/20/06		
2) 3A2-20	↓			
3) 3A3-21	↓			
4) 7A1-22	Caulking			
5) 7A2-23	↓			
6) 7A3-24	↓			
7) 7B1-25	Covebase mastic			
8) 7B2-26	↓			
9) 7B3-27	↓			
10) 7C1-28	Sink Undercoating (white)			
11) 7C2-29	↓			
12) 7C3-30	↓			
13) 7D1-31	Sink Undercoating (Gray)			
14) 7D2-32	↓			
15) 7D3-33	↓			
16)				
17)				
18)				
19)				
20)				

1)Relinquished by:	Date: 1/4/07	Time: 3:30p	3)Relinquished by:	Date:	Time:
2)Received by: Kathy Knowles	Date: 1-4-07	Time: 3:30	4)Received by:	Date:	Time:
* TEM Water: Sampler's name Required by State of Arizona	Print Name				

APPENDIX 4
Certification
Donovan Denny
Landon Johnson
Stephanie Jury
James Marshall
Suzette Numkena

D 7497

THE ASBESTOS INSTITUTE

Certifies that

Donavan Denny

has attended the EPA approved course

AHERA
Building Inspector
March 6-8, 2006

and successfully passed the competency exam.

Date of Examination: ***March 8, 2006***

Date of Expiration: ***March 8, 2007***

Director

Approved Instructor

THE ASBESTOS INSTITUTE
8102 North 23rd Avenue
Suite A
Phoenix, AZ 85021-4962
602-864-6564

This training meets all requirements for asbestos accreditation under TSCA Title II.

D 7503

THE ASBESTOS INSTITUTE

Certifies that

Landon Johnson

has attended the EPA approved course

AHERA
Building Inspector
March 6-8, 2006

and successfully passed the competency exam.

Date of Examination: ***March 8, 2006***

Date of Expiration: ***March 8, 2007***

Director

Approved Instructor

THE ASBESTOS INSTITUTE
8102 North 23rd Avenue
Suite A
Phoenix, AZ 85021-4962
602-864-6564

This training meets all requirements for asbestos accreditation under TSCA Title II.

D 8017

THE ASBESTOS INSTITUTE

Certifies that

Stephanie Jury

has attended the EPA approved course

AHERA Refresher
Building Inspector
June 9, 2006

and successfully passed the competency exam.

Date of Examination: ***June 9, 2006***

Date of Expiration: ***June 9, 2007***

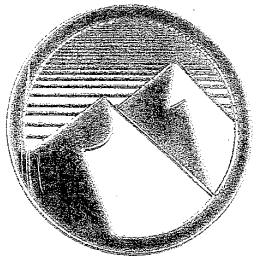
Director

Approved Instructor

THE ASBESTOS INSTITUTE
8102 North 23rd Avenue
Suite A
Phoenix, AZ 85021-4962
602-864-6564

This training meets all requirements for asbestos accreditation under TSCA Title II.

ROCKY MOUNTAIN CENTER FOR
OCCUPATIONAL AND
ENVIRONMENTAL HEALTH
Department of Family & Preventive Medicine
University of Utah
391 Chipeta Way, Suite C
Salt Lake City UT 84108
Phone: (801) 581-4055
Fax: (801) 585-5275



THIS CERTIFIES THAT

James Marshall

HAS COMPLETED THE REQUISITE TRAINING FOR ASBESTOS ACCREDITATION UNDER TSCA TITLE II

ATTENDED AN ANNUAL REFRESHER COURSE IN

PRACTICES AND PROCEDURES IN
ASBESTOS ABATEMENT

Asbestos Inspector Refresher

California Course Approval Number for Asbestos Inspector Refresher
#CA-004-06

DATE: December 2, 2005
NUMBER: 250702
EXPIRES: December 2, 2006
CREDITS: 0.350 CEUs / .50 ABIH CM points

Connie Crandall, MBA, MA
Continuing Education Director

D 8208

THE ASBESTOS INSTITUTE

Certifies that

Suzette Valencia Numkena

has attended the EPA approved course

AHERA
Building Inspector
July 10-12, 2006

and successfully passed the competency exam.

Date of Examination: *July 12, 2006*

Date of Expiration: *July 12, 2007*

Director

Approved Instructor

THE ASBESTOS INSTITUTE
8102 North 23rd Avenue
Suite A
Phoenix, AZ 85021-4962
602-864-6564

This training meets all requirements for asbestos accreditation under TSCA Title II.